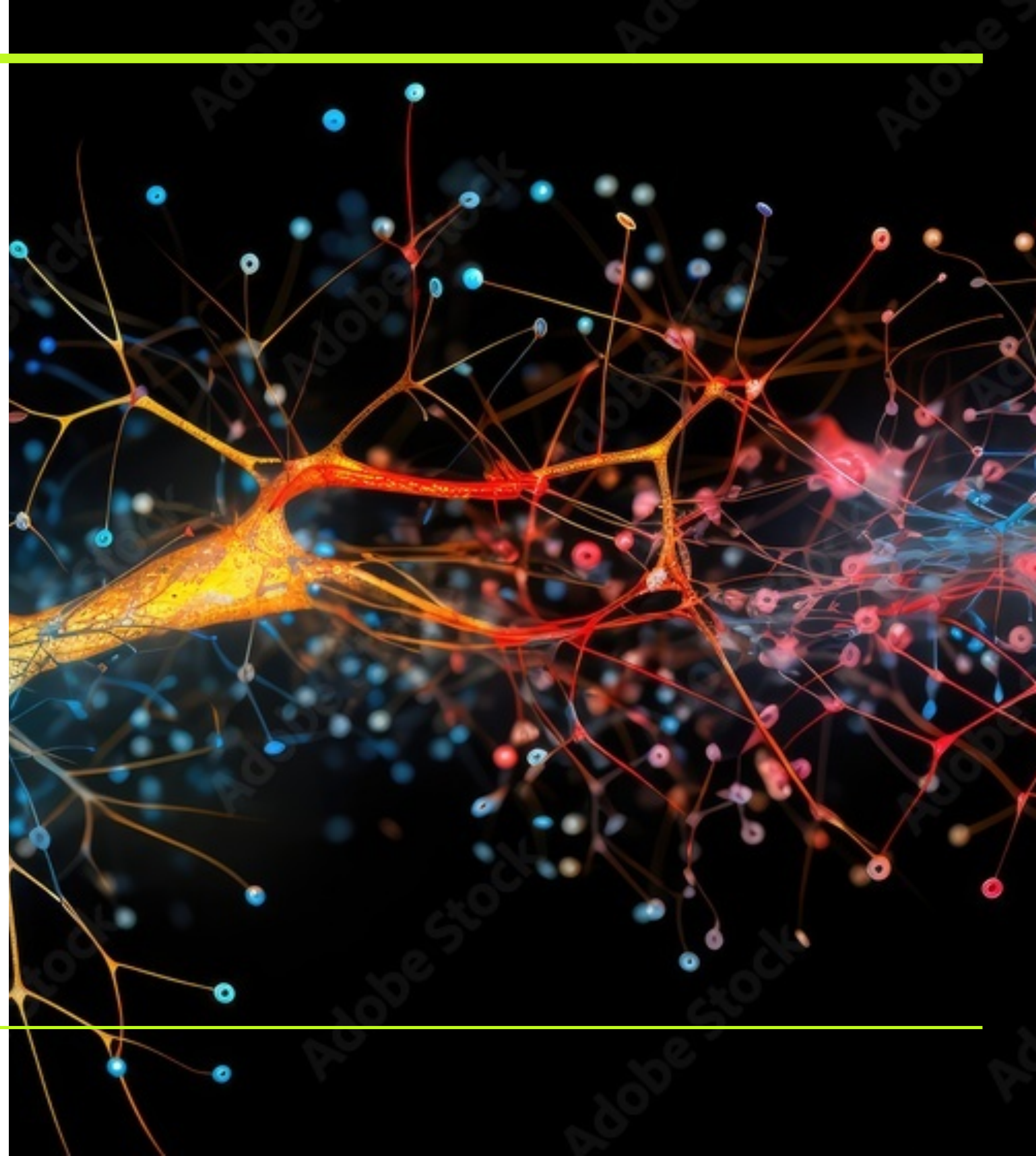


系统和计算神经科学导论

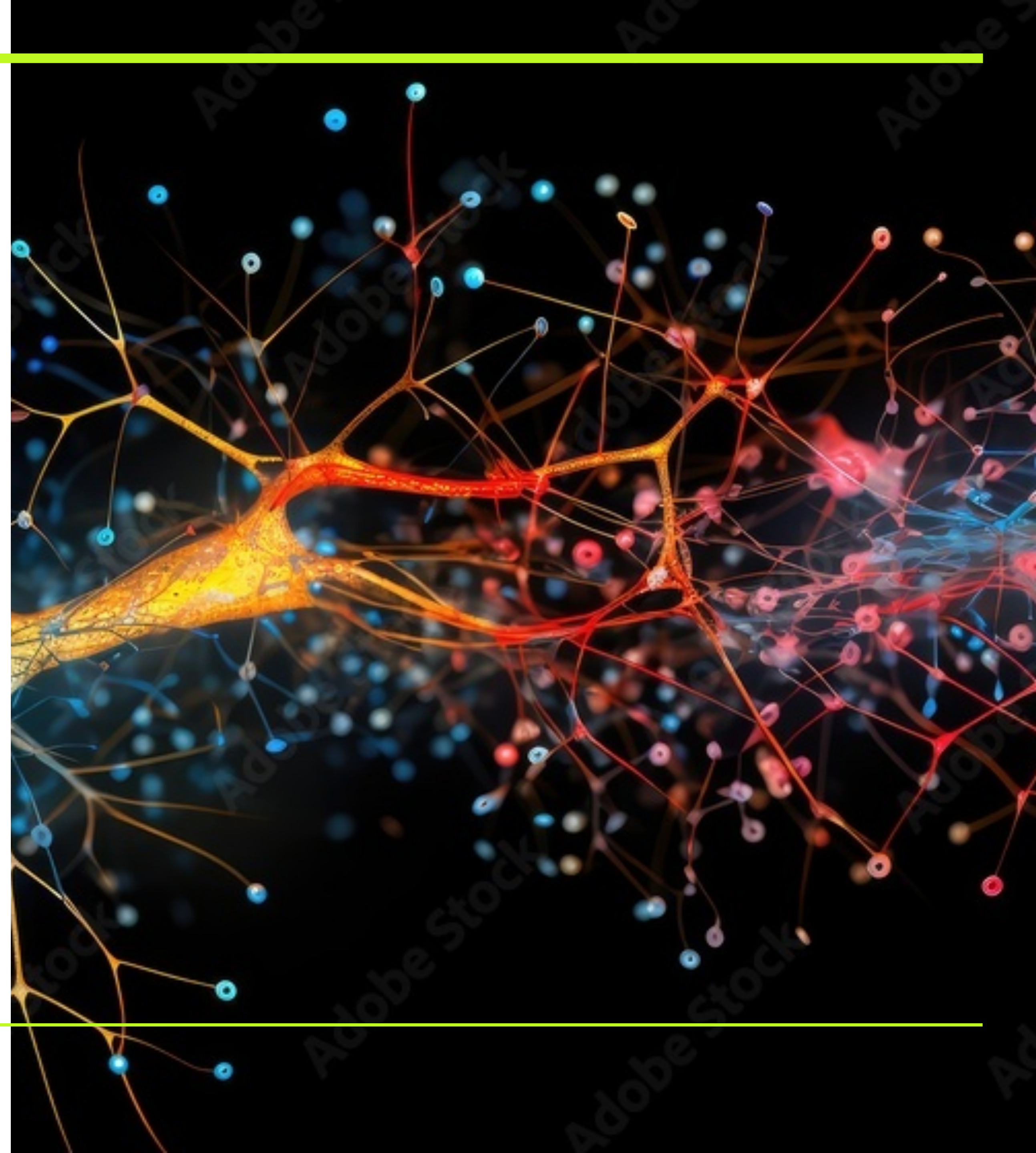
四、神经编解码

中国科学院脑科学与智能技术卓越创新中心（神经所） 杨天明 TYANG@ION.AC.CN

-
- ❖ **Basics**
 - ❖ **Encoding**
 - ❖ **Decoding**
 - ❖ **Information Theory**
 - ❖ **Population Analysis**
-



-
- ❖ **Basics**
 - ❖ Encoding
 - ❖ Decoding
 - ❖ Information Theory
 - ❖ Population Analysis
-



Poisson

$$P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}, \quad k = 0, 1, 2, \dots$$

Spikes are:

- discrete, sparse events
 - stochastic
 - memoryless (no refractory period)
 - constant rate within a certain time window
-

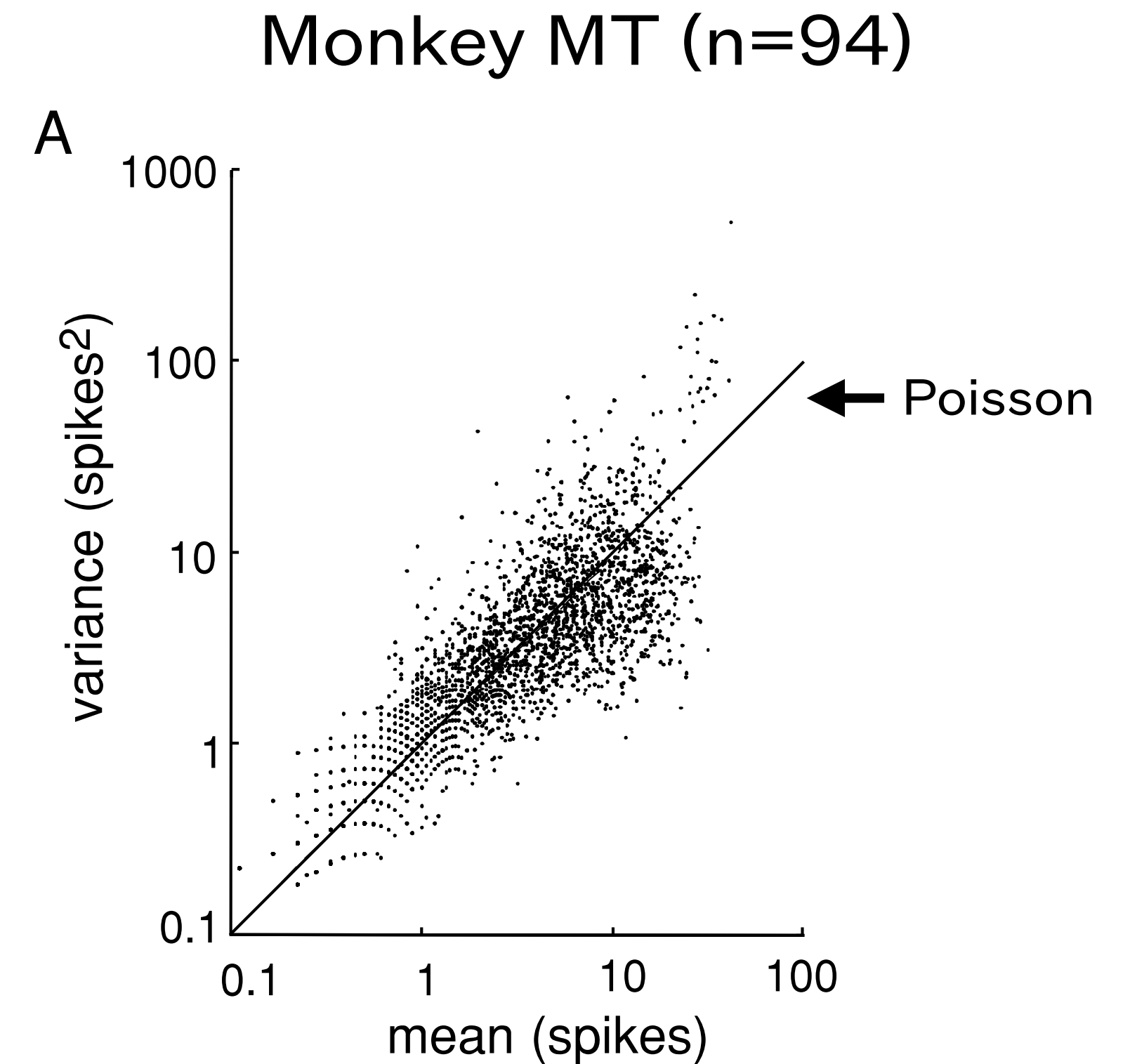
Poisson

$$P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}, \quad k = 0, 1, 2, \dots$$

Fano Factor $\frac{\sigma_n^2}{\langle n \rangle}$ Poisson:1; Real neurons <1

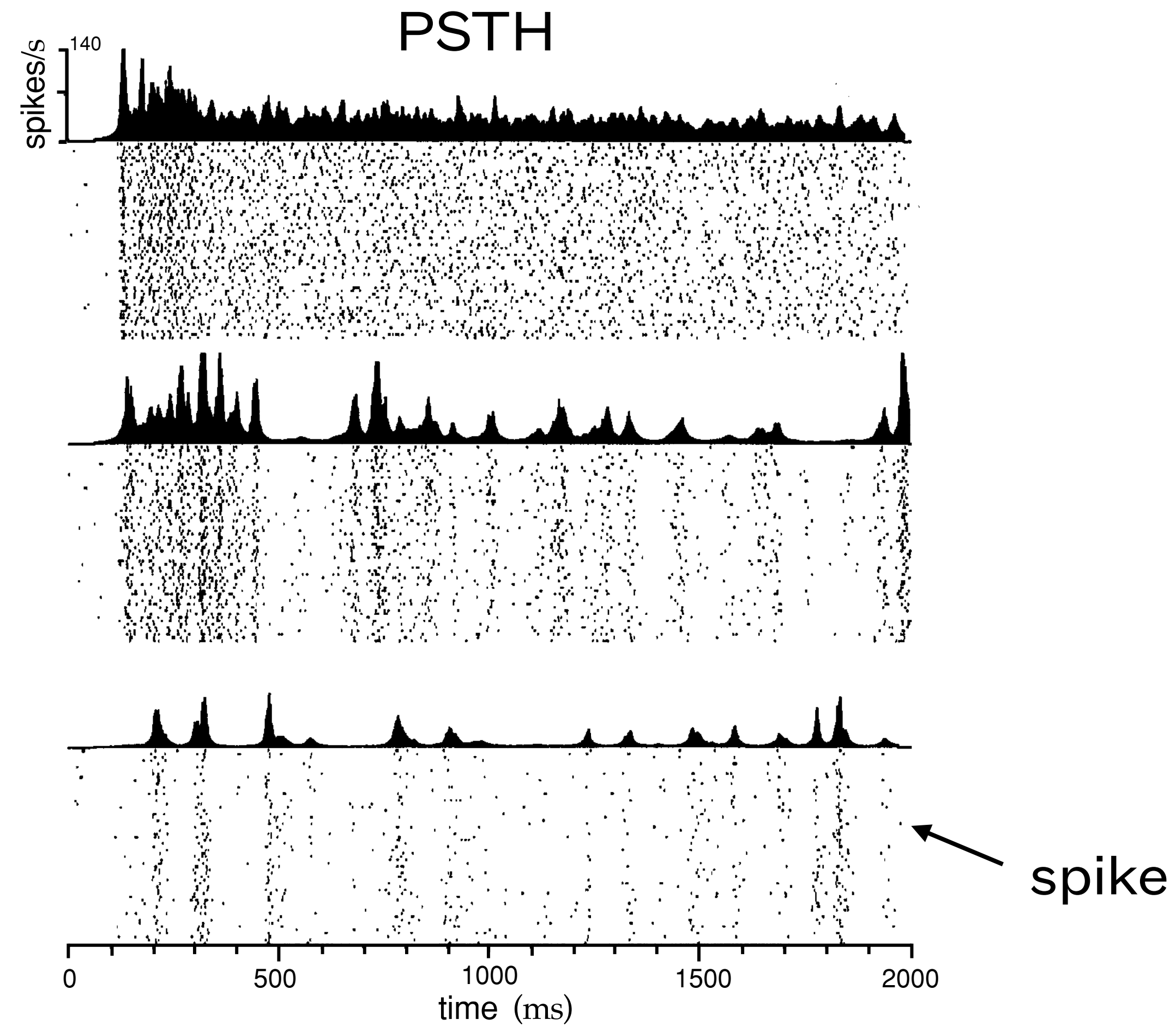
Inhomogeneous Poisson Process

$$P(X = k) = \frac{\lambda(t)^k e^{-\lambda(t)}}{k!}, \quad k = 0, 1, 2, \dots$$



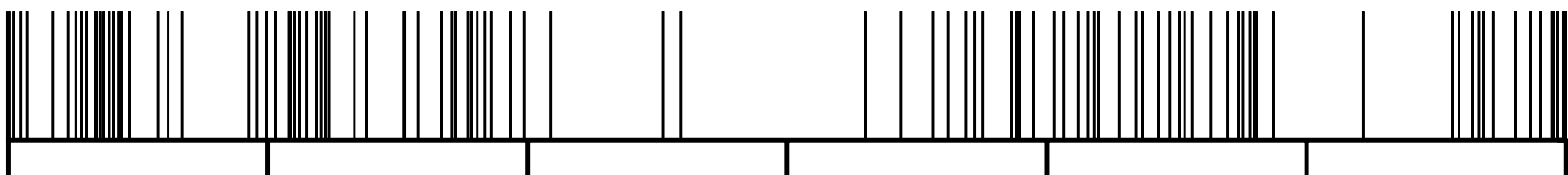
O'Keefe et al., 1997

Peri-Stimulus Time Histogram



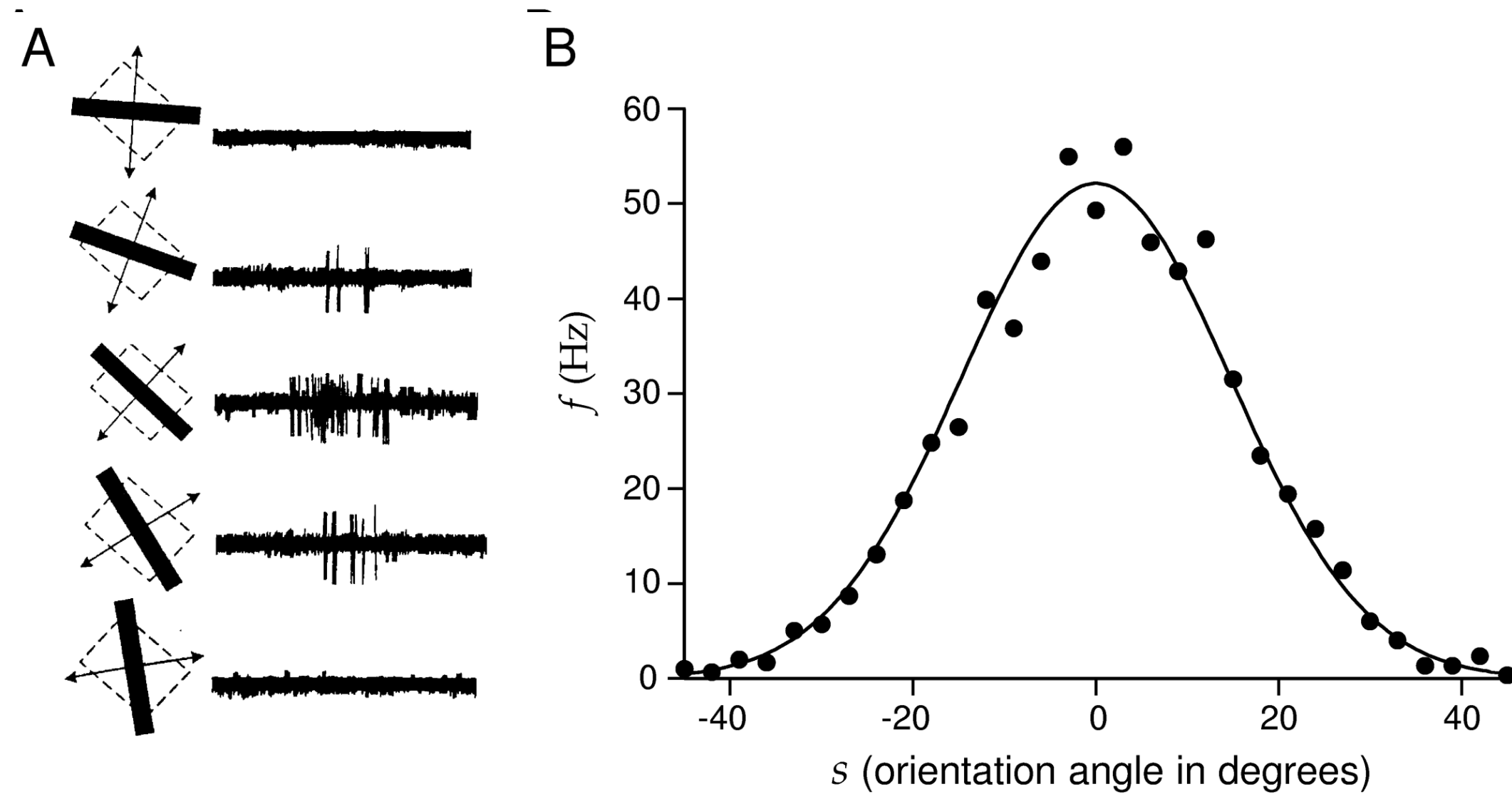
A

spikes



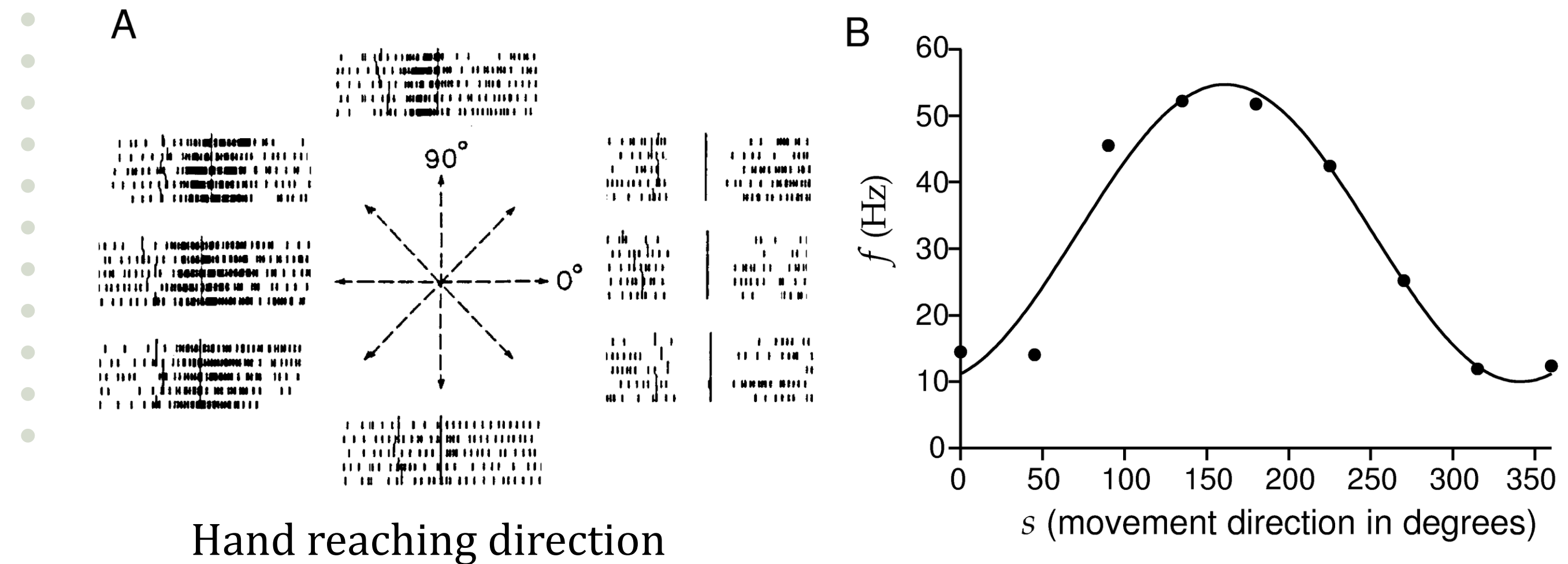
原始spike序列

Tuning curve



A V1 neuron's direction tuning

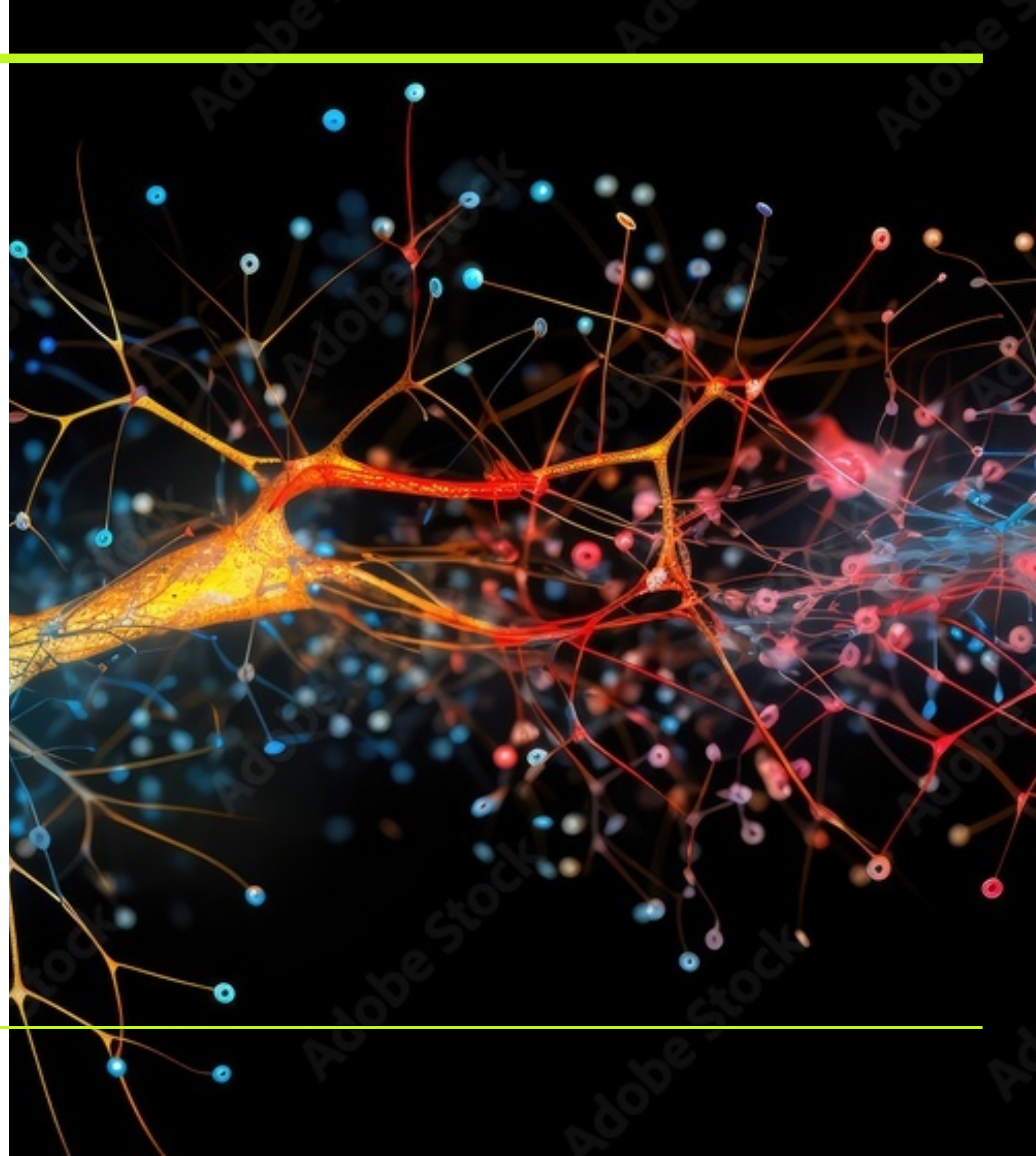
$$\langle r | s \rangle = f(s)$$



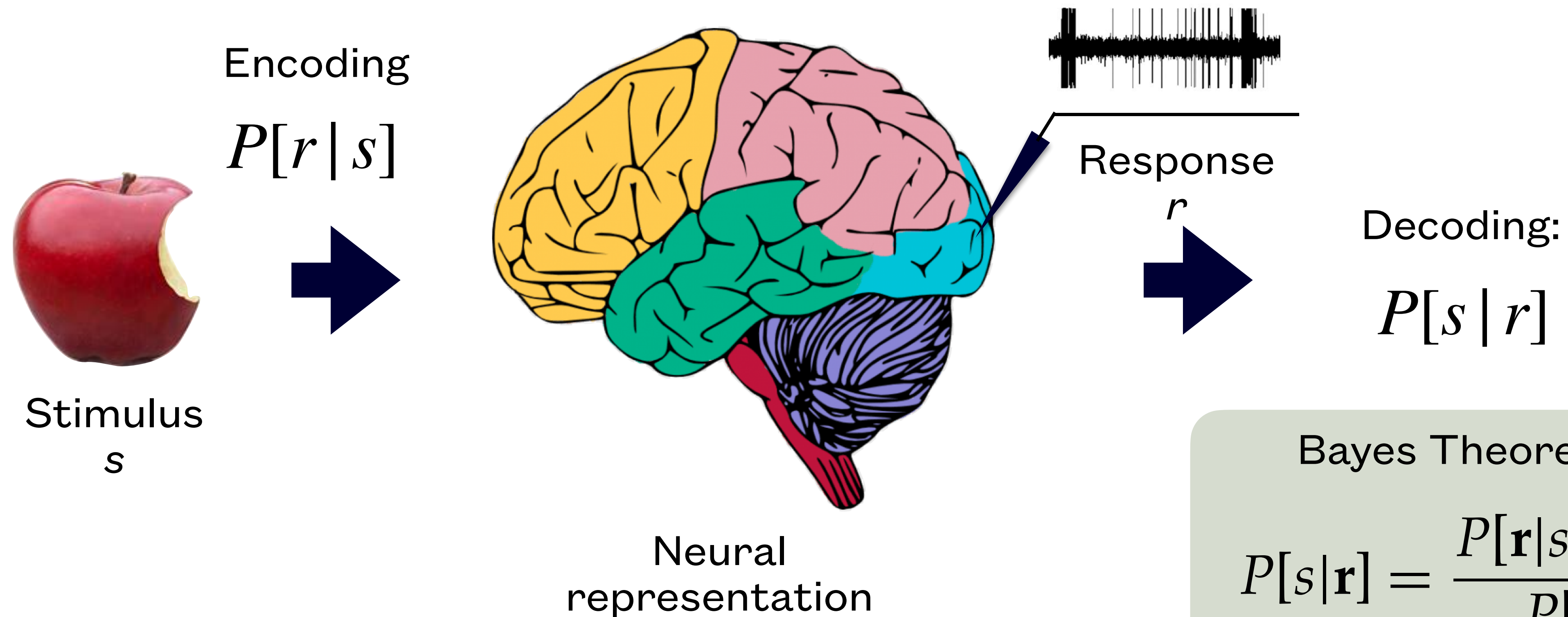
An M1 neuron's movement direction tuning

$$r \sim \text{Poisson}(f(s))$$

-
- ❖ Basics
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 - ❖ Information Theory
 - ❖ Population Analysis
-



Encoding and Decoding

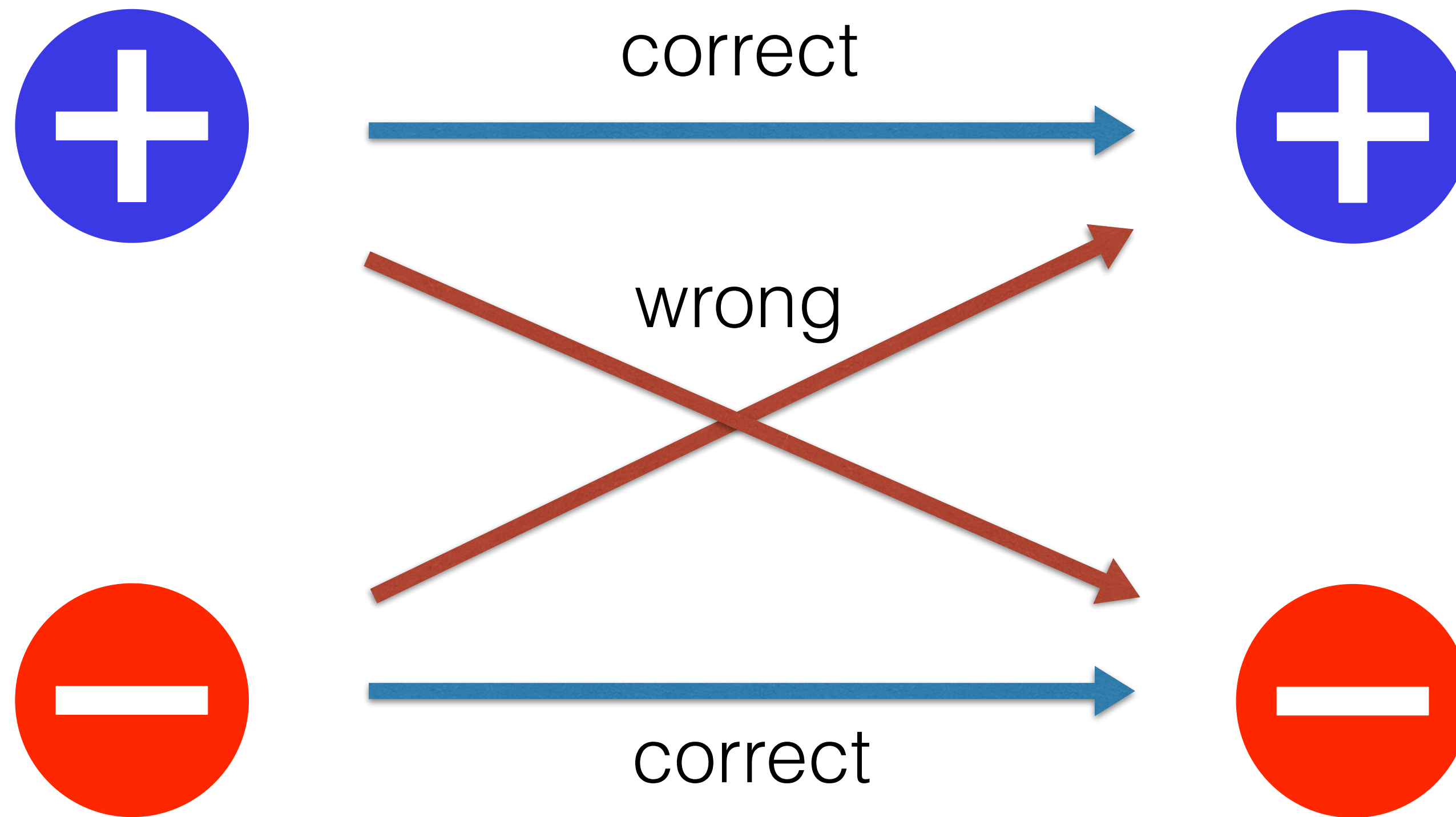


Bayes Theorem

$$P[s|\mathbf{r}] = \frac{P[\mathbf{r}|s]P[s]}{P[\mathbf{r}]}$$

Signal Detection Theory

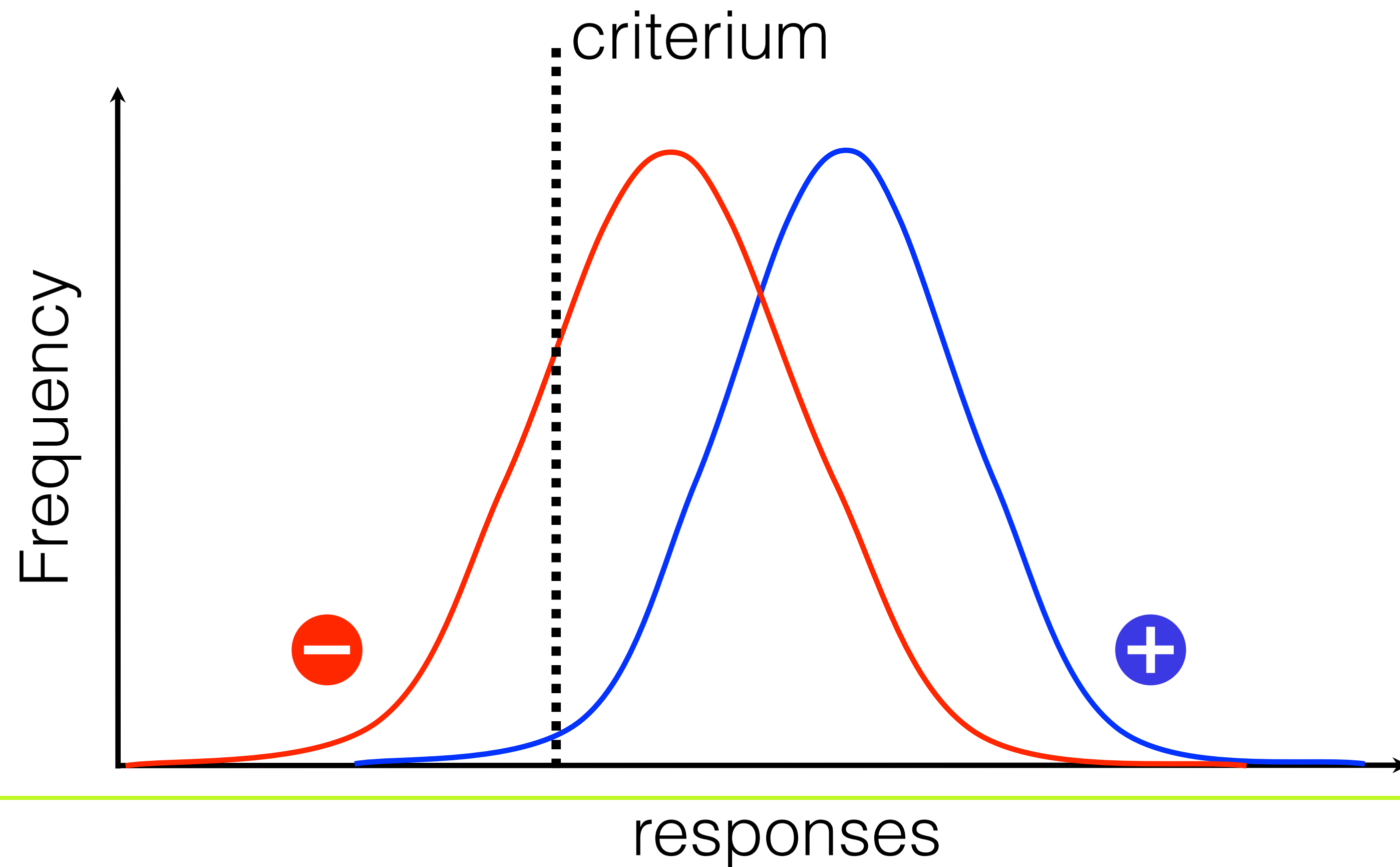
Green and Swets, 1966



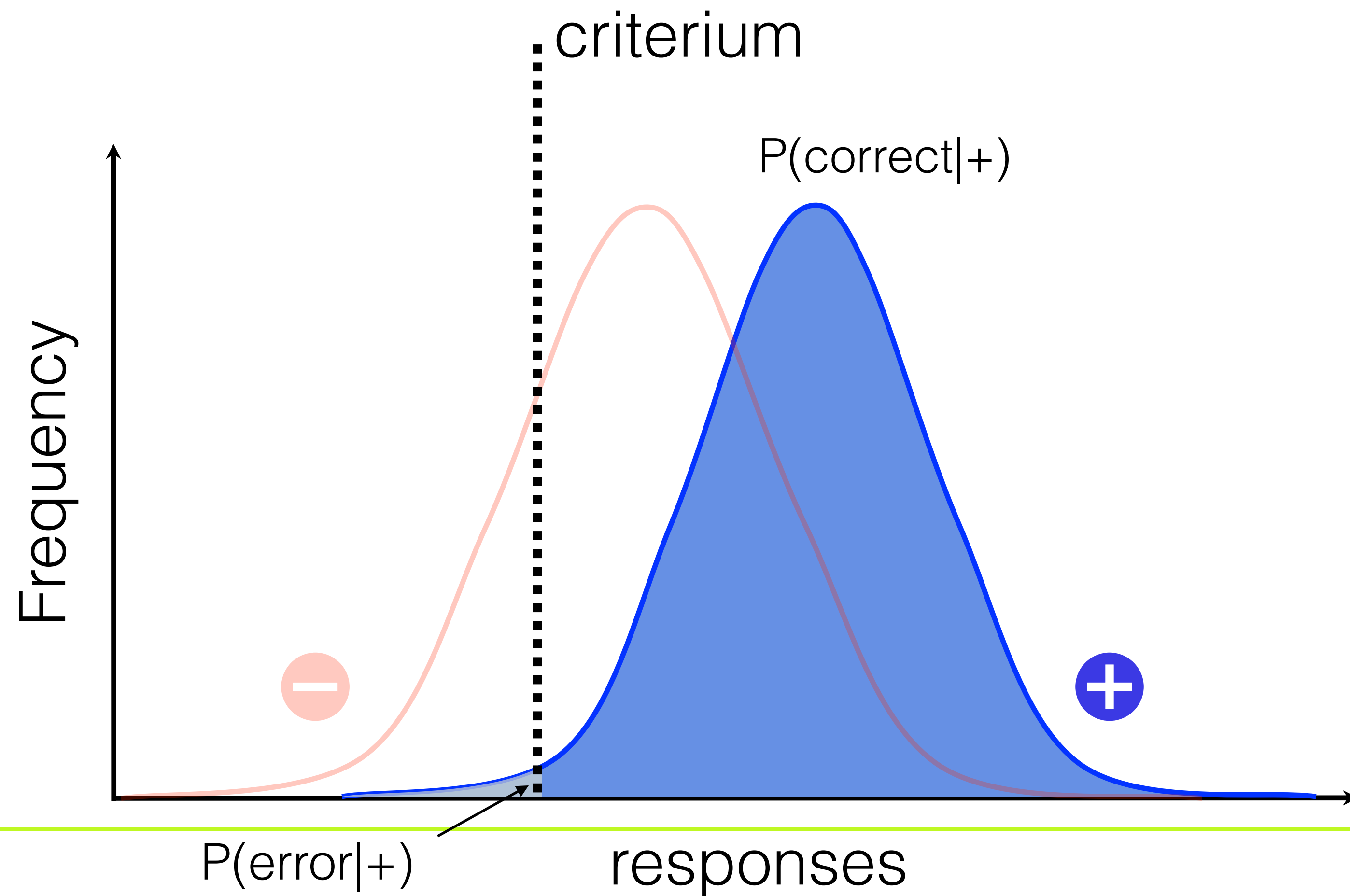
Ground Truth

Decoder
Output

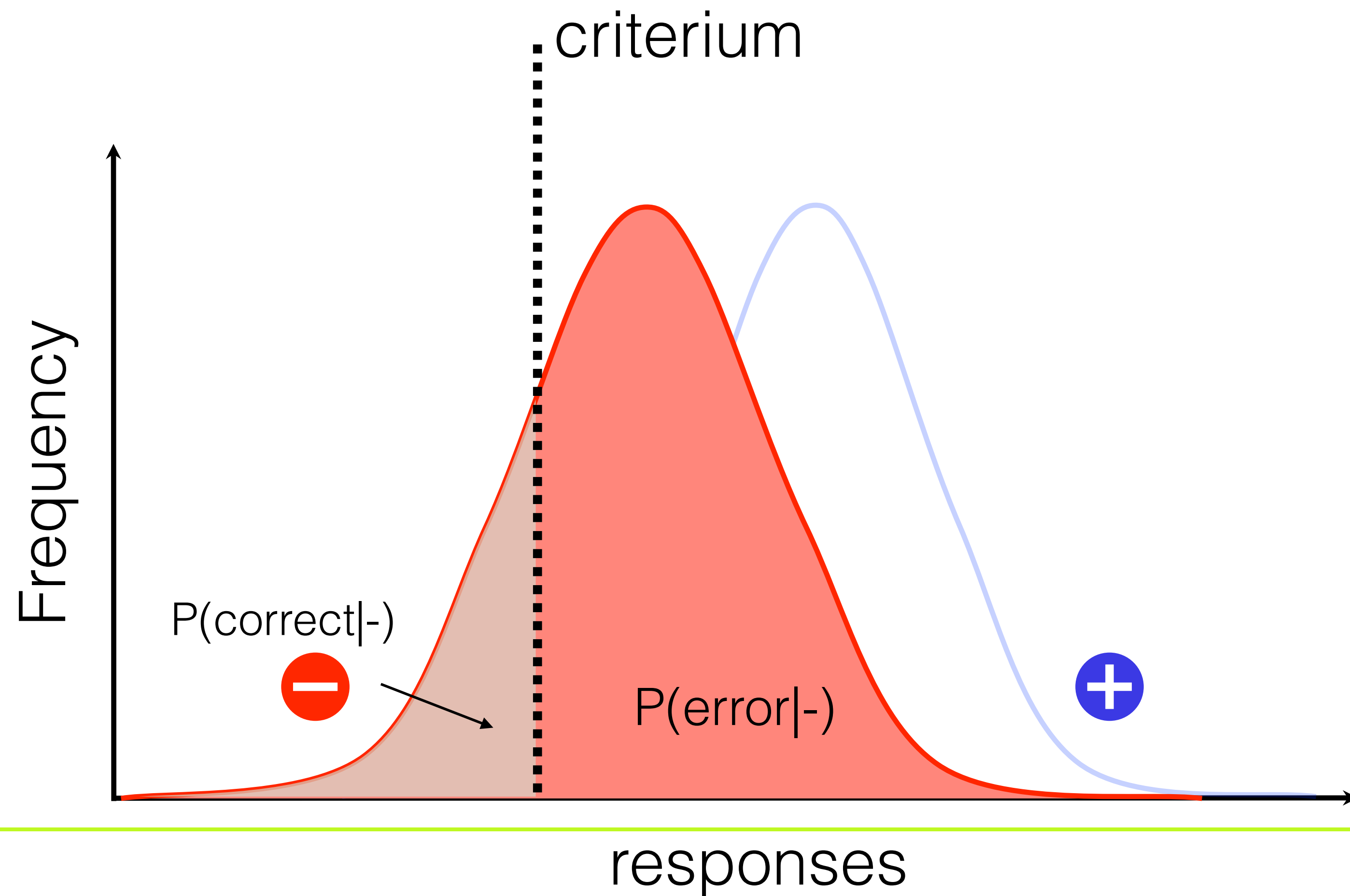
Signal Detection Theory



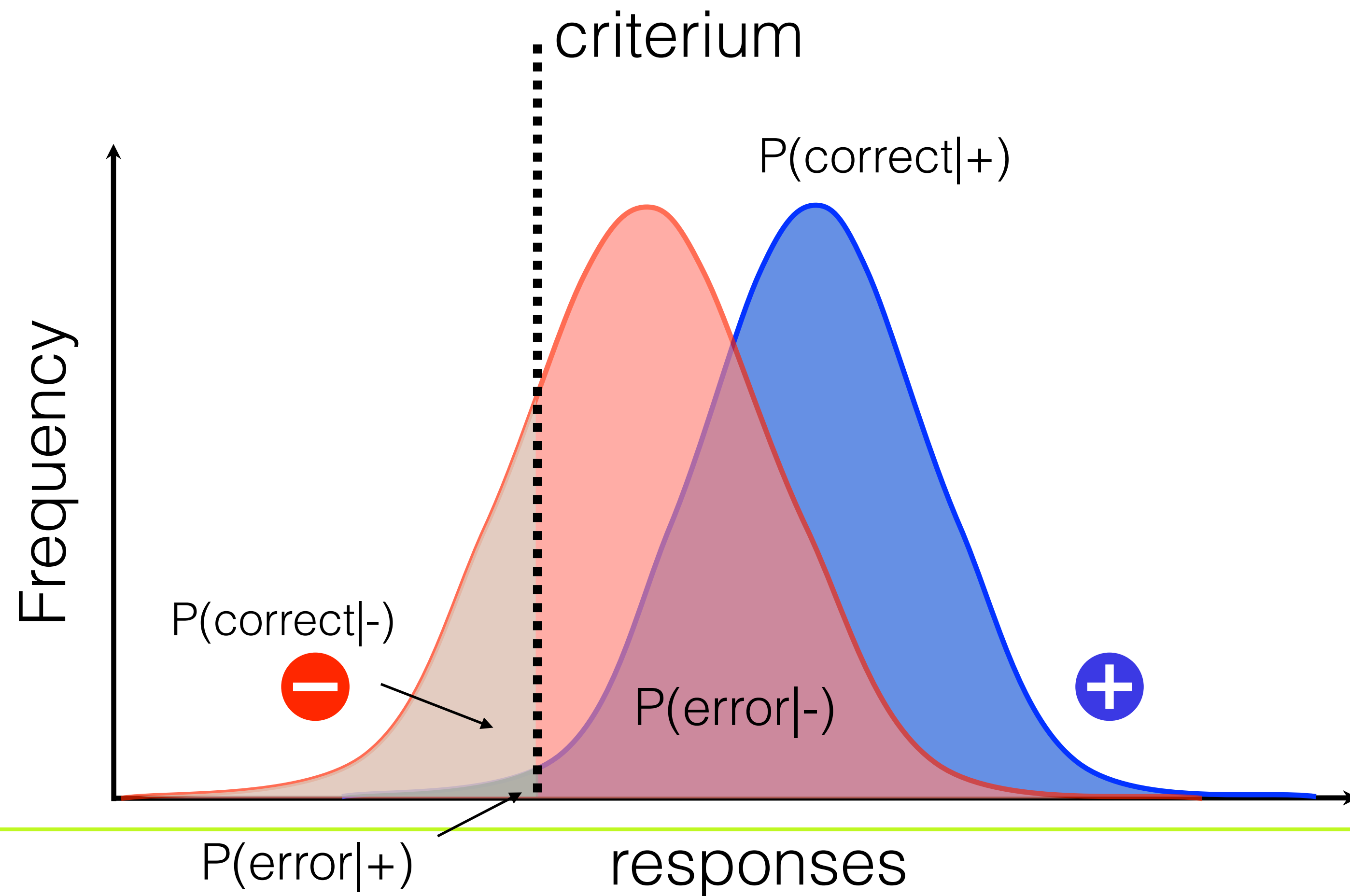
Signal Detection Theory



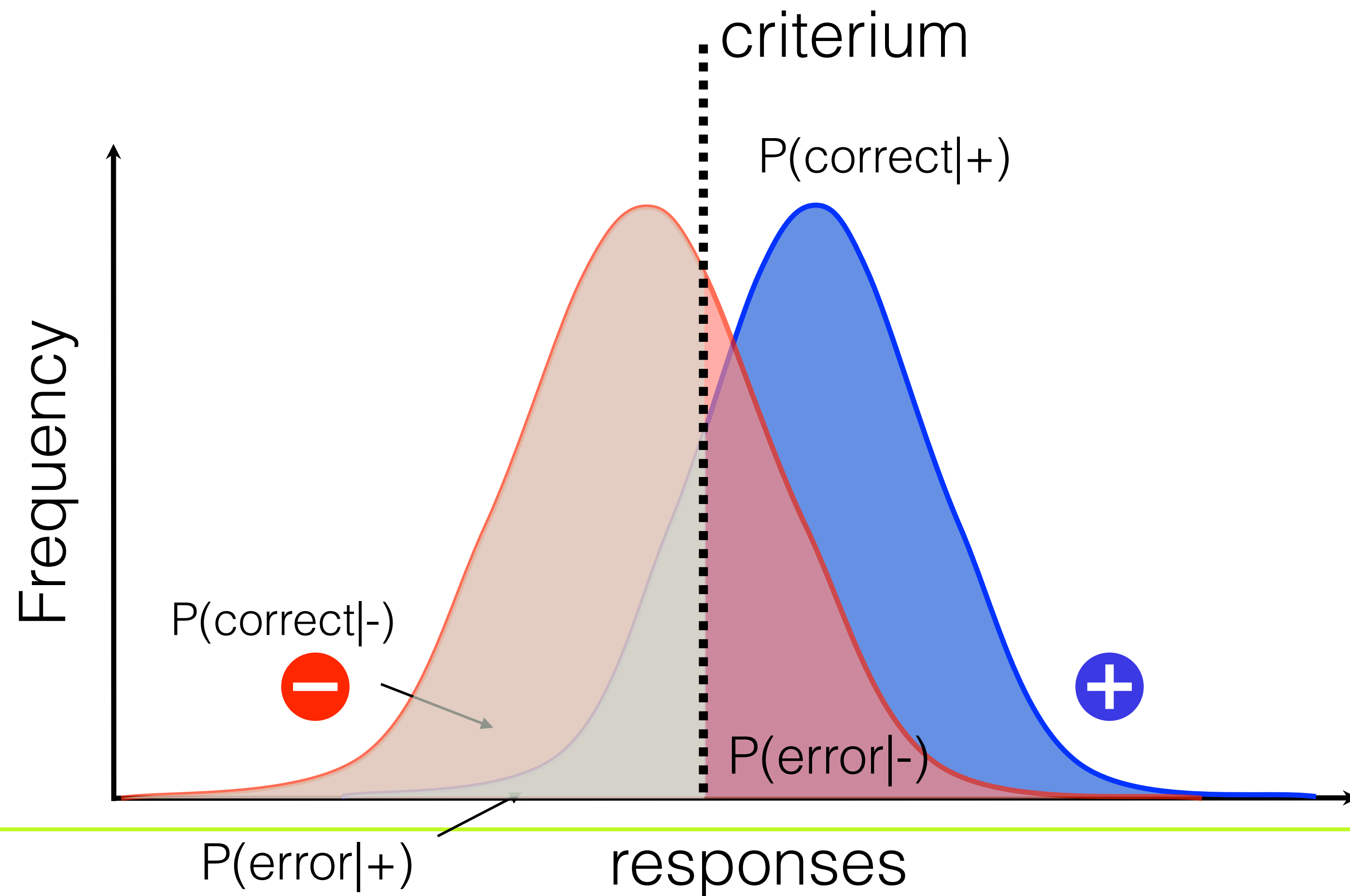
Signal Detection Theory



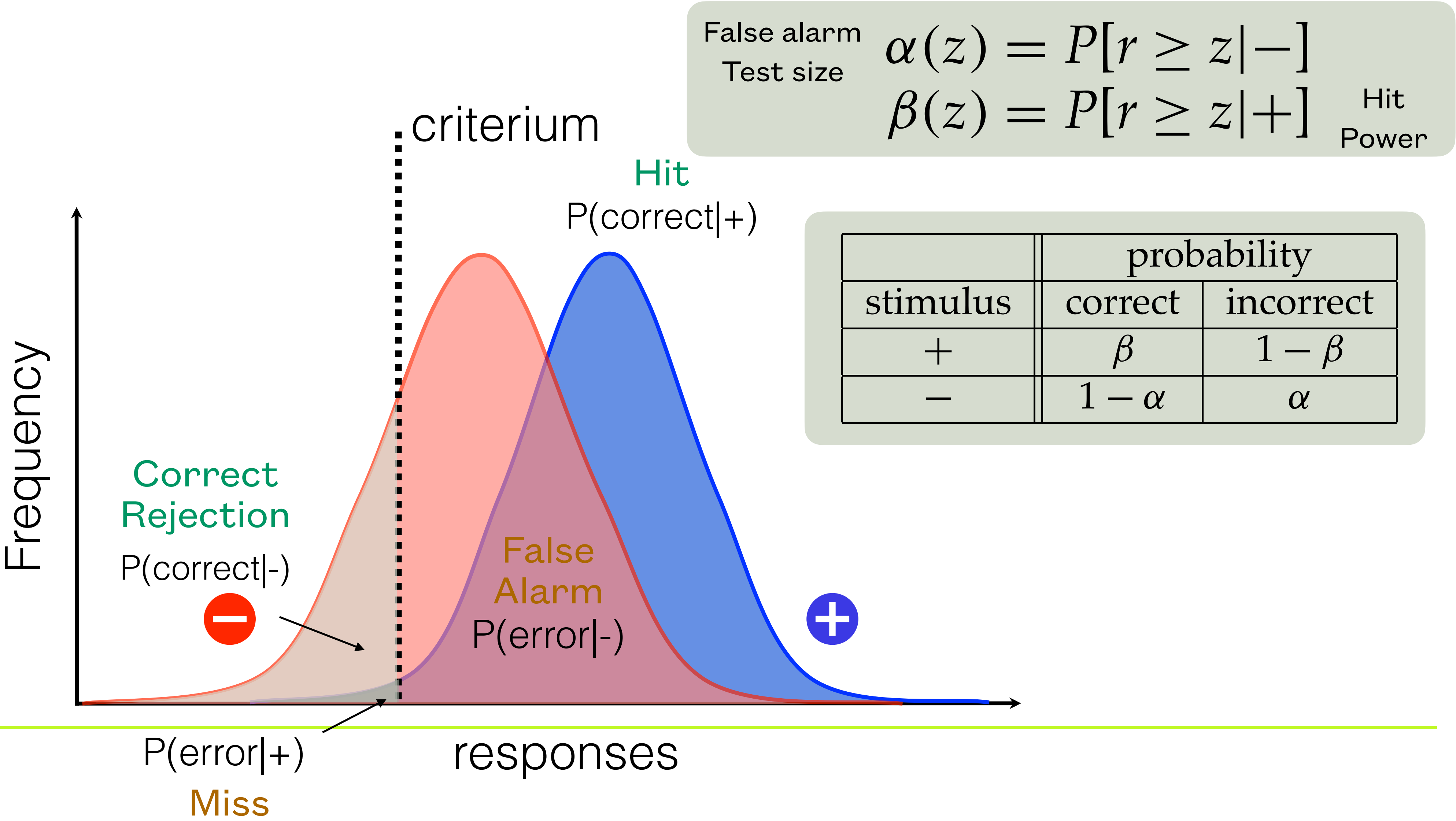
Signal Detection Theory



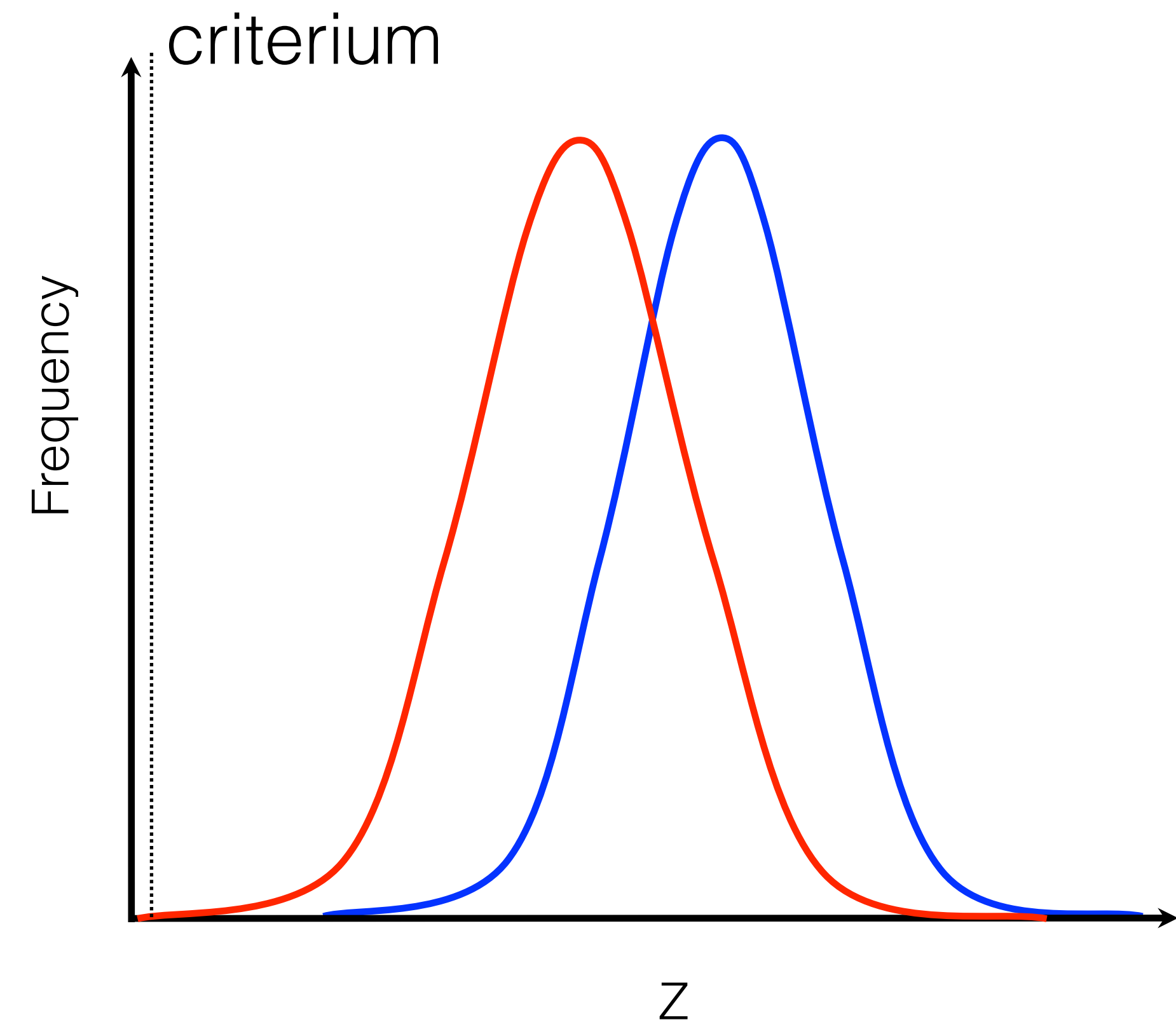
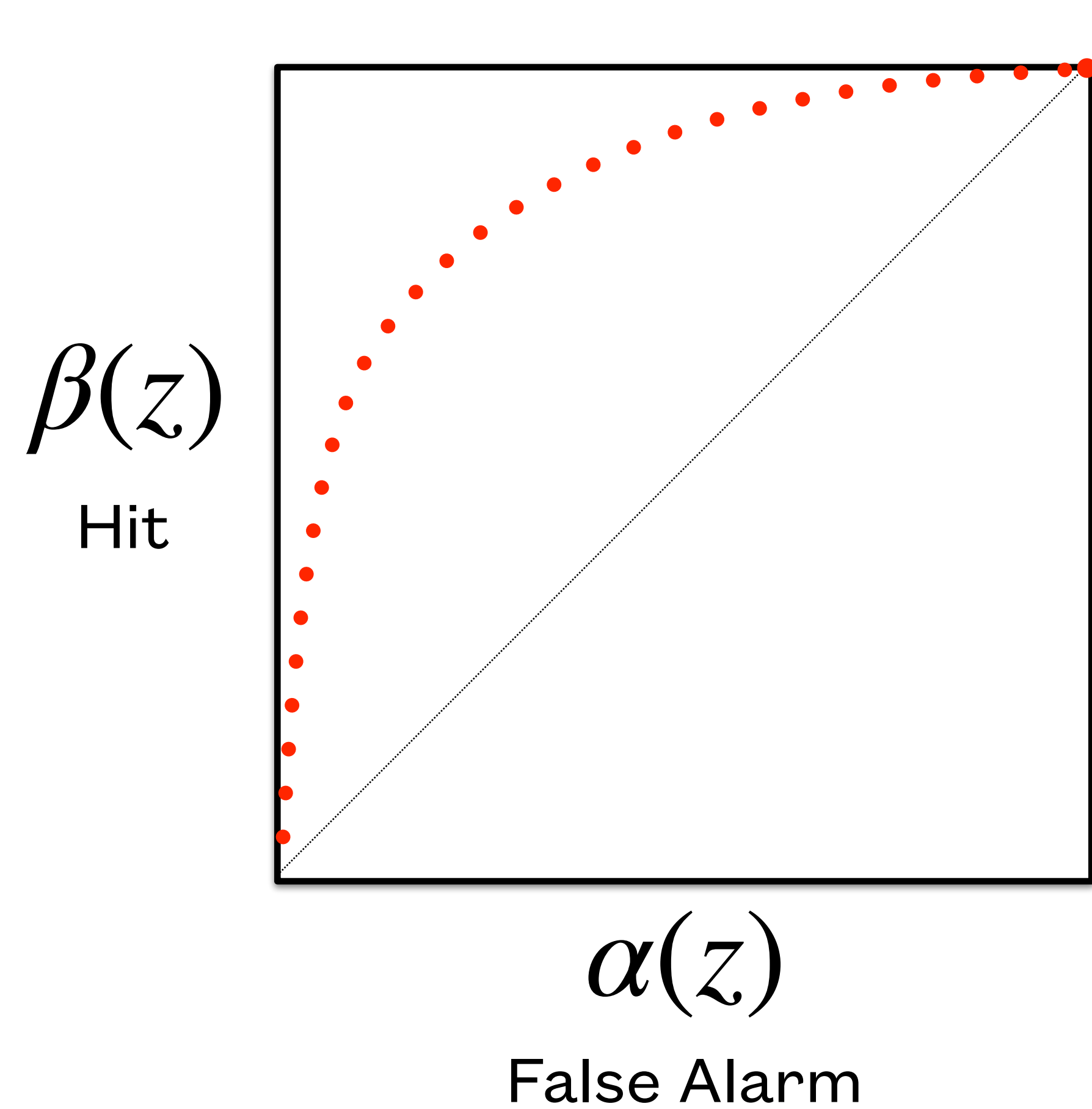
Signal Detection Theory



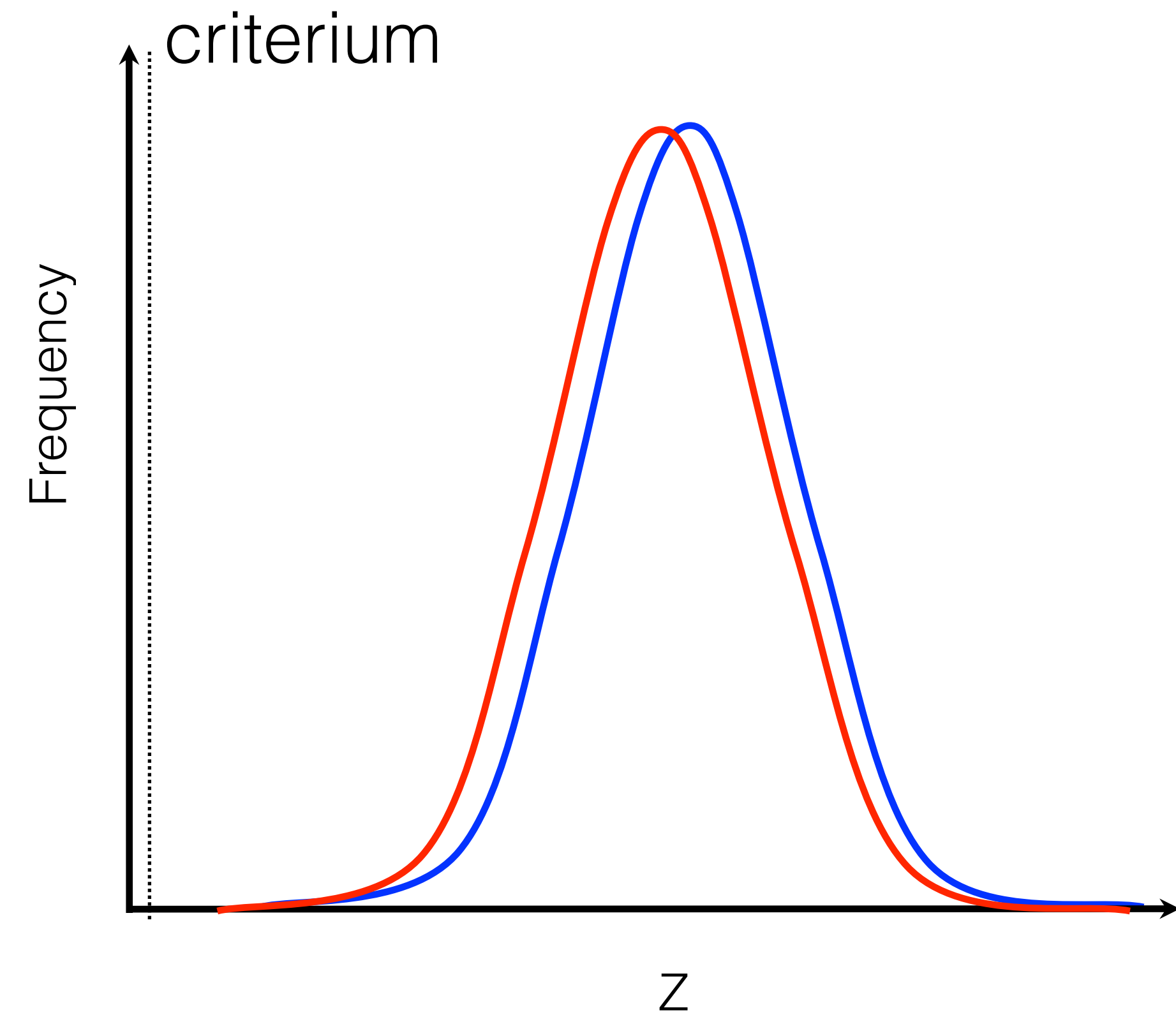
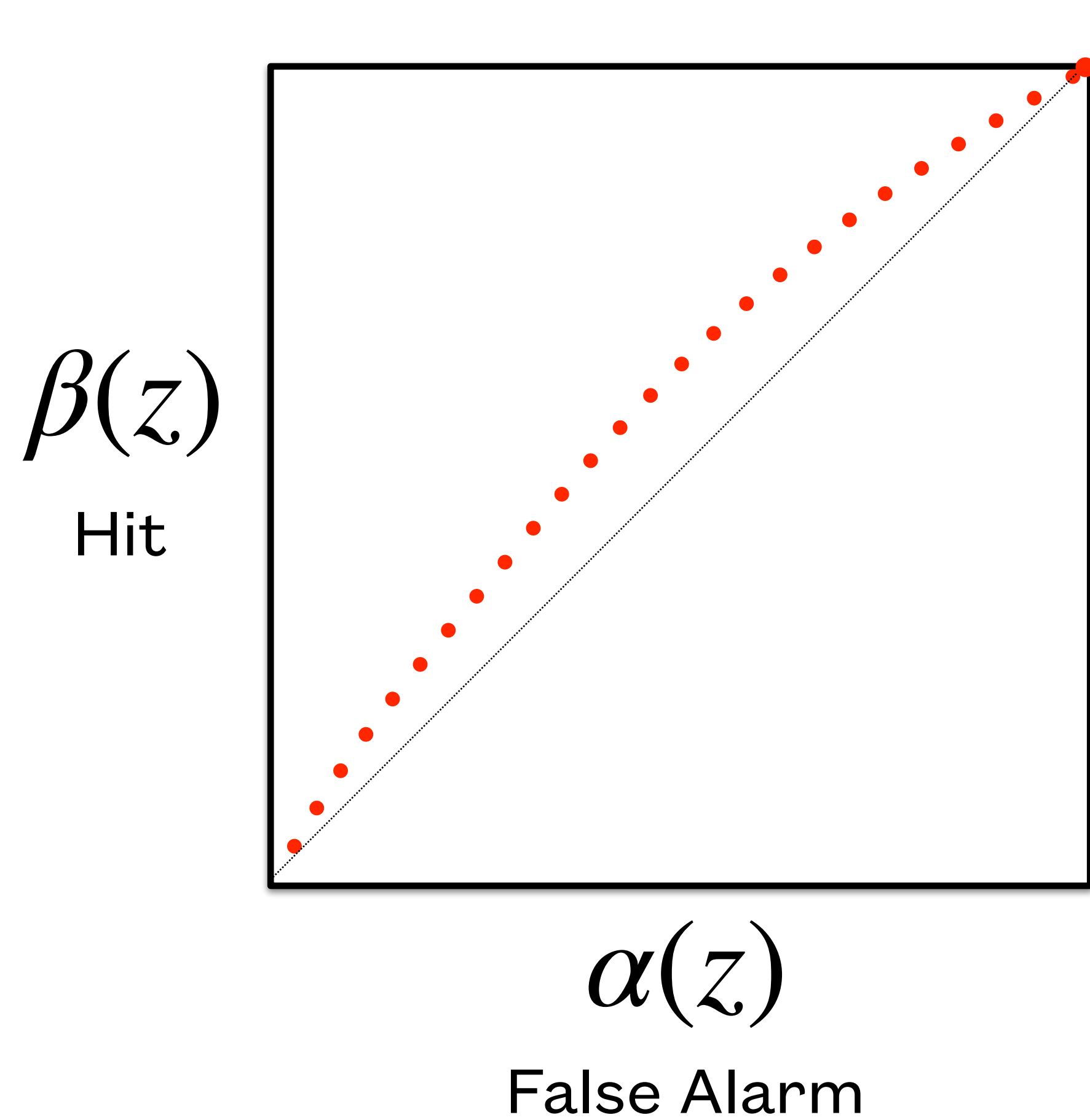
Signal Detection Theory



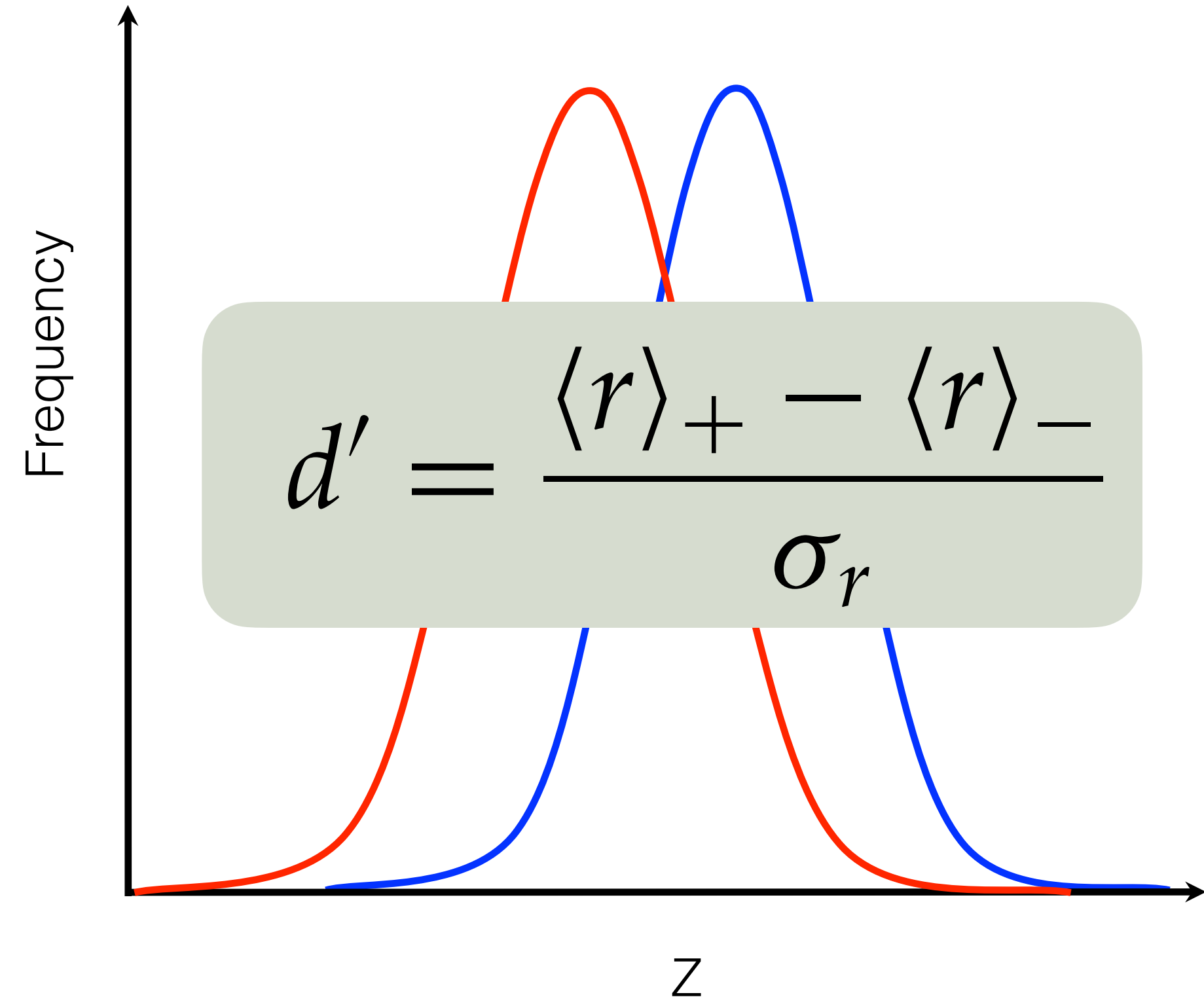
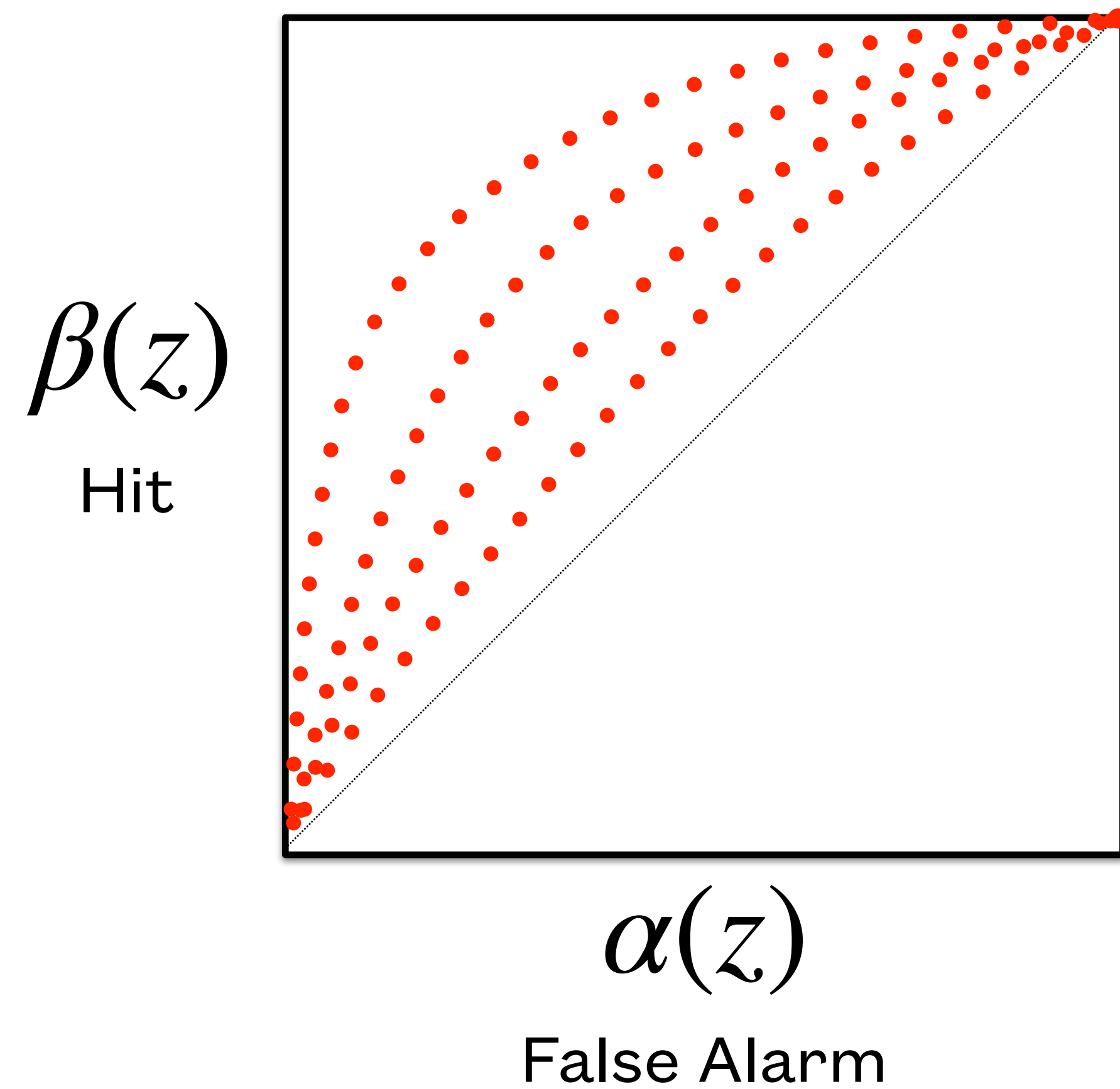
ROC (Receiver Operating Characteristic)



ROC (Receiver Operating Characteristic)

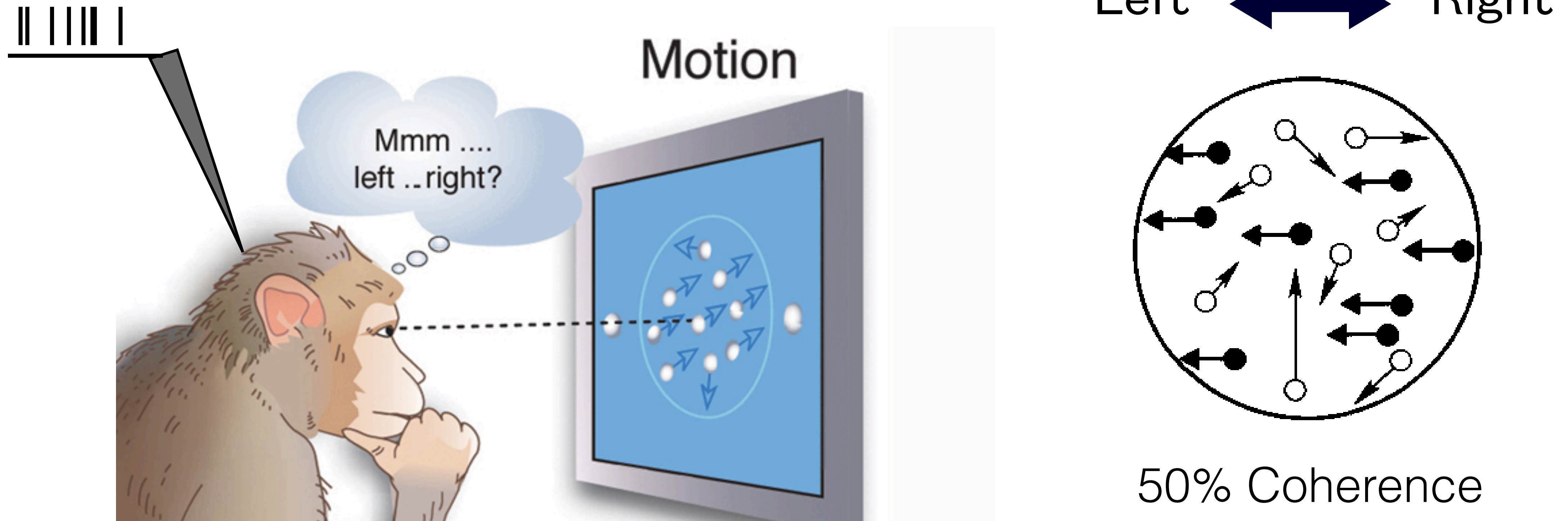


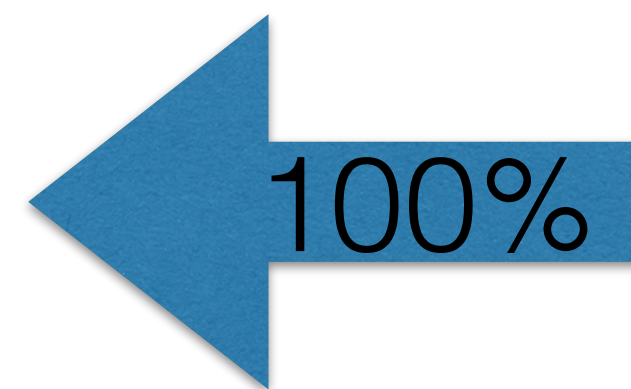
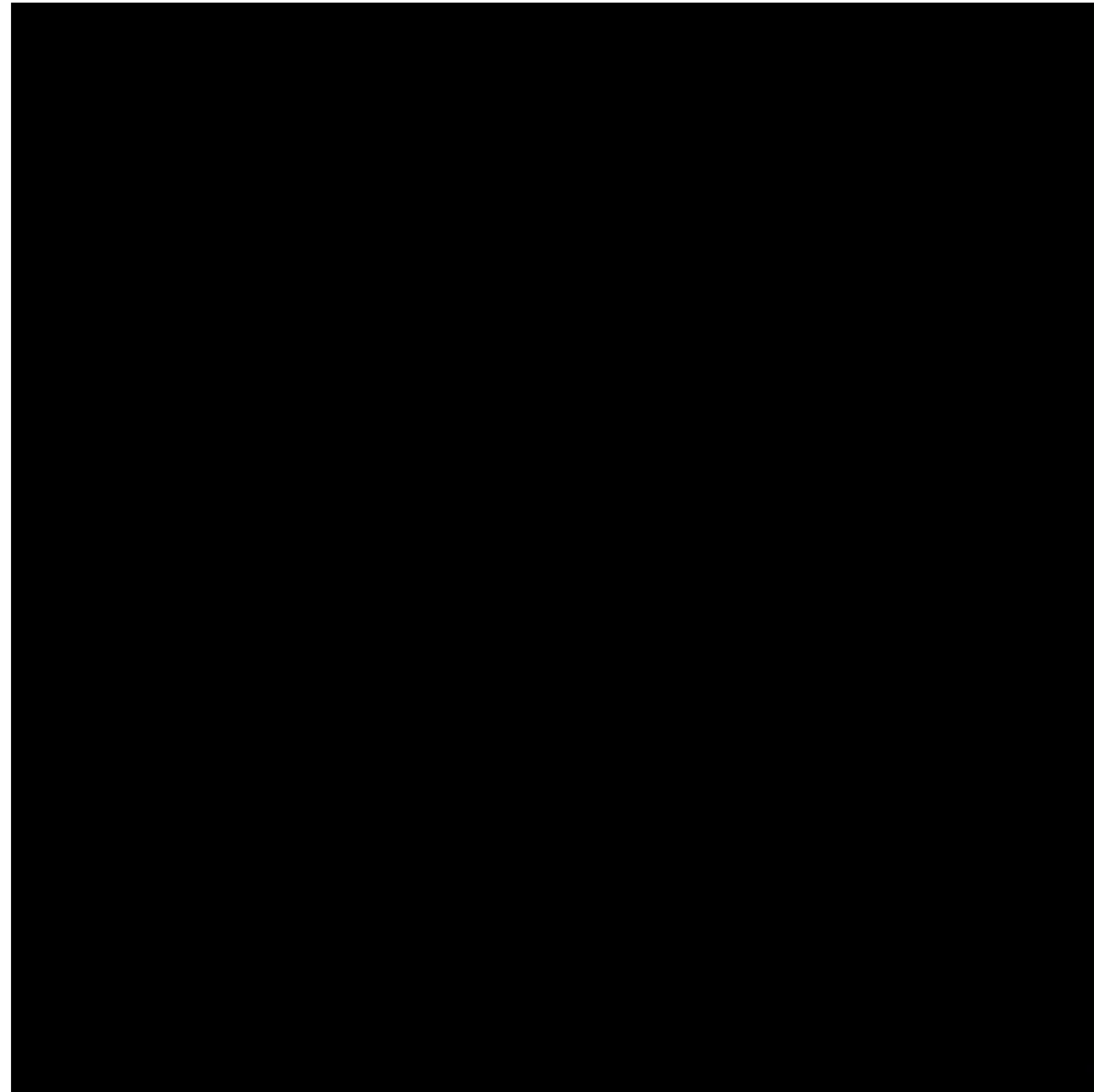
ROC (Receiver Operating Characteristic)

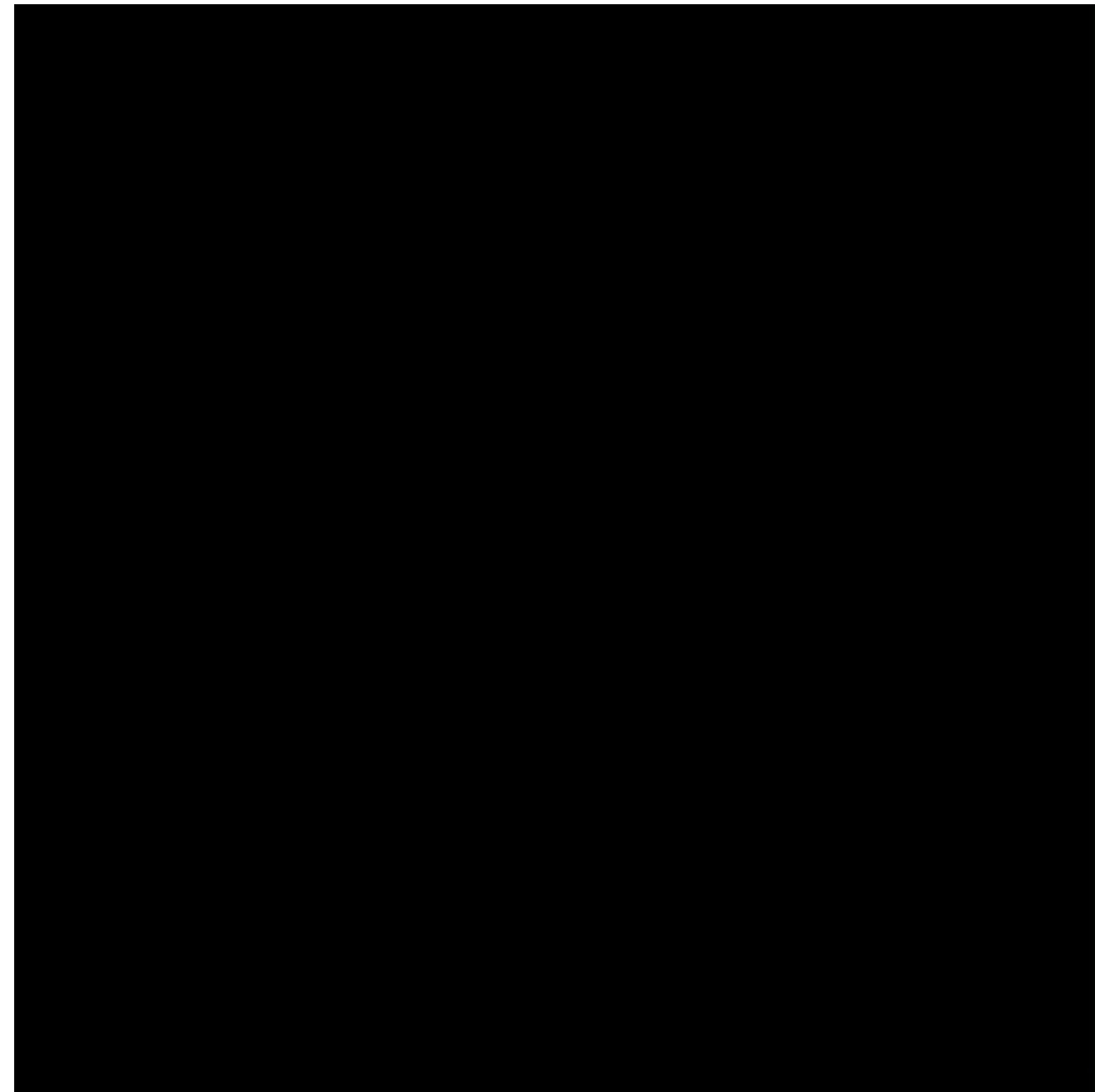


Random Dots Motion Direction Discrimination

2-Alternative Forced Choice

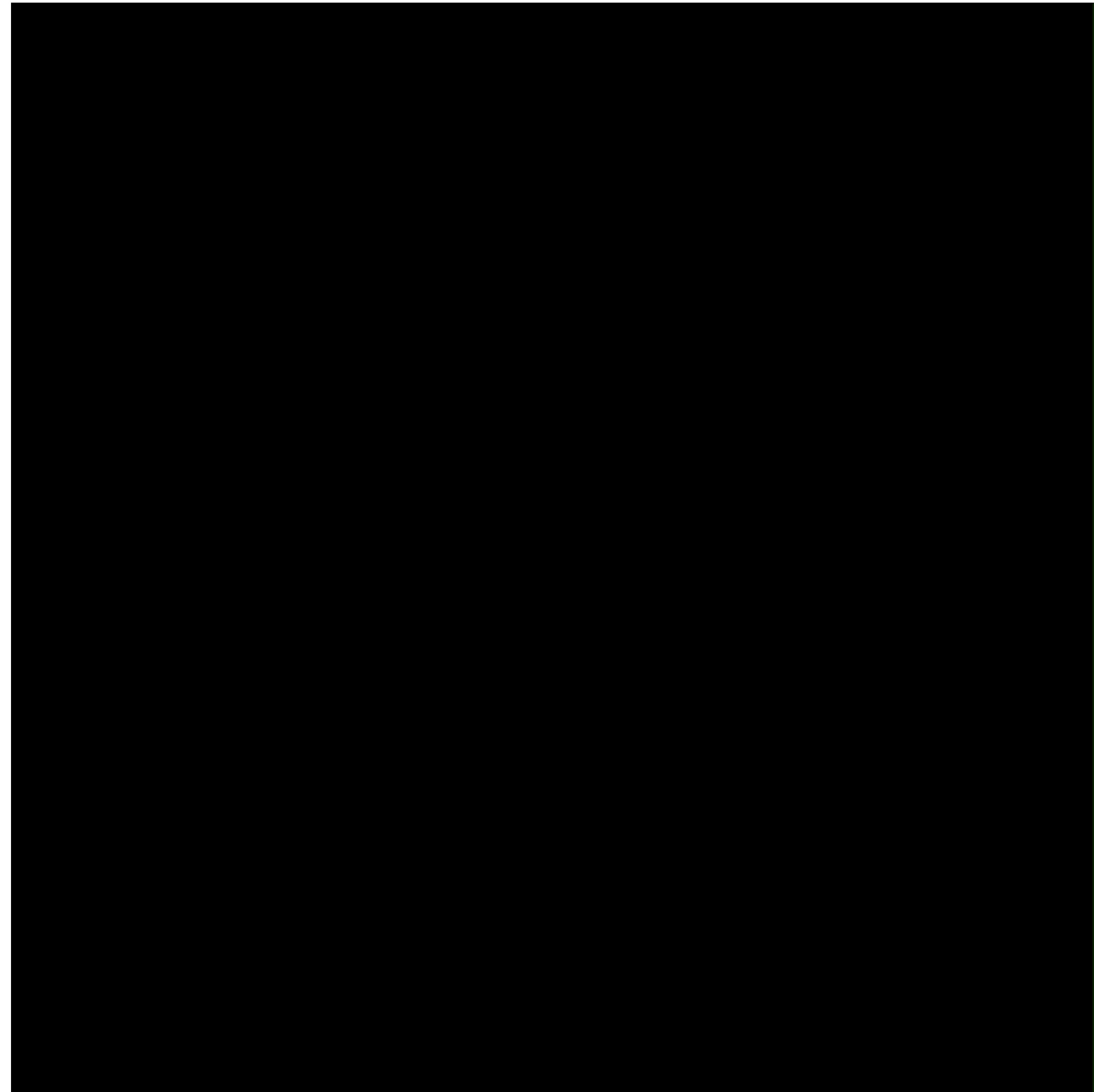






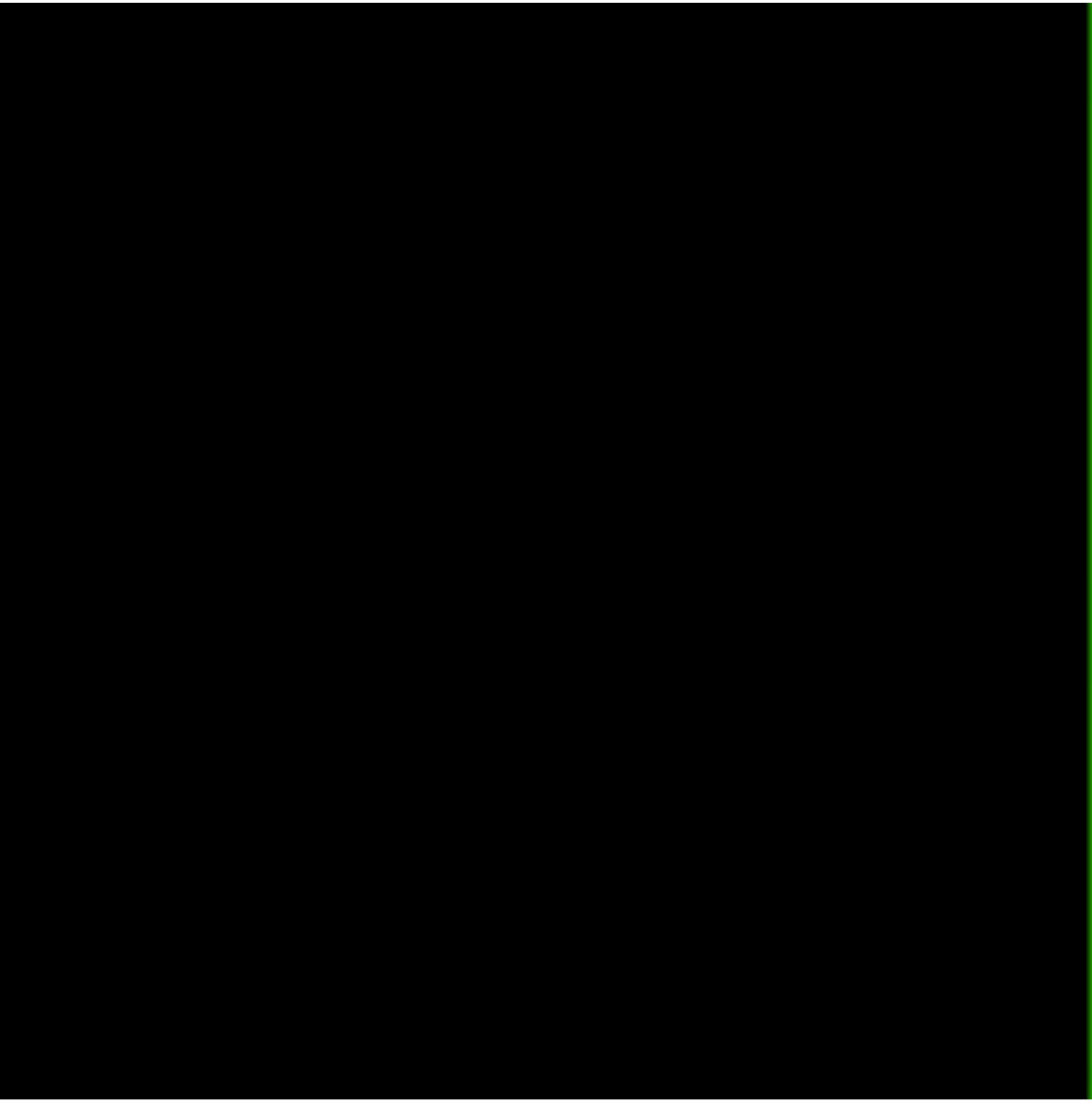
51.2%

A blue arrow pointing to the right, positioned below the percentage text.



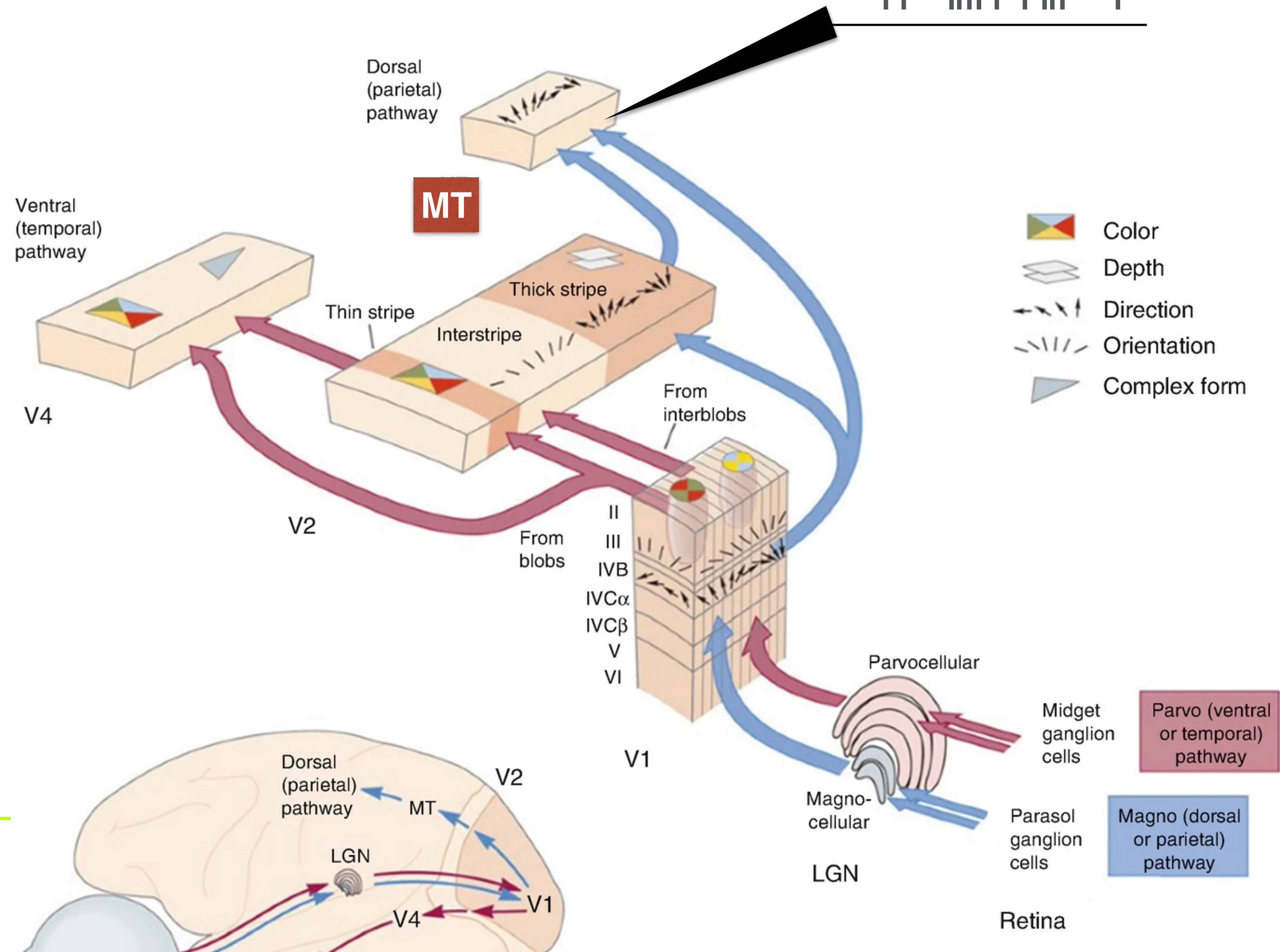
12.8%

A blue arrow pointing to the right, positioned below the percentage text.

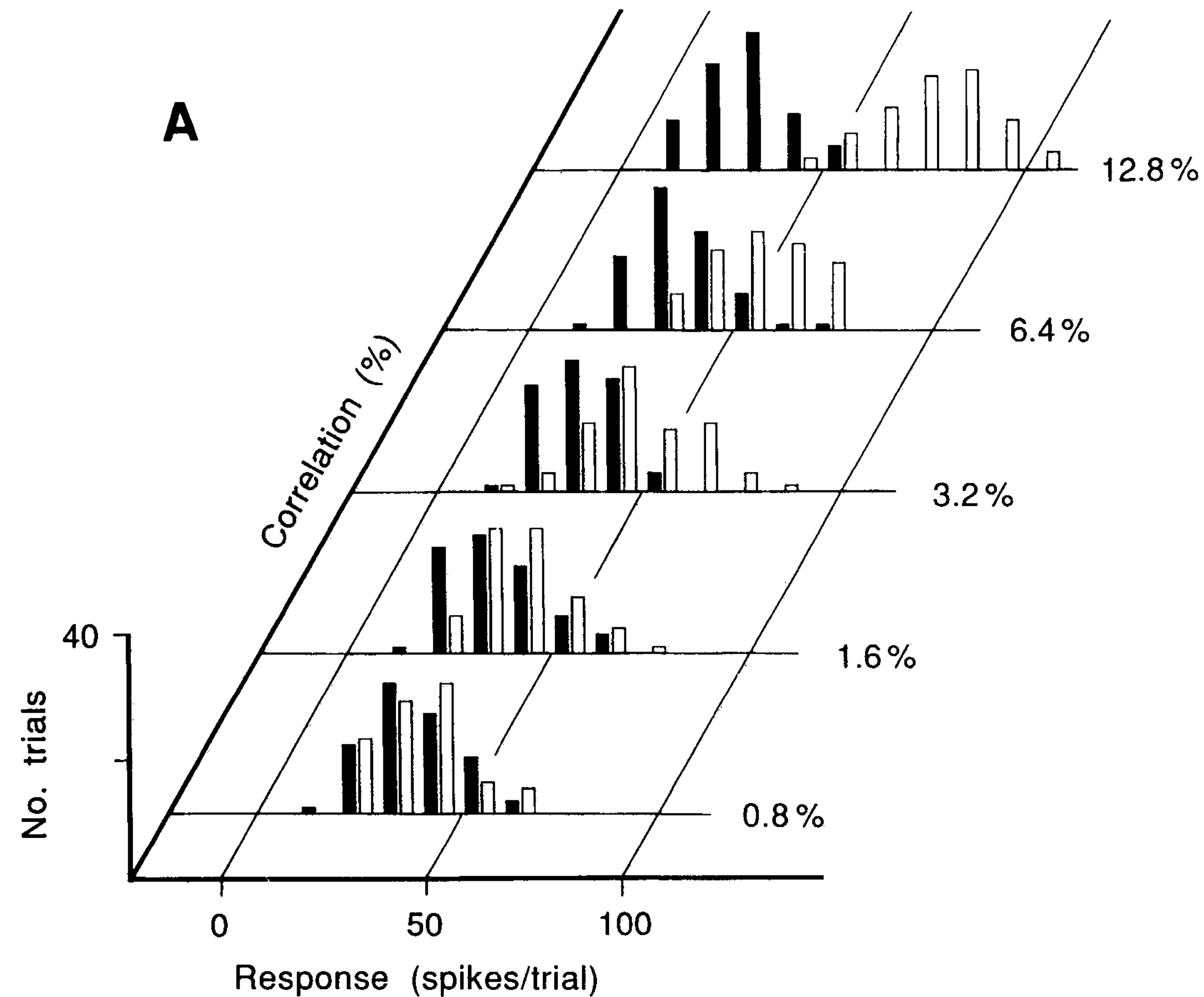


0%

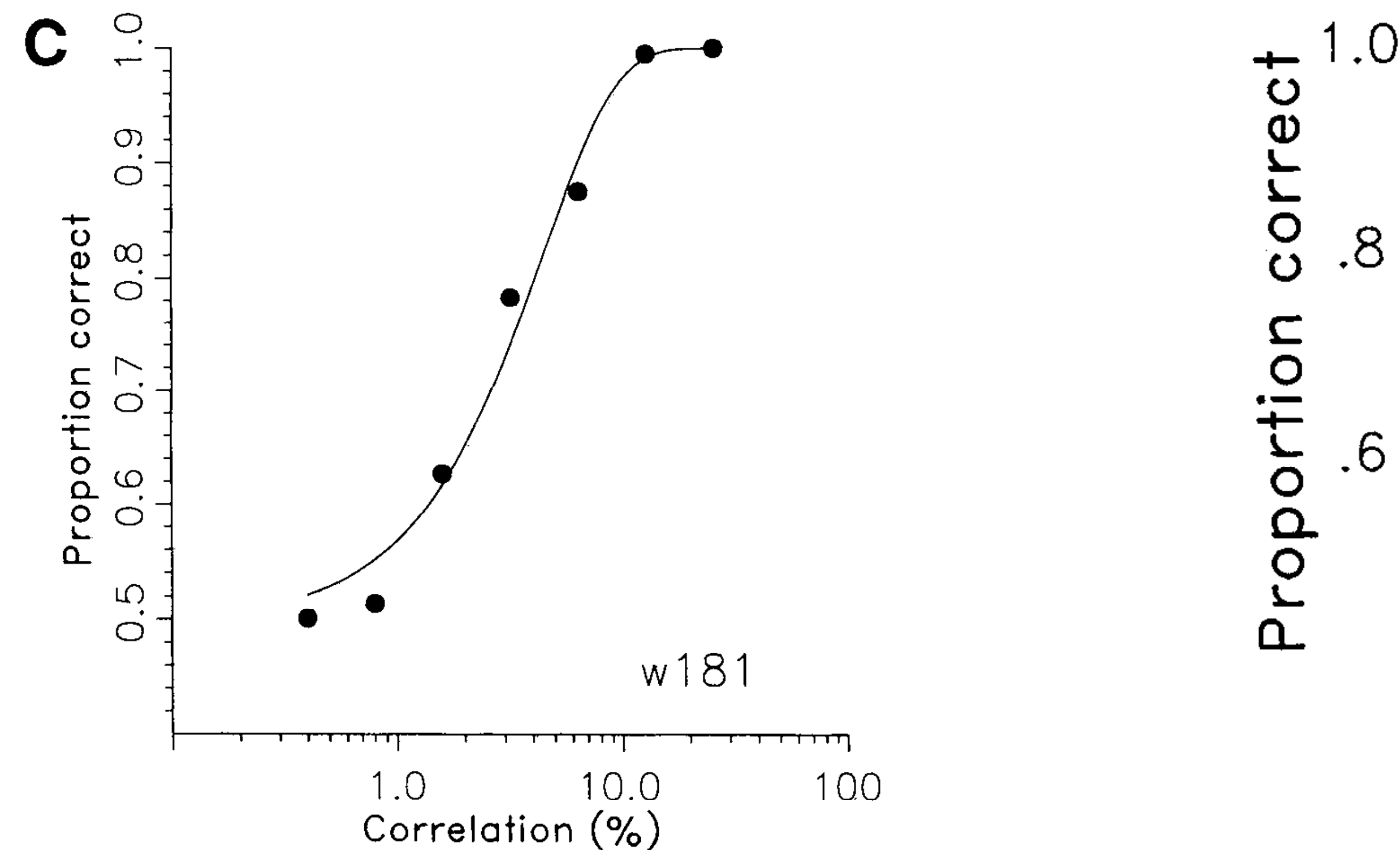
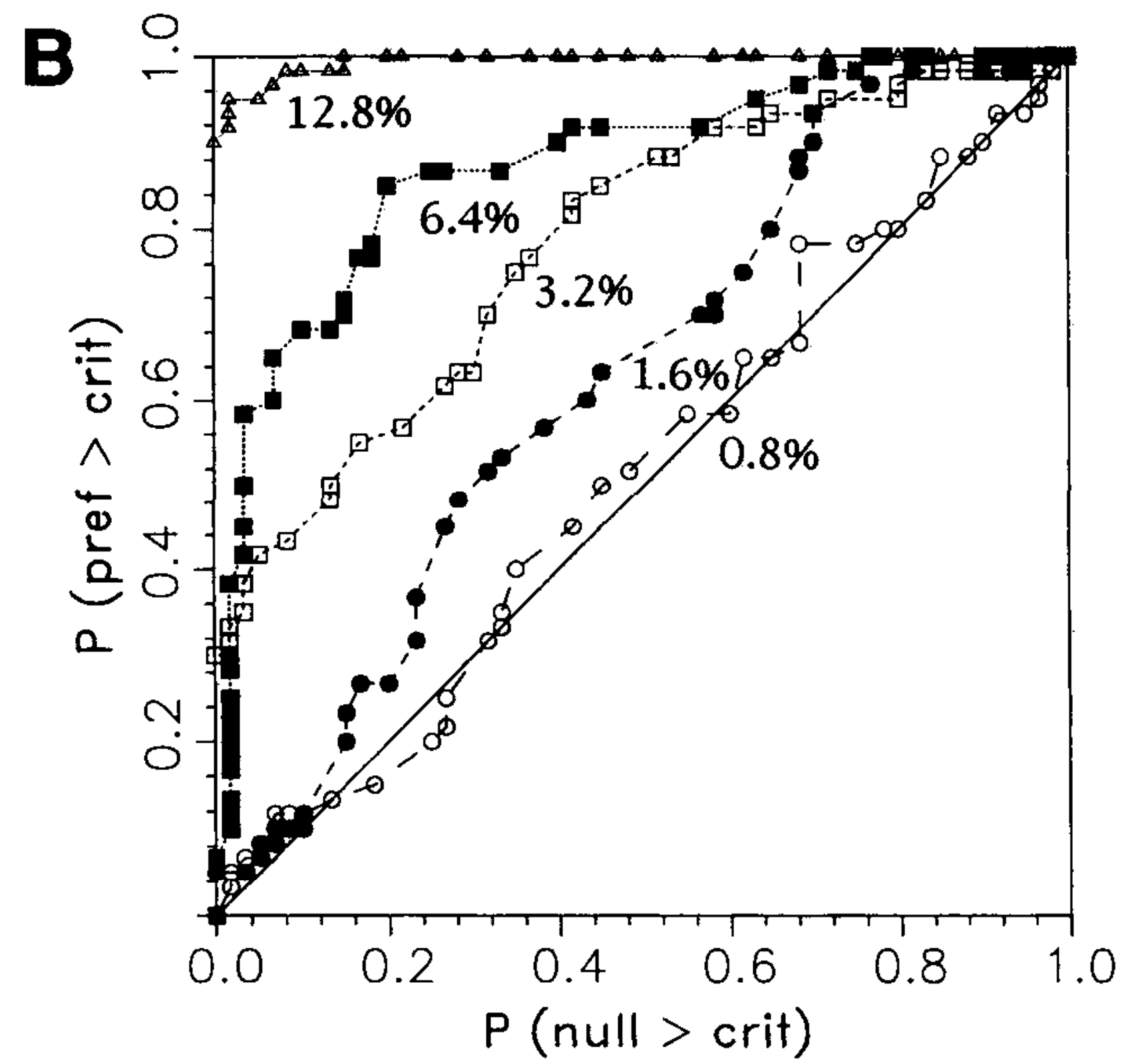
Target Brain Area: MT



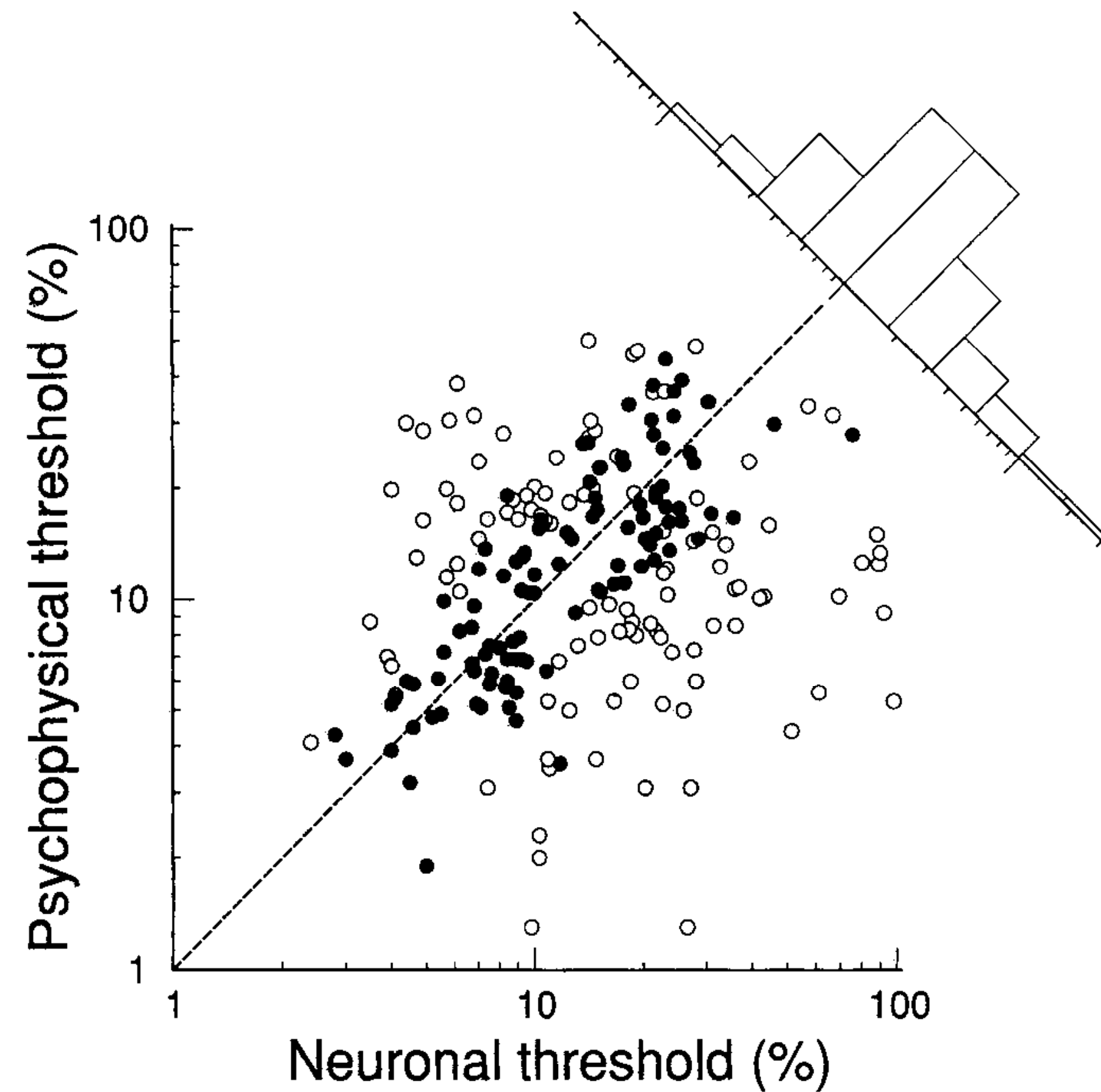
MT Neuron's responses to different directions



Neural Performance Estimated using SDT



Single Neuron's Performance Matches Behaviour



Britten, Shadlen, Newsome and Movshon, J Neurosci., 1992

Dynamic Stimulus?

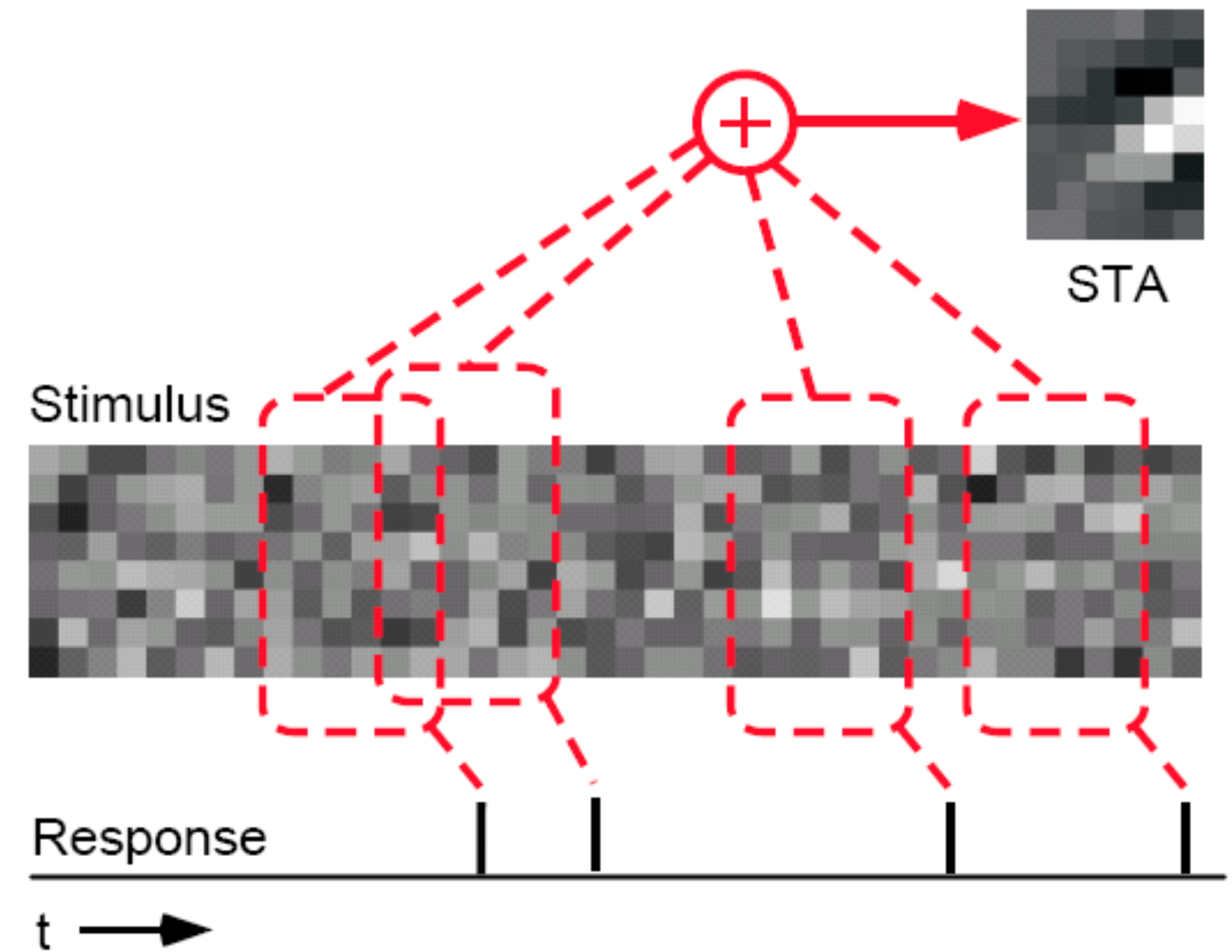
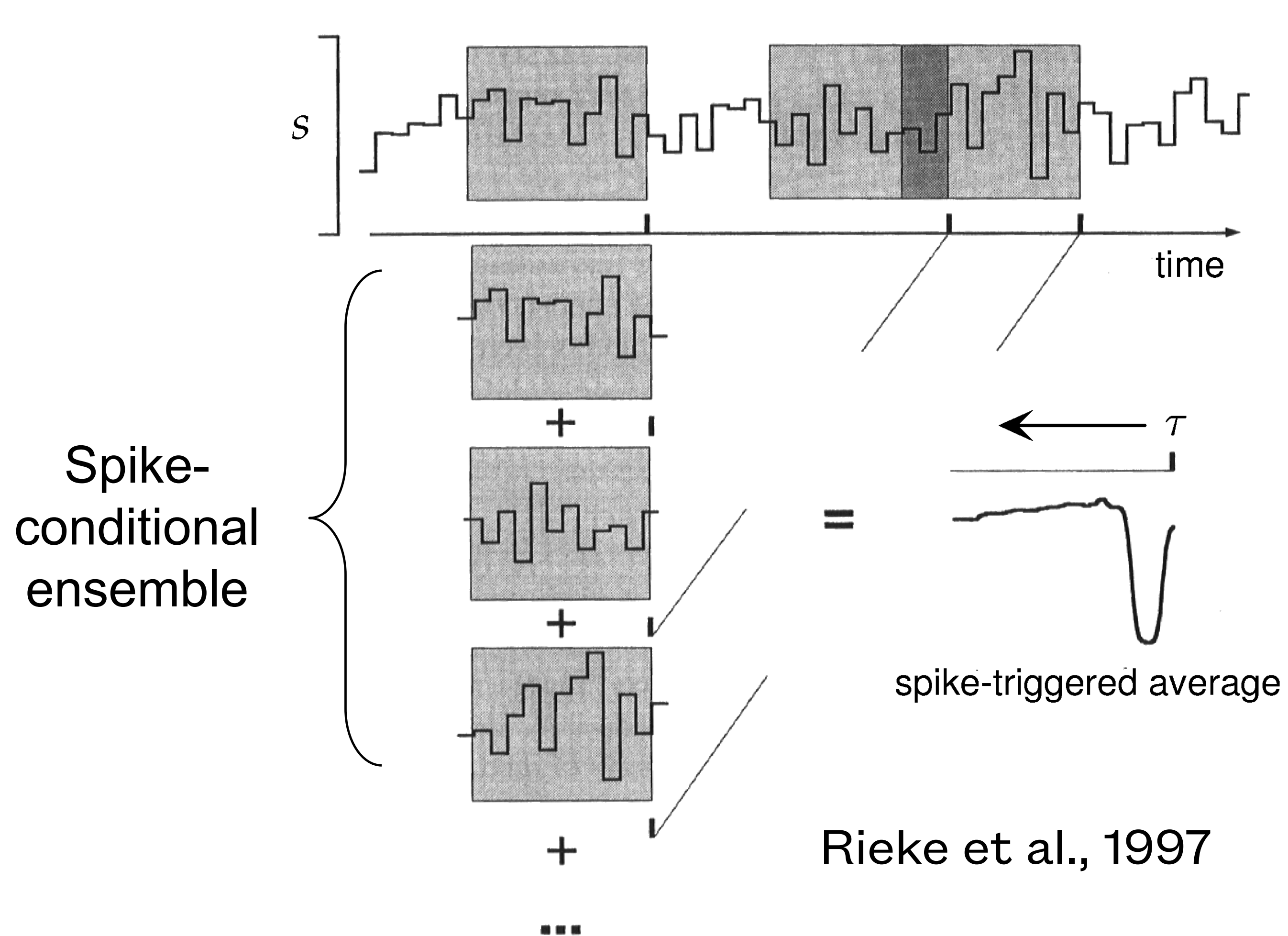


Linear System: Temporal filtering

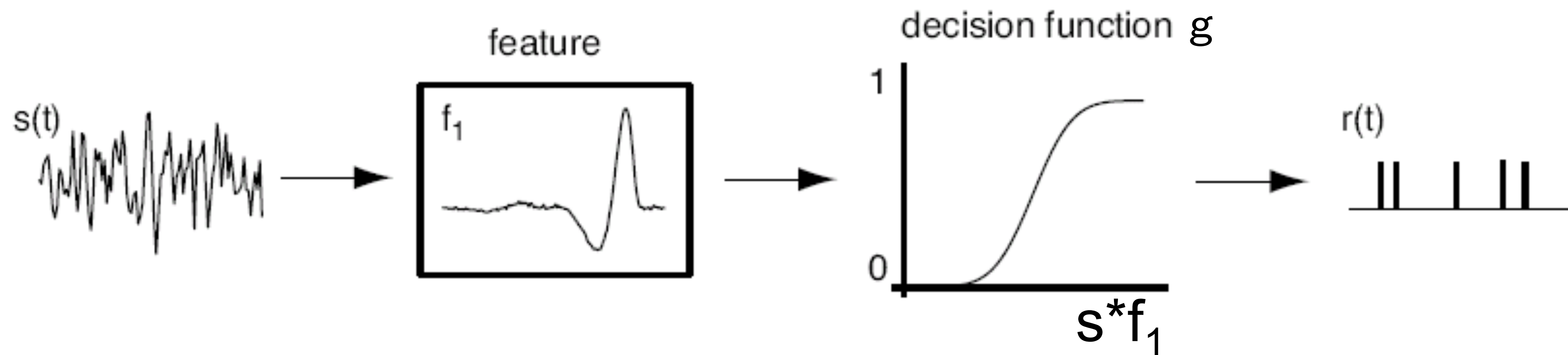


$$r(t) = s(t) * f(t) = \int s(t - \tau) f(\tau) d\tau$$

Reverse correlation

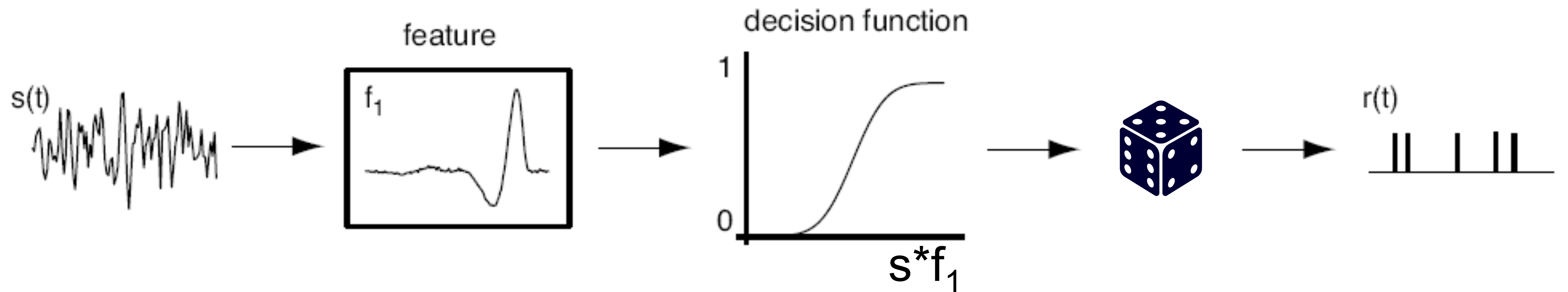


Temporal filtering + Nonlinearity



$$r(t) = g[s(t) * f(t)] = g\left[\int s(t - \tau)f(\tau)d\tau\right]$$

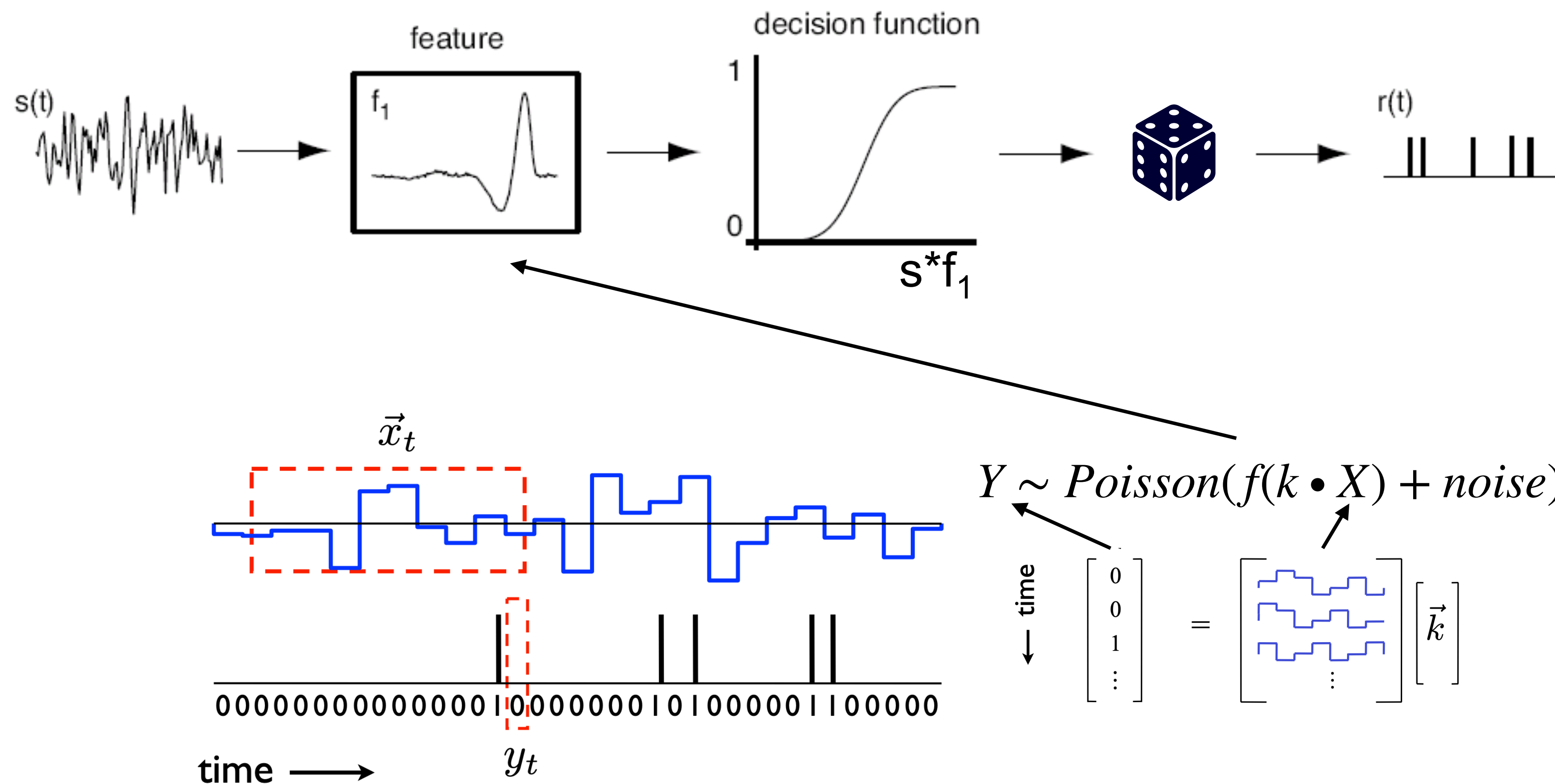
Temporal filtering + Nonlinearity + Stochasticity



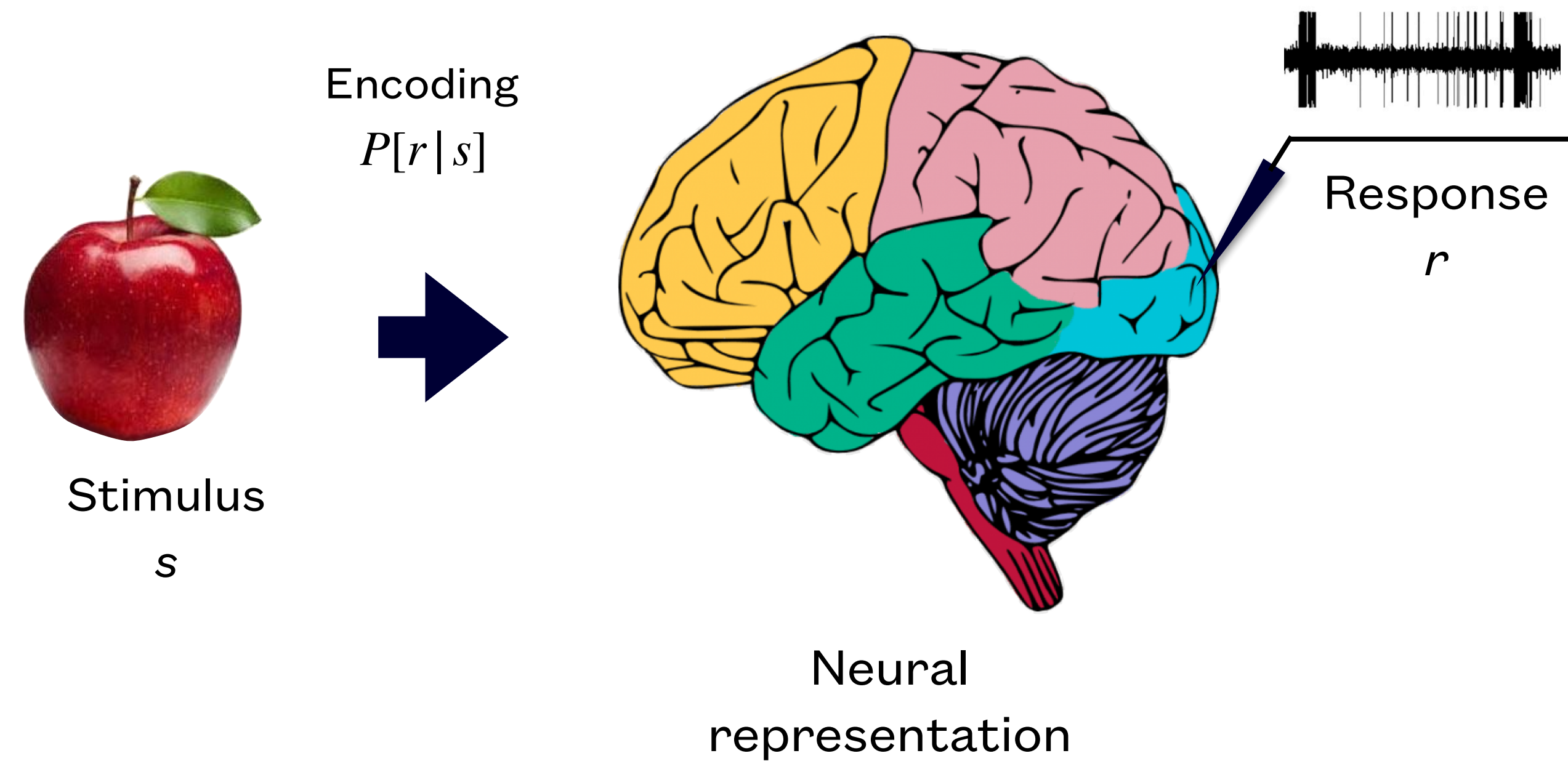
$$r \sim \text{Poisson}(\lambda)$$

$$\lambda = g[s(t) * f(t)] = g\left[\int s(t - \tau)f(\tau)dt\right]$$

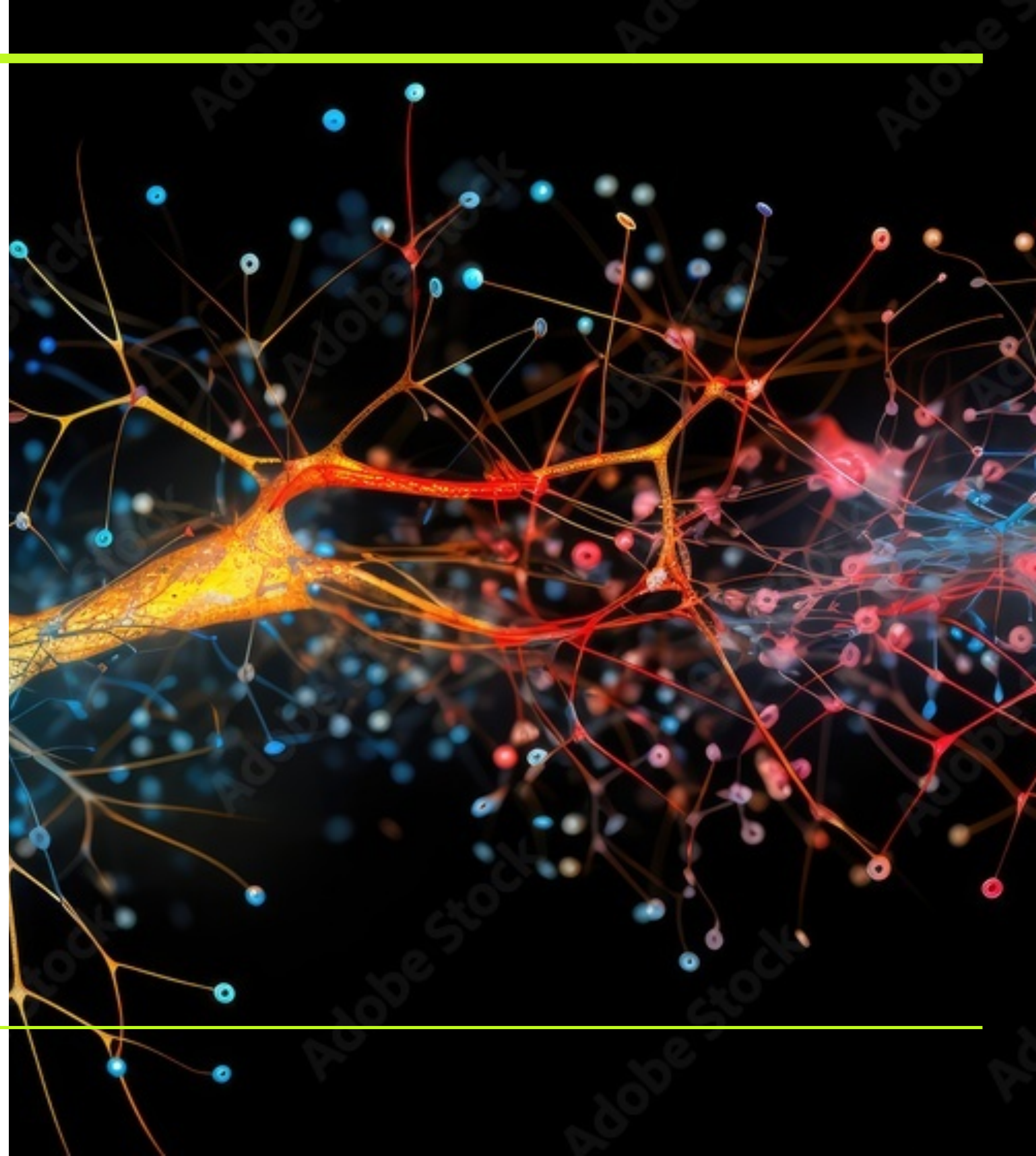
Generalized Linear Model



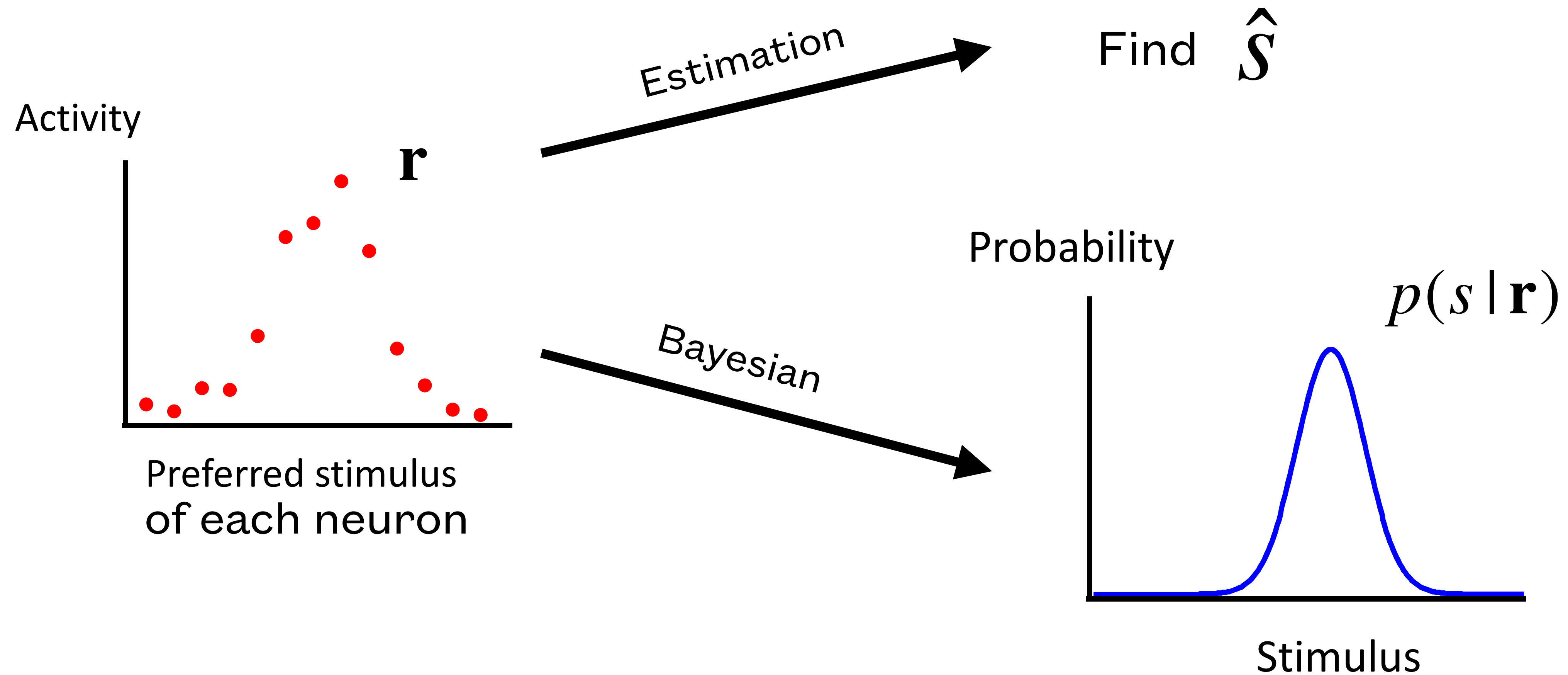
One More Thing...



-
- ❖ Basics
 - ❖ Encoding
 - ❖ **Decoding**
 - ❖ Information Theory
 - ❖ Population Analysis
-



Decoding



Maximum Likelihood Estimation

$$\hat{s} = \operatorname{argmax}_s P[r | s]$$

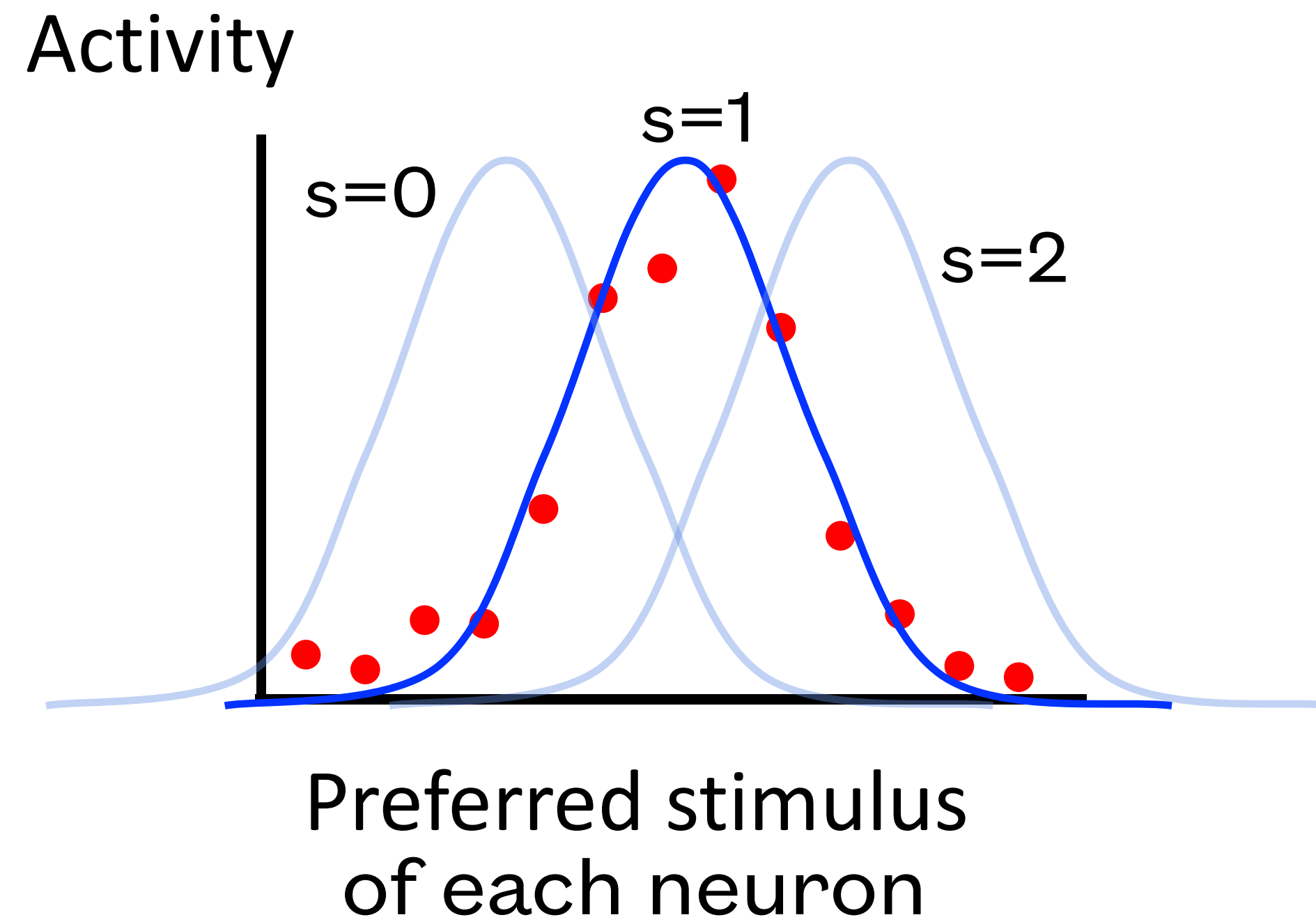
$$r \sim \text{Poisson}(f(s))$$

$$P[r | s] = \prod_i \frac{e^{f_i(s)} f_i(s)^{r_i}}{r_i!} = \frac{\overset{\text{CONSTANT}}{e^{\sum_i f_i(s) + \sum_i r_i \ln f_i(s)}}}{\prod_i r_i!}$$

$$\operatorname{argmax}_s P[r | s] = \operatorname{argmax}_s e^{\sum_i r_i \ln f_i(s)} = \operatorname{argmax}_s \sum_i r_i \ln f_i(s)$$

MLE: Template Matching

$$\operatorname{argmax}_s P[r | s] = \operatorname{argmax}_s \sum_i r_i \ln f_i(s)$$



Cramér-Rao bound

$$\sigma^2(s) \geq \frac{1}{\left\langle -\frac{\partial^2 \ln P[r | s]}{\partial s^2} \right\rangle}$$

Fisher Information

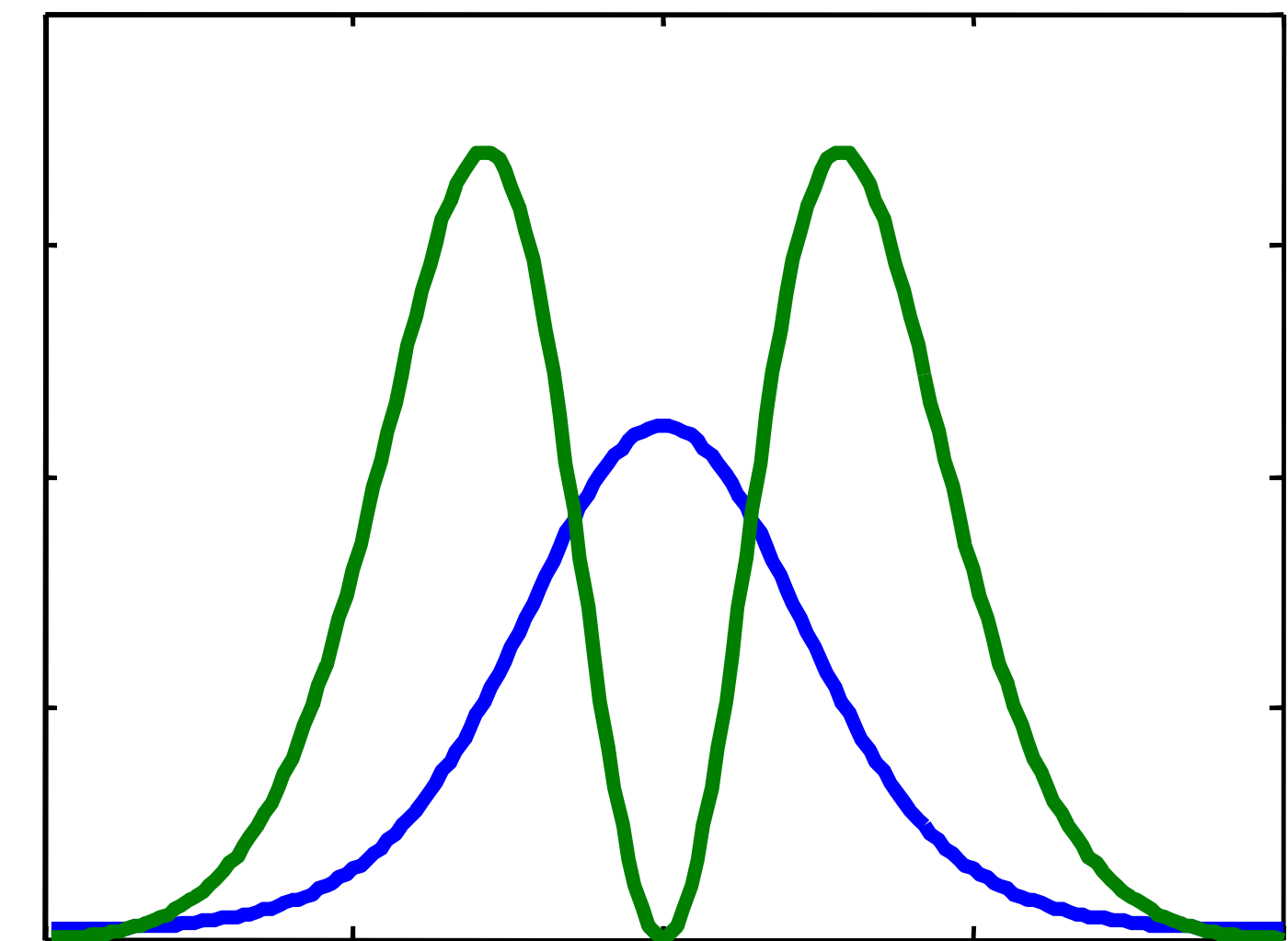
Fisher Information

Gaussian tuning curve, Poisson neuron

$$I_i(s) = \frac{f_i'(s)^2}{f_i(s)}$$

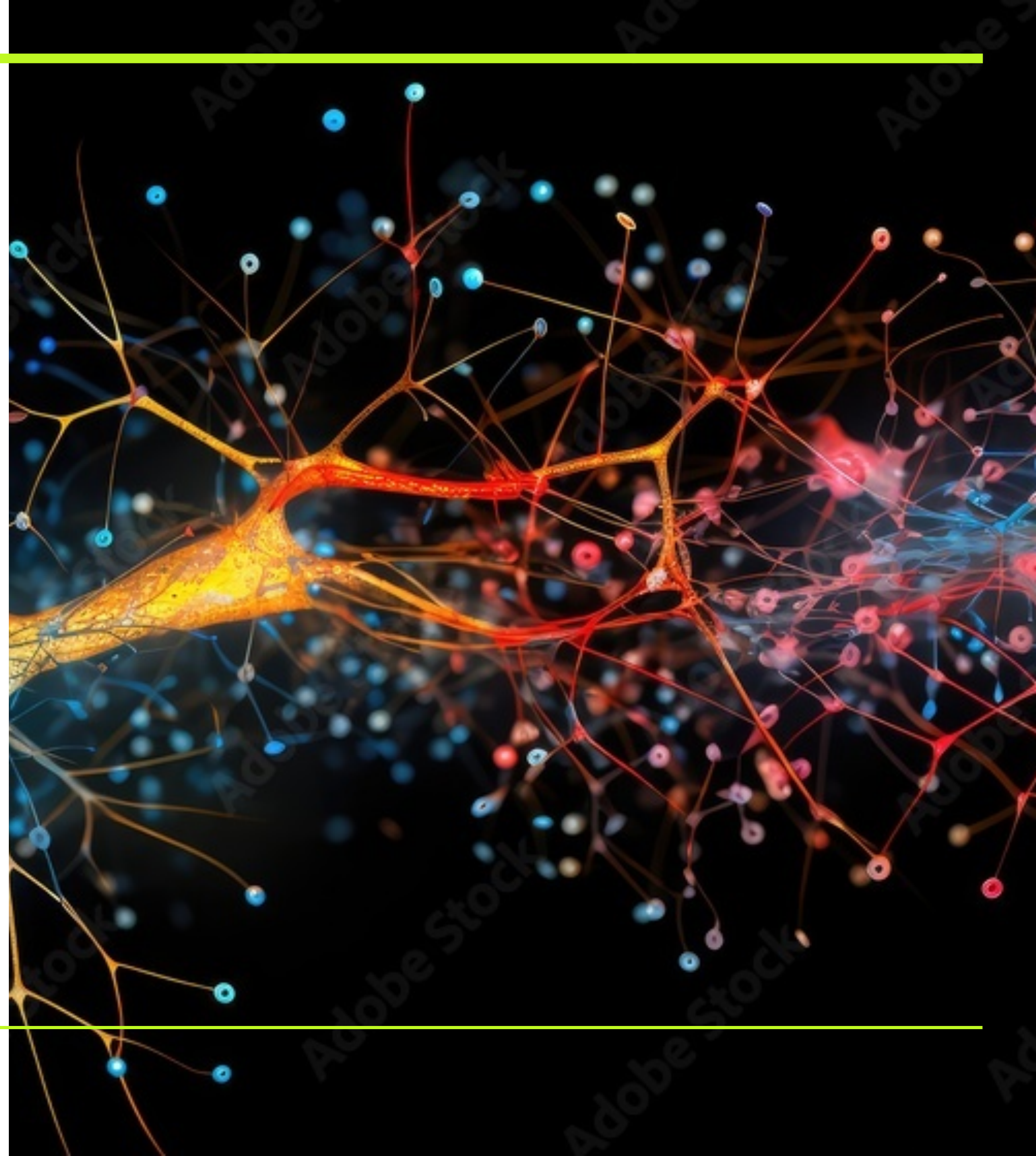
↙ $\langle r(s) \rangle$

— Mean activity
— Fisher information per neuron



Preferred stimulus

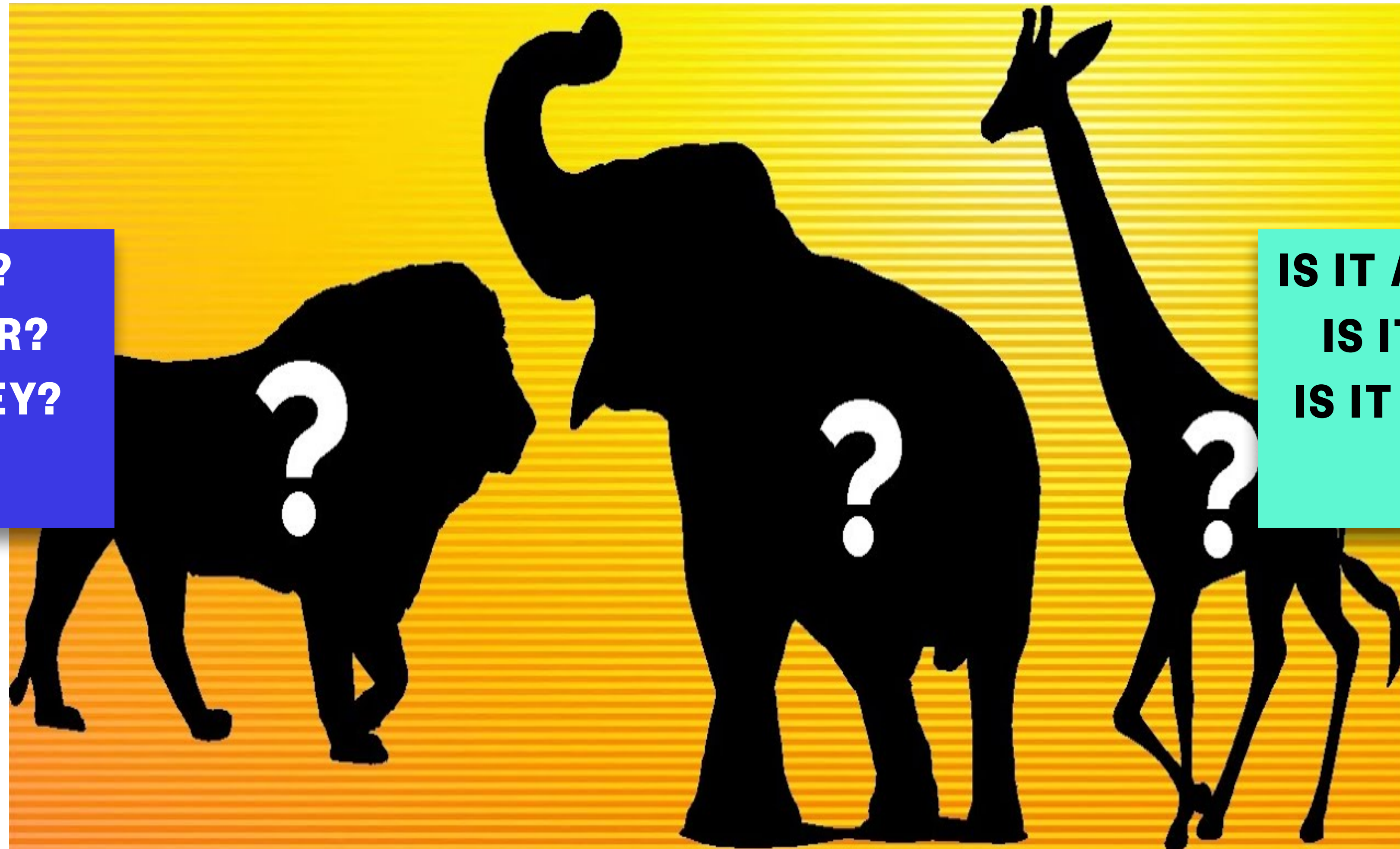
-
- ❖ Basics
 - ❖ Encoding
 - ❖ Decoding
 - ❖ **Information Theory**
 - ❖ Population Analysis
-



10 Questions

**IS IT A CAT?
IS IT A SPIDER?
IS IT A MONKEY?**

...



**IS IT A VERTEBRATE?
IS IT A MAMMAL?
IS IT A CARNIVORE?**

...

10 Questions

Ideal questions can double the number of categories

$$2^Q = N$$

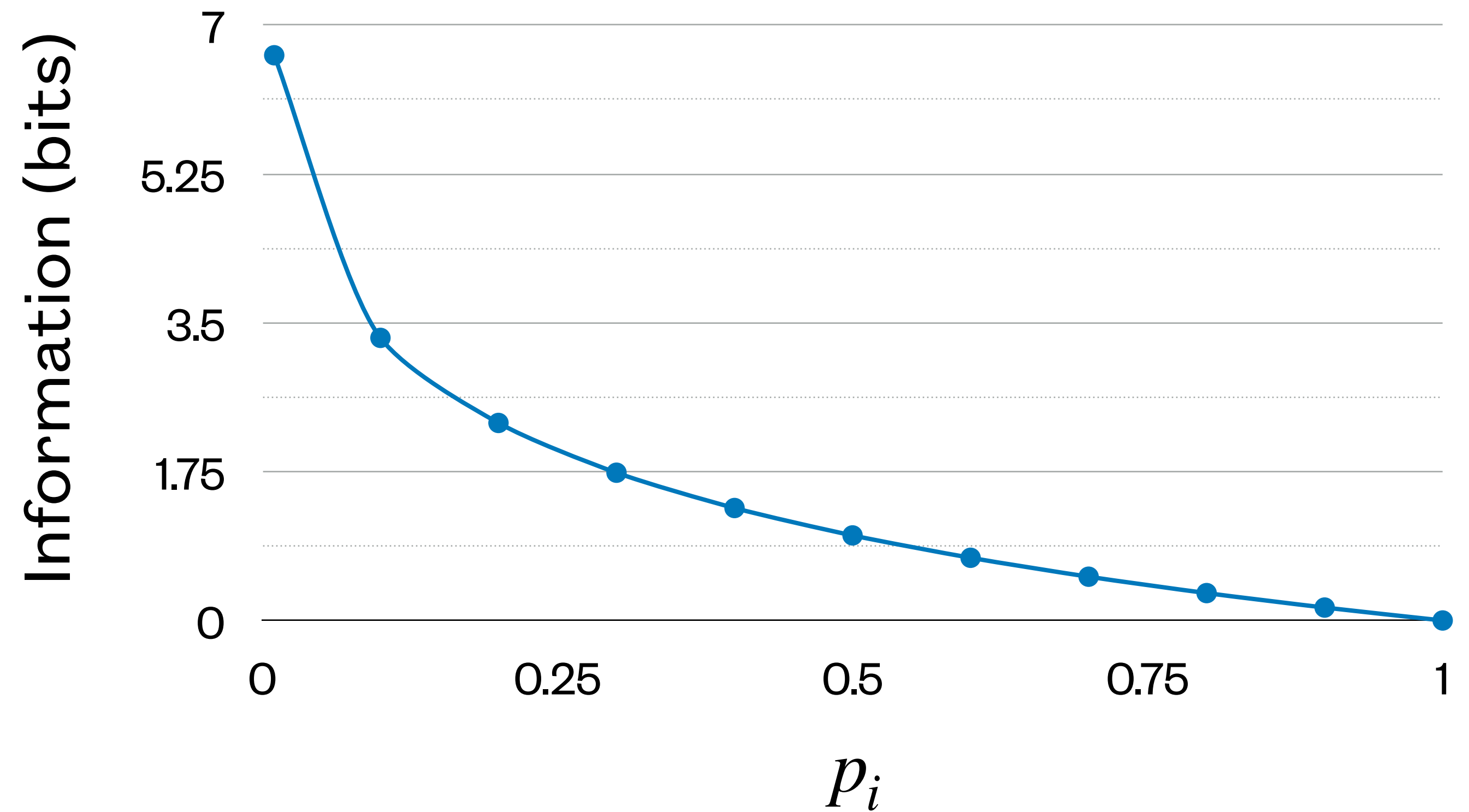
$$p = 1/N$$

$$Q = \log N = \log 1/p = -\log p$$

Information

$$I = -\log_2 p_i$$

- Continuous w.r.t. p
- Additive
- Greater when p is smaller



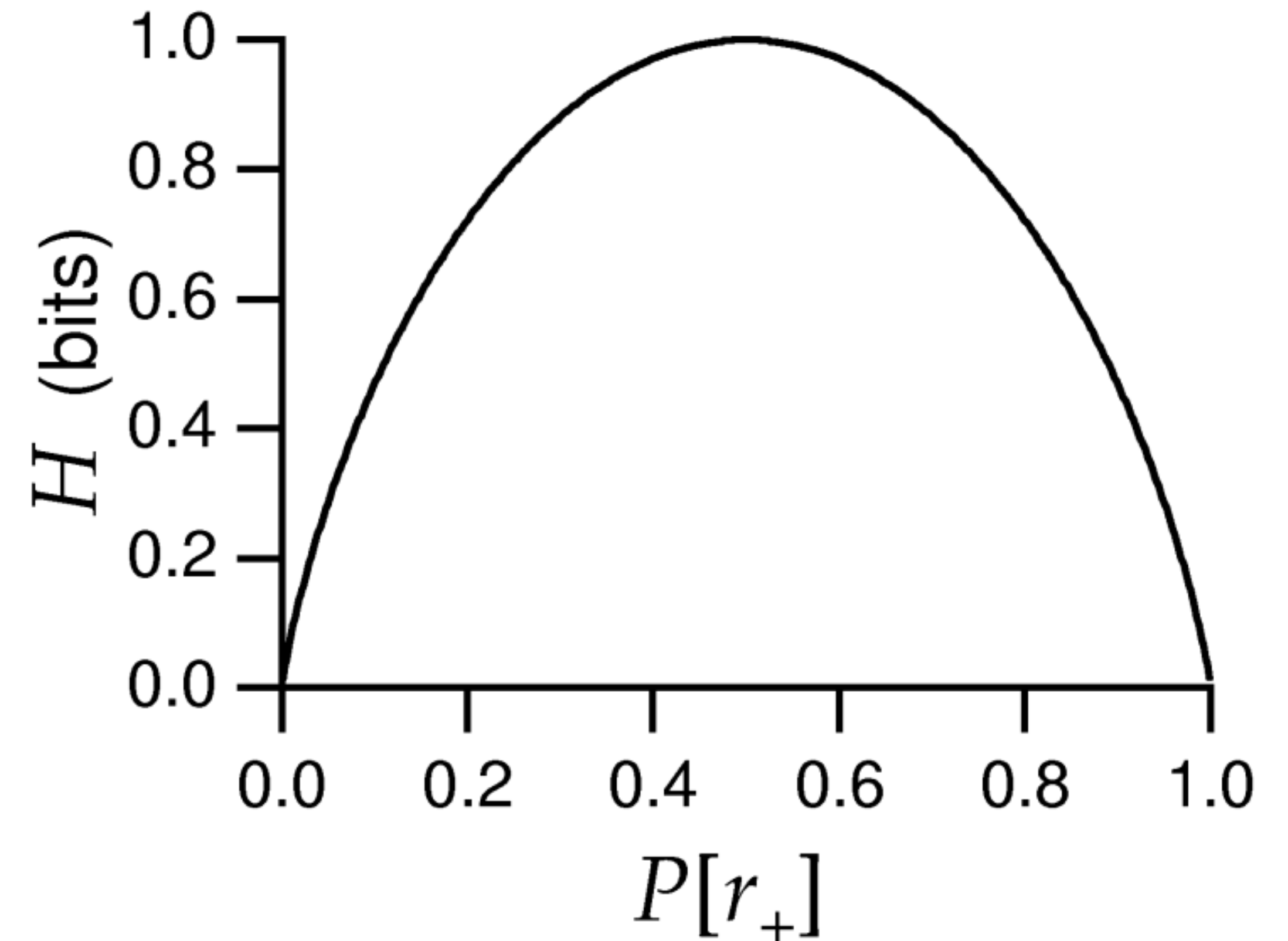
Entropy

$$H = - \sum_i p_i \log_2 p_i$$

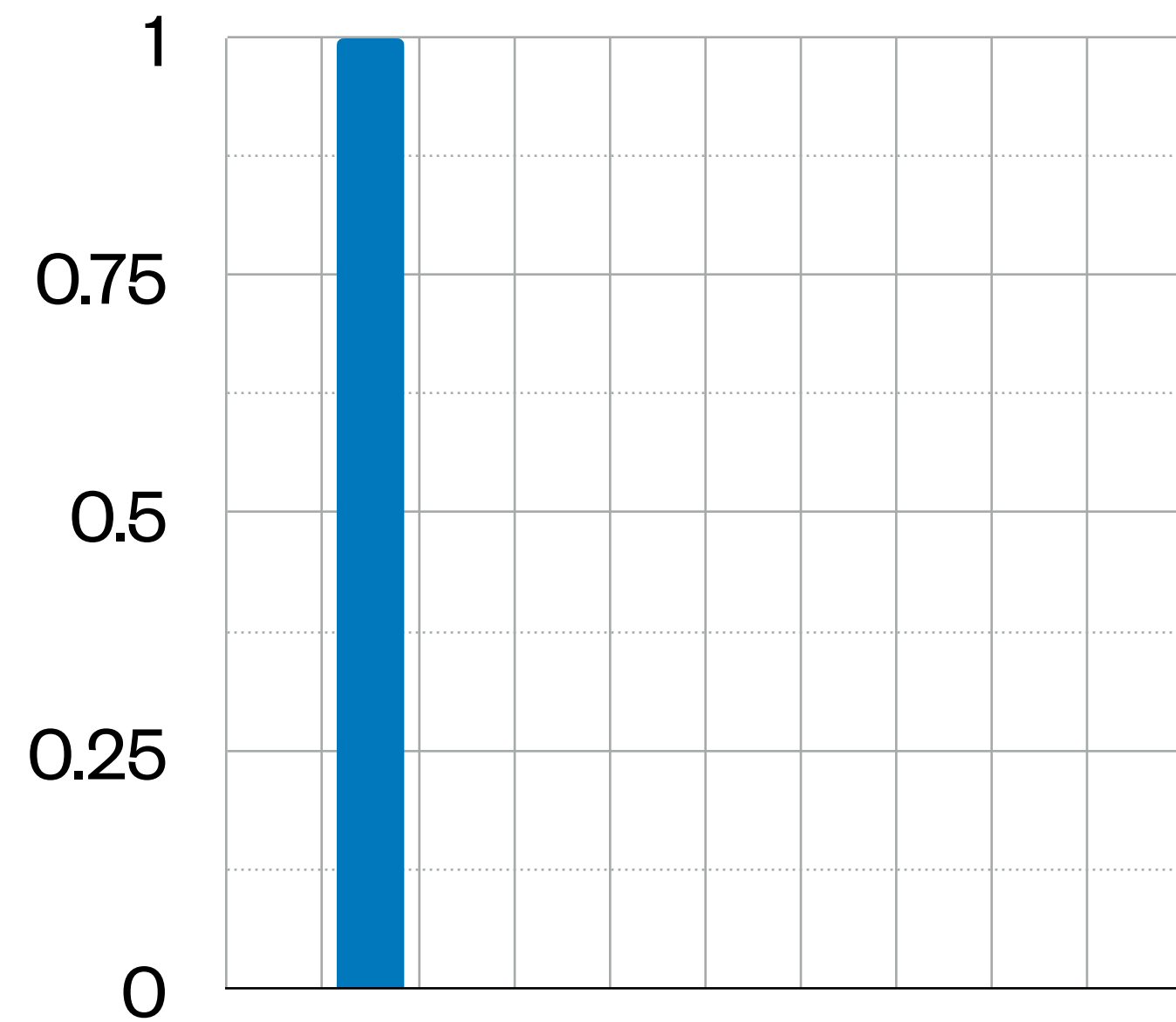
$$H[r] = -p_+ \log p_+ - (1-p_+) \log(1-p_+)$$

When $p_+ = 1/2$,

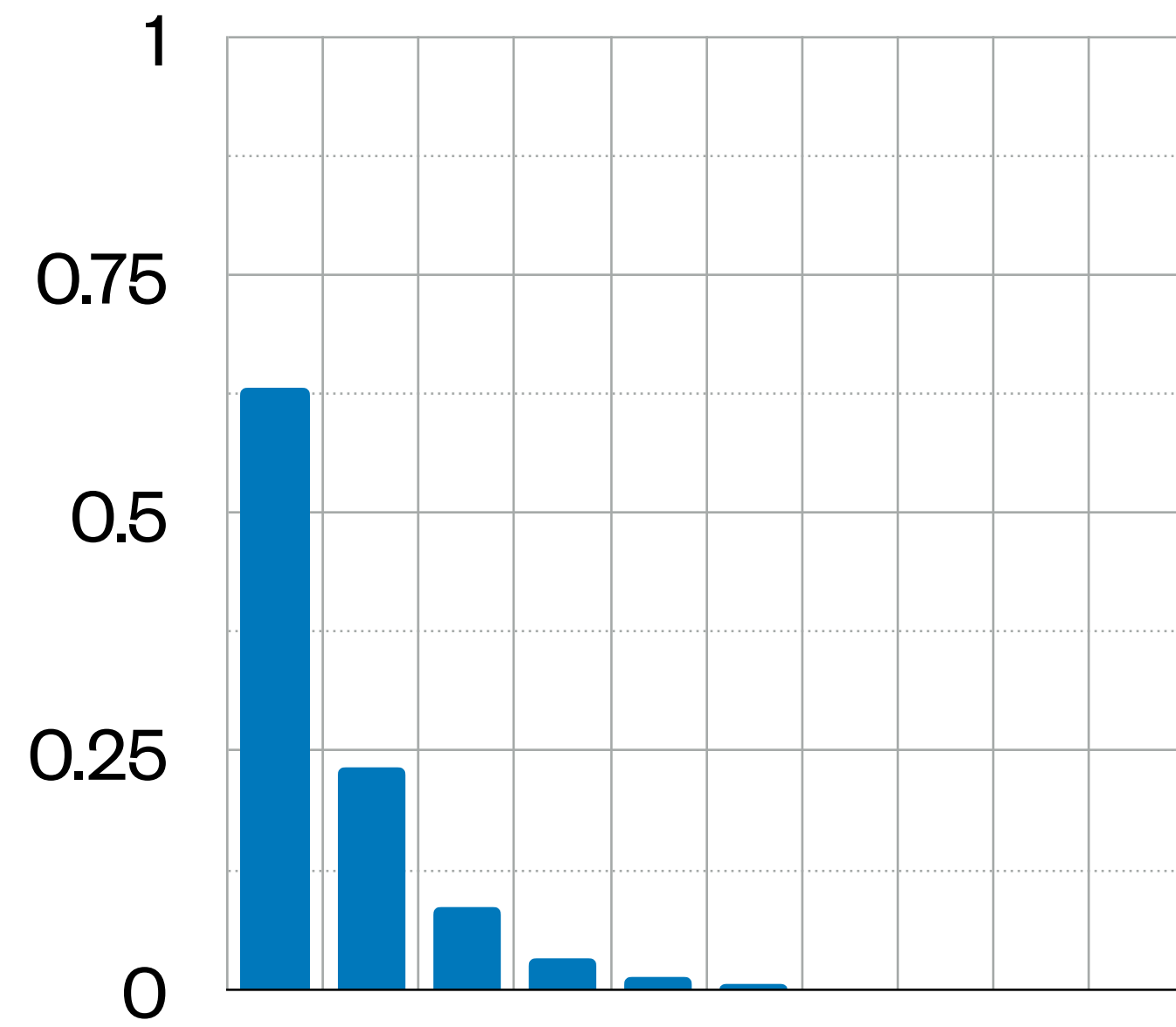
$$H[r|s] = -p \log p - (1-p) \log(1-p)$$



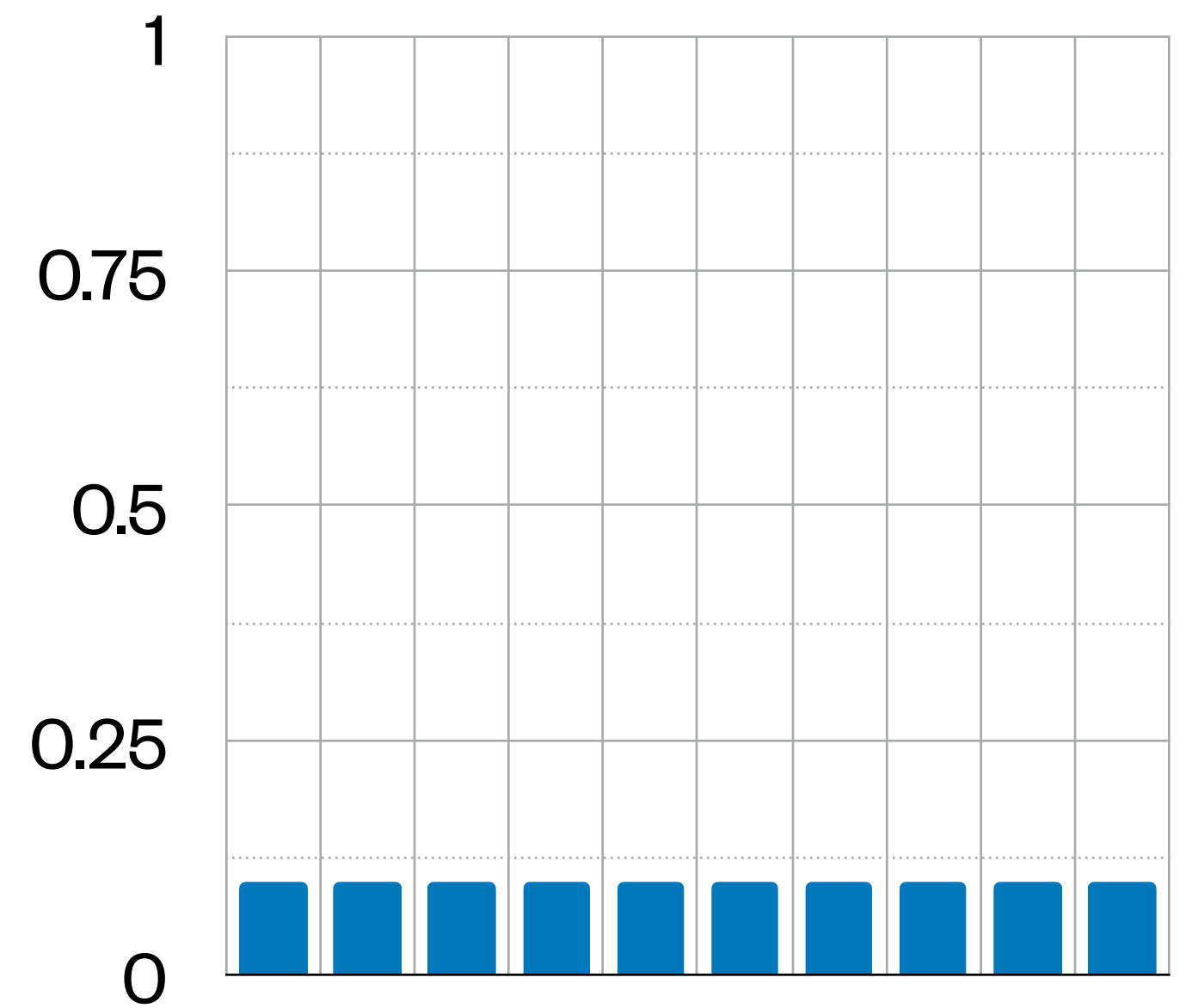
Entropy



0



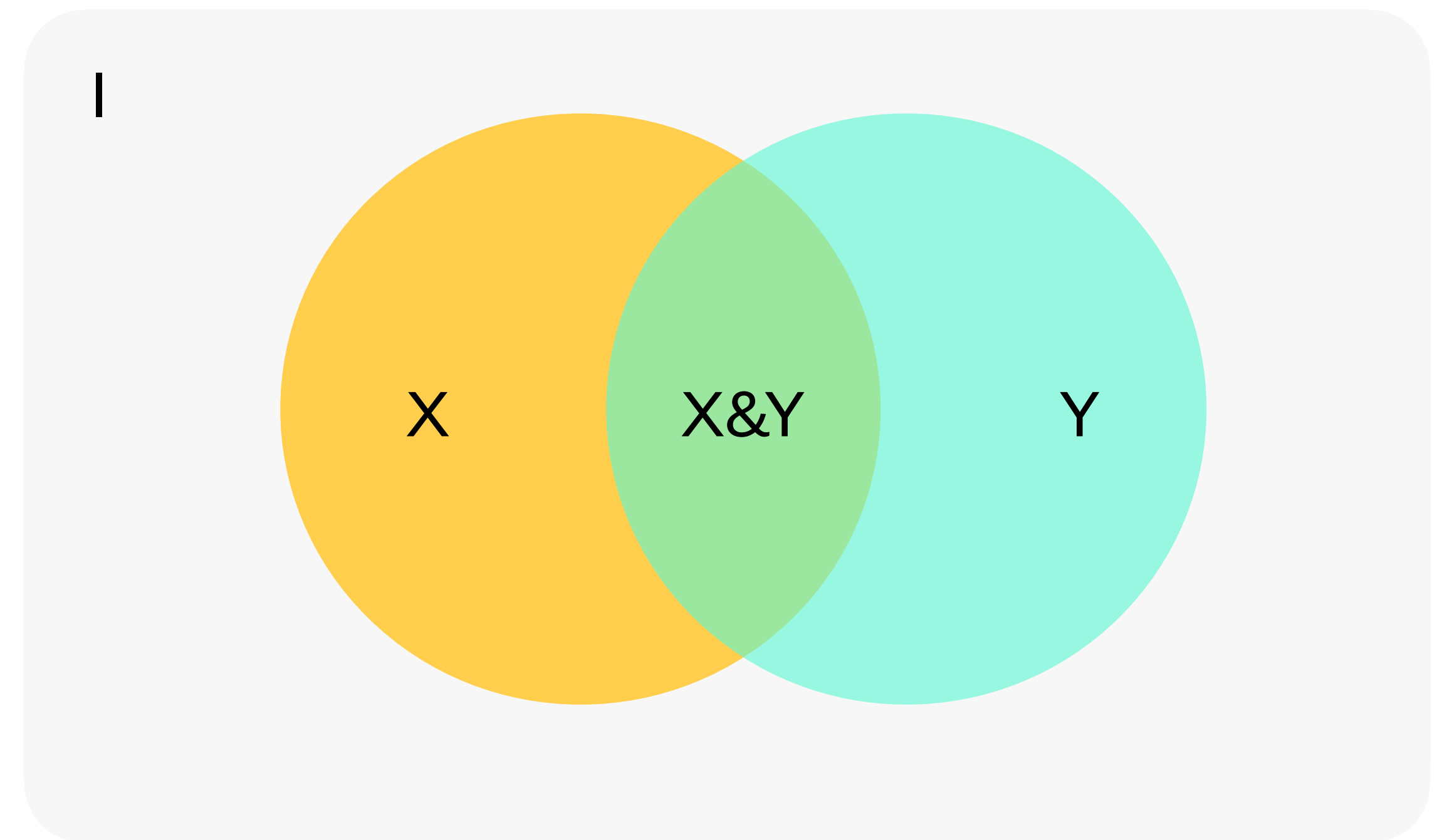
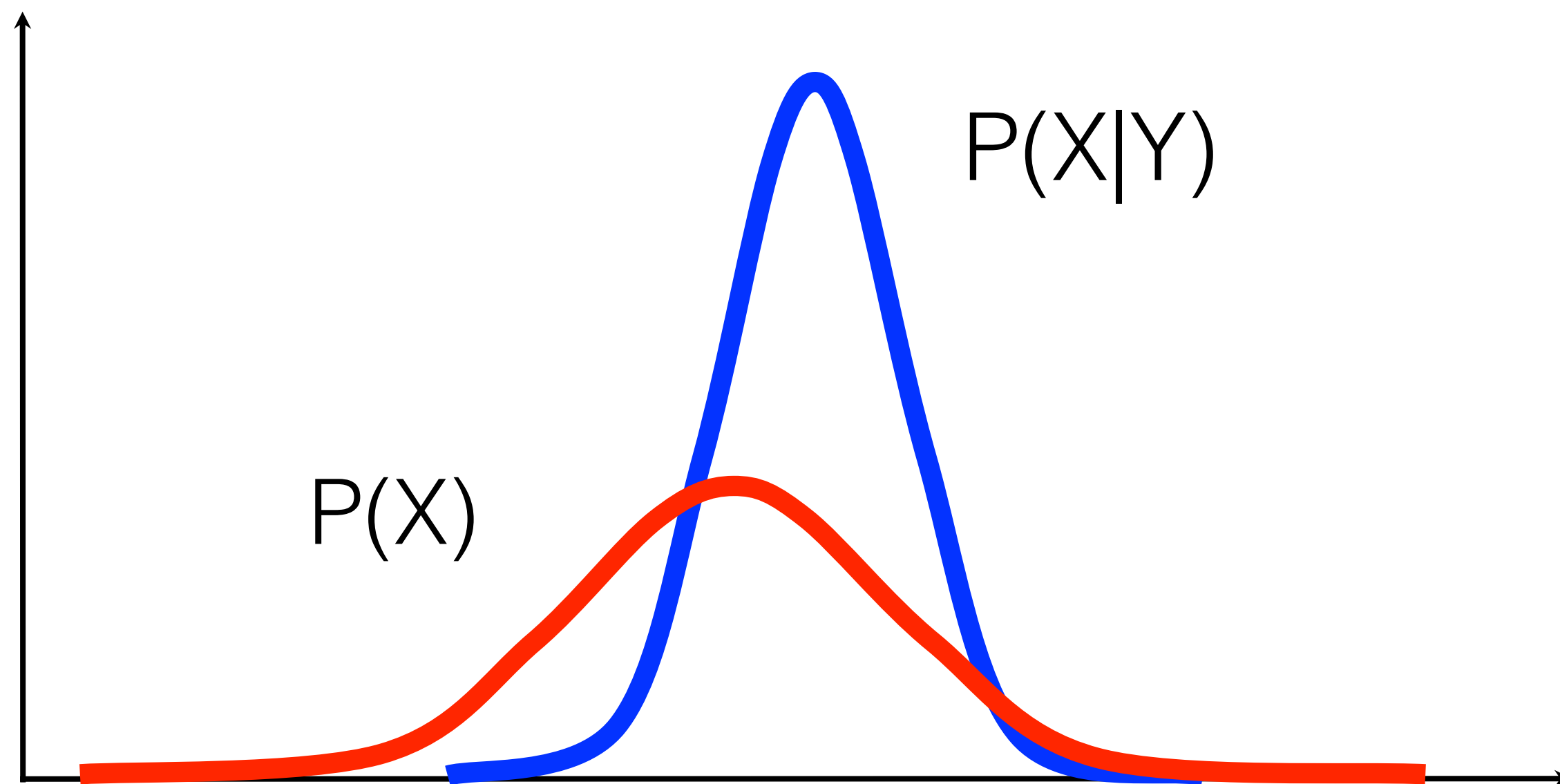
1.5
(bit)



0.33

Mutual Information

$$I(X; Y) = H(X) - H(X|Y) = H(Y) - H(Y|X)$$



Mutual Information

$$I(X; Y) = H(X) - H(X | Y) = H(Y) - H(Y | X)$$

$$I_m = H(r) - \sum_s P[s] H(r | s)$$

response $\uparrow$ stimulus $\swarrow$ (neural) noise entropy $\leftarrow$

$$I_m = - \sum_r P[r] \log_2 P[r] + \sum_{s,r} P[s] P[r|s] \log_2 P[r|s] \quad P[r] = \sum_s P[s] P[r|s]$$
$$= \sum_{s,r} P[s] P[r|s] \log_2 \left(\frac{P[r|s]}{P[r]} \right) = \sum_{s,r} P[r, s] \log_2 \left(\frac{P[r, s]}{P[r] P[s]} \right)$$

Mutual Information

$$I_m = \sum_{s,r} \underbrace{P[s]P[r|s]} \log_2 \left(\frac{P[r|s]}{P[r]} \right) = \sum_{s,r} \underbrace{P[r,s]} \log_2 \left(\frac{P[r,s]}{P[r]P[s]} \right)$$

$$P[r|s] \leftrightarrow P[r]$$

$$P[r,s] \leftrightarrow P[r]P[s]$$

Kullback-Leibler divergence

Mutual Information

$$H = -P[r_+] \log P[r_+] - (1 - P[r_+]) \log(1 - P[r_+])$$

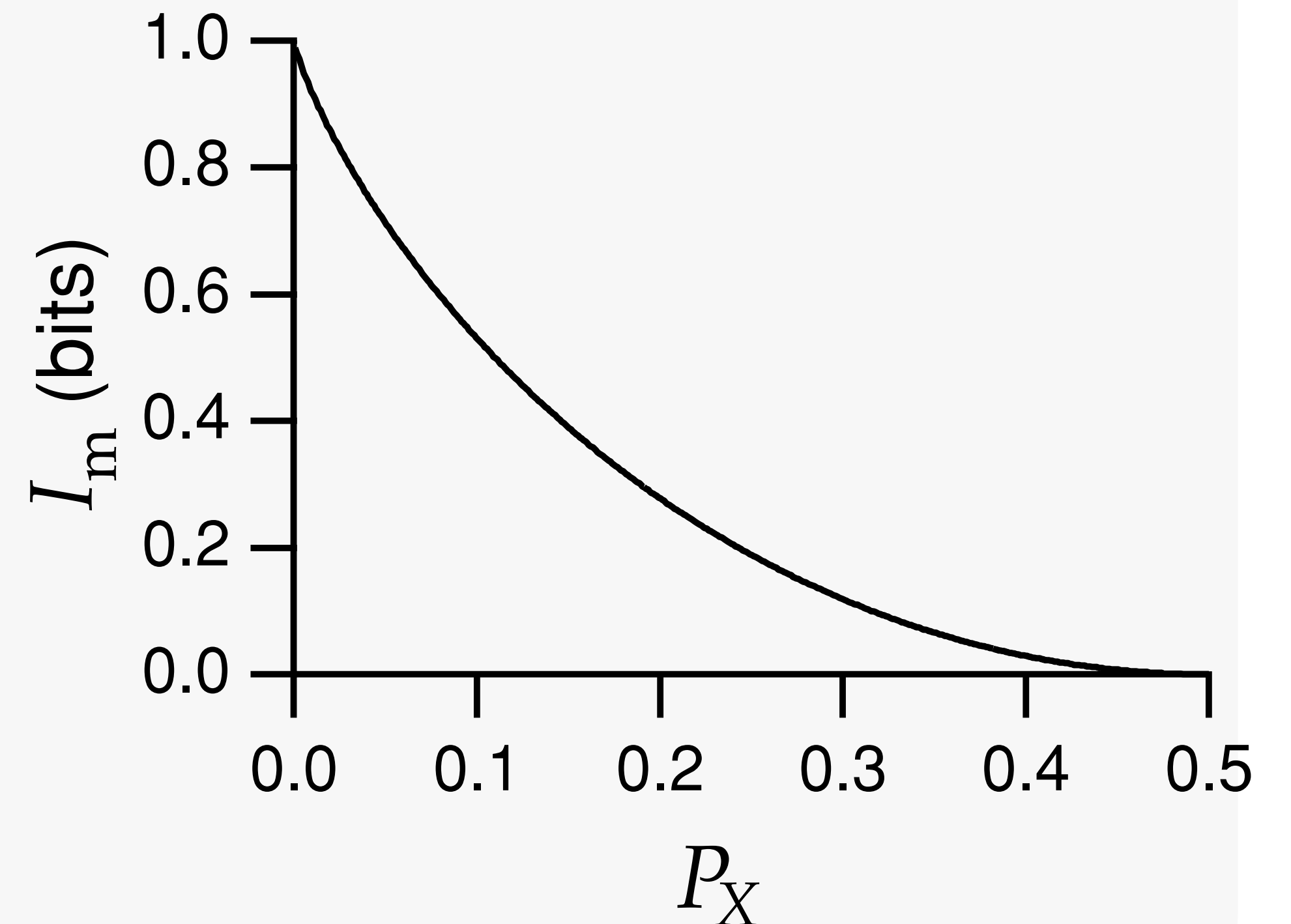
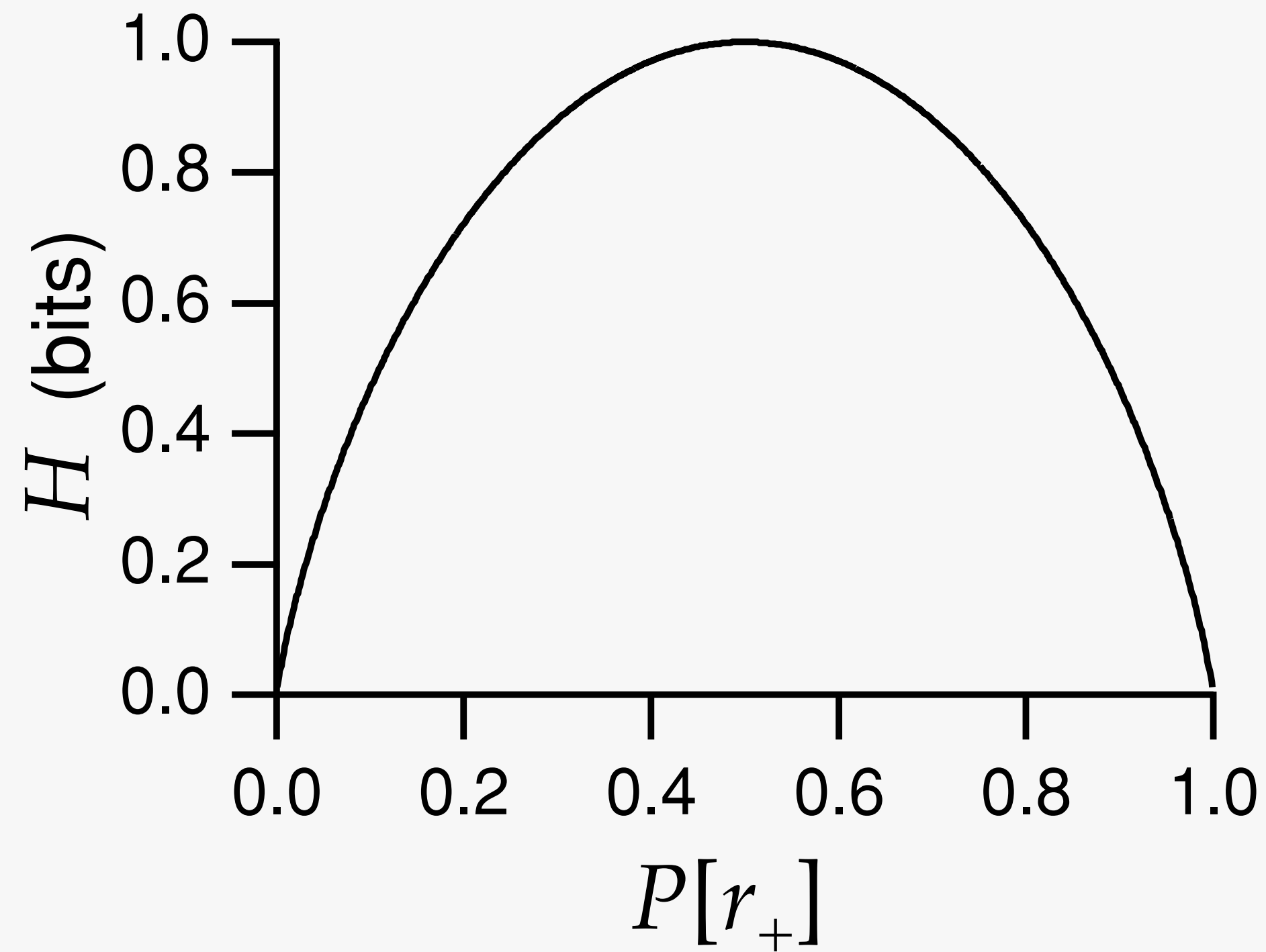
$$P_x = P[r_+ | +]$$

$$I_m = 1 + (1 - P_X) \log_2(1 - P_X) + P_X \log_2 P_X$$

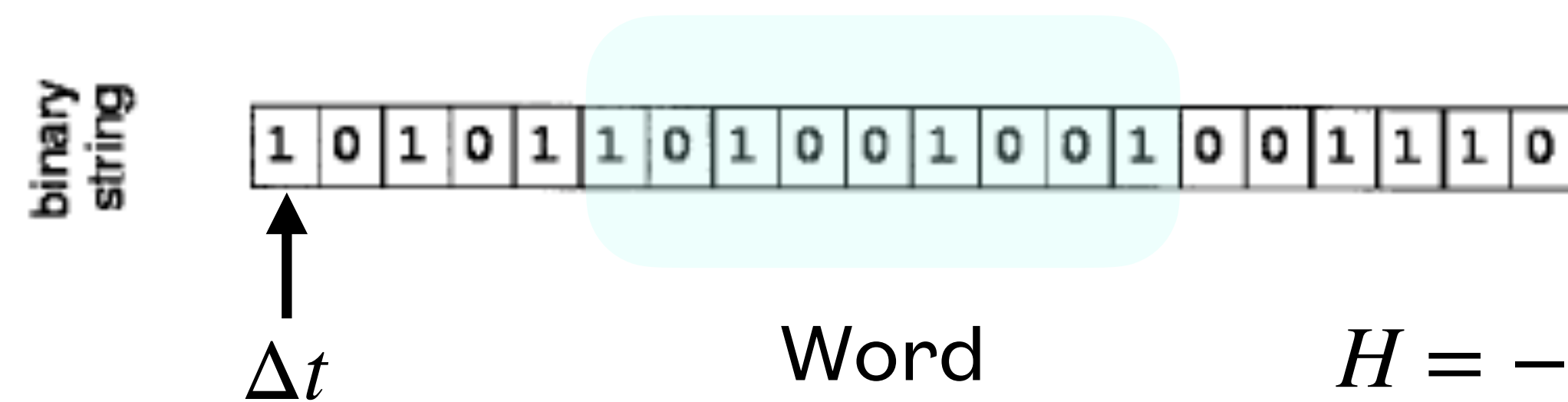
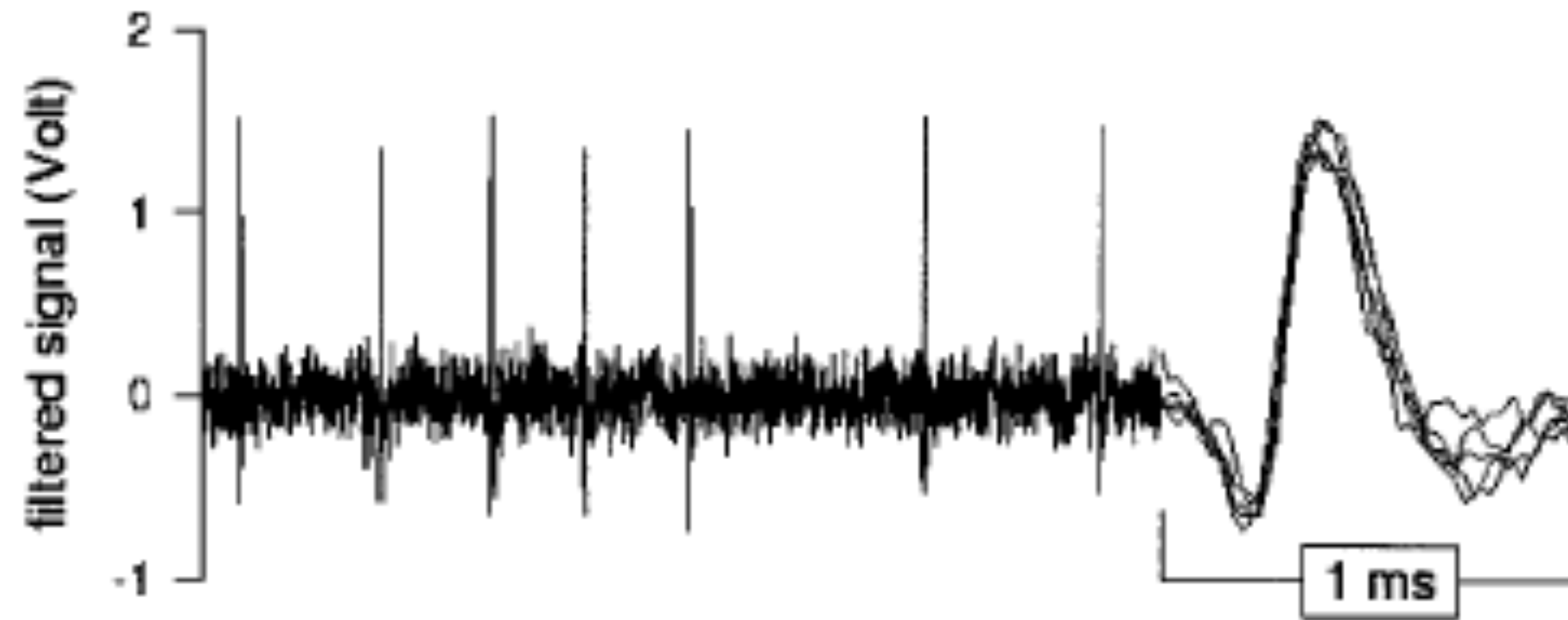
Stim Response

+ r_+

- r_-



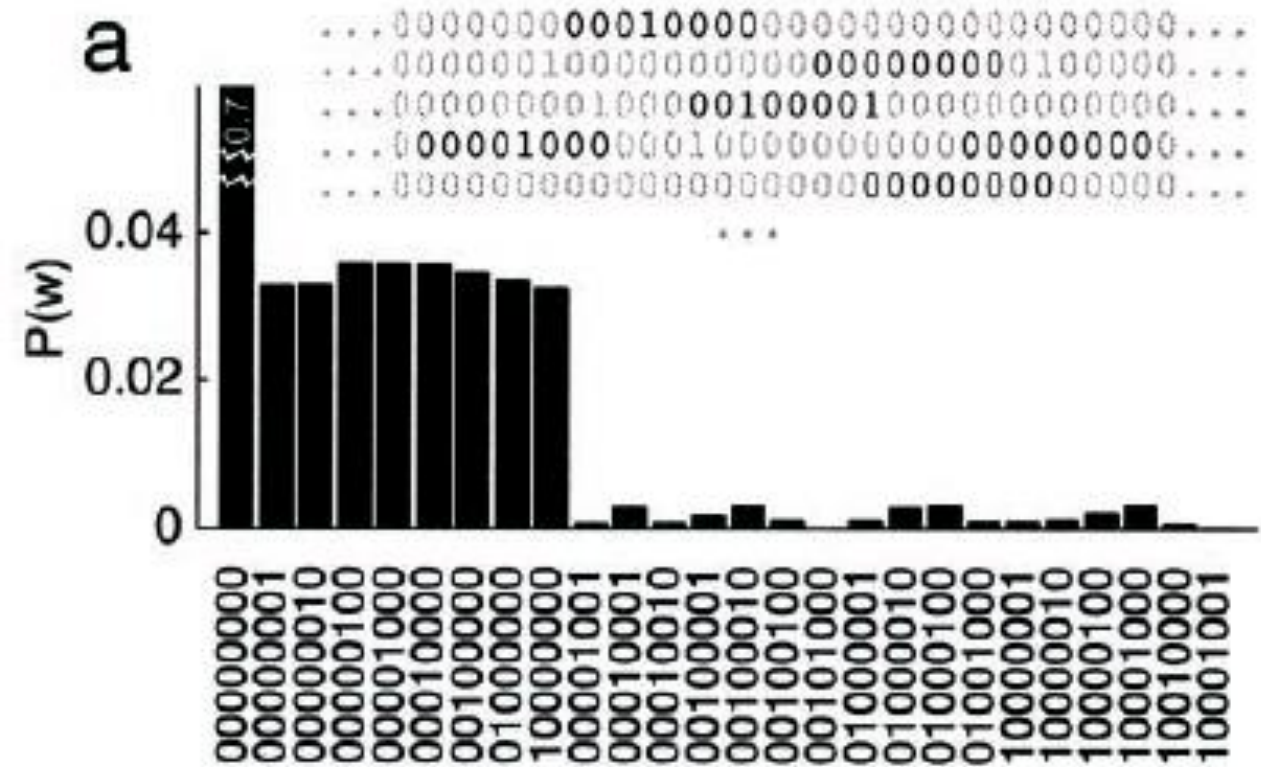
Spike train information



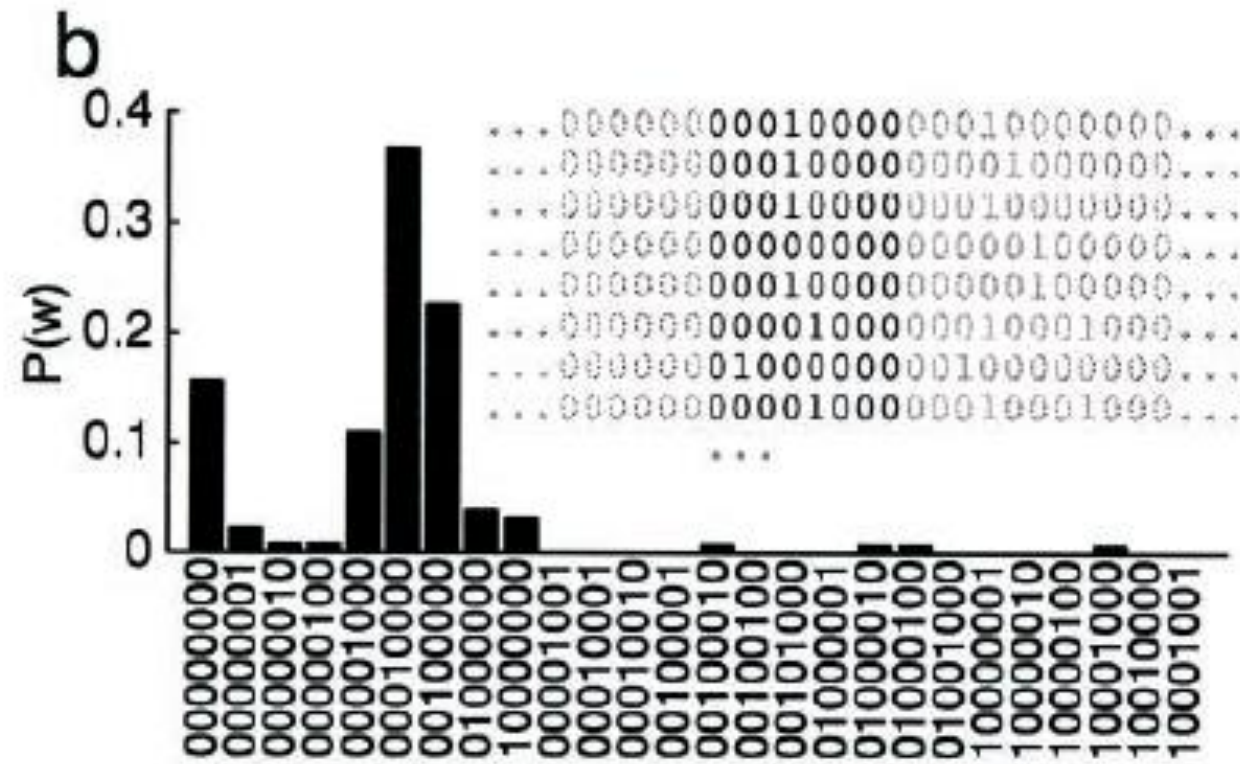
$$H = - \sum_i p(w_i) \log_2 p(w_i)$$

Spike train information

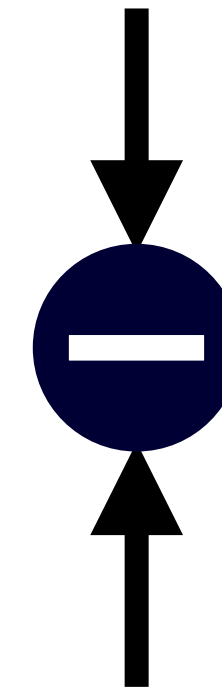
Noise



Stimuli



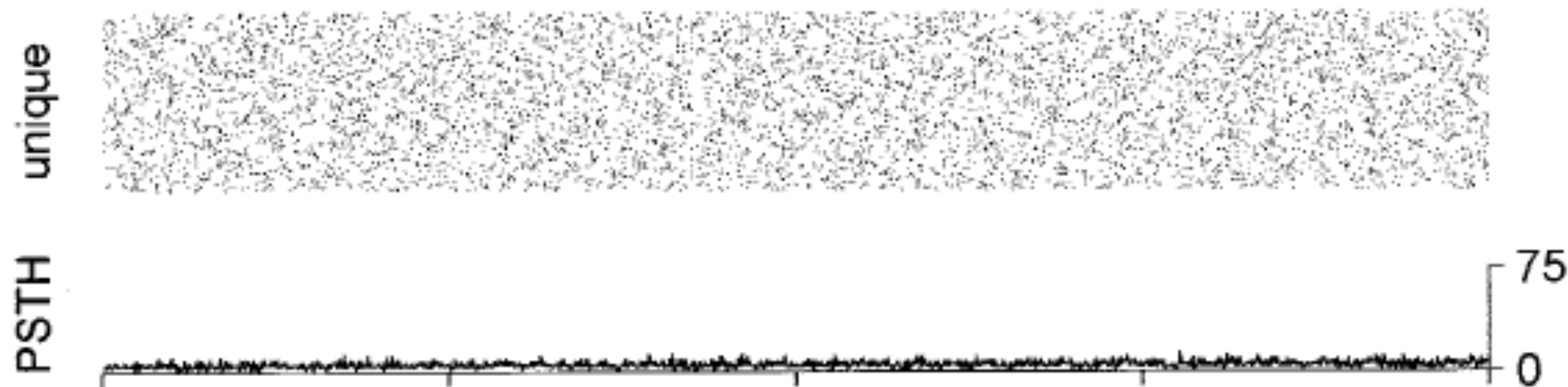
$$H = - \sum_i p(w_i | noise) \log_2 p(w_i | noise)$$



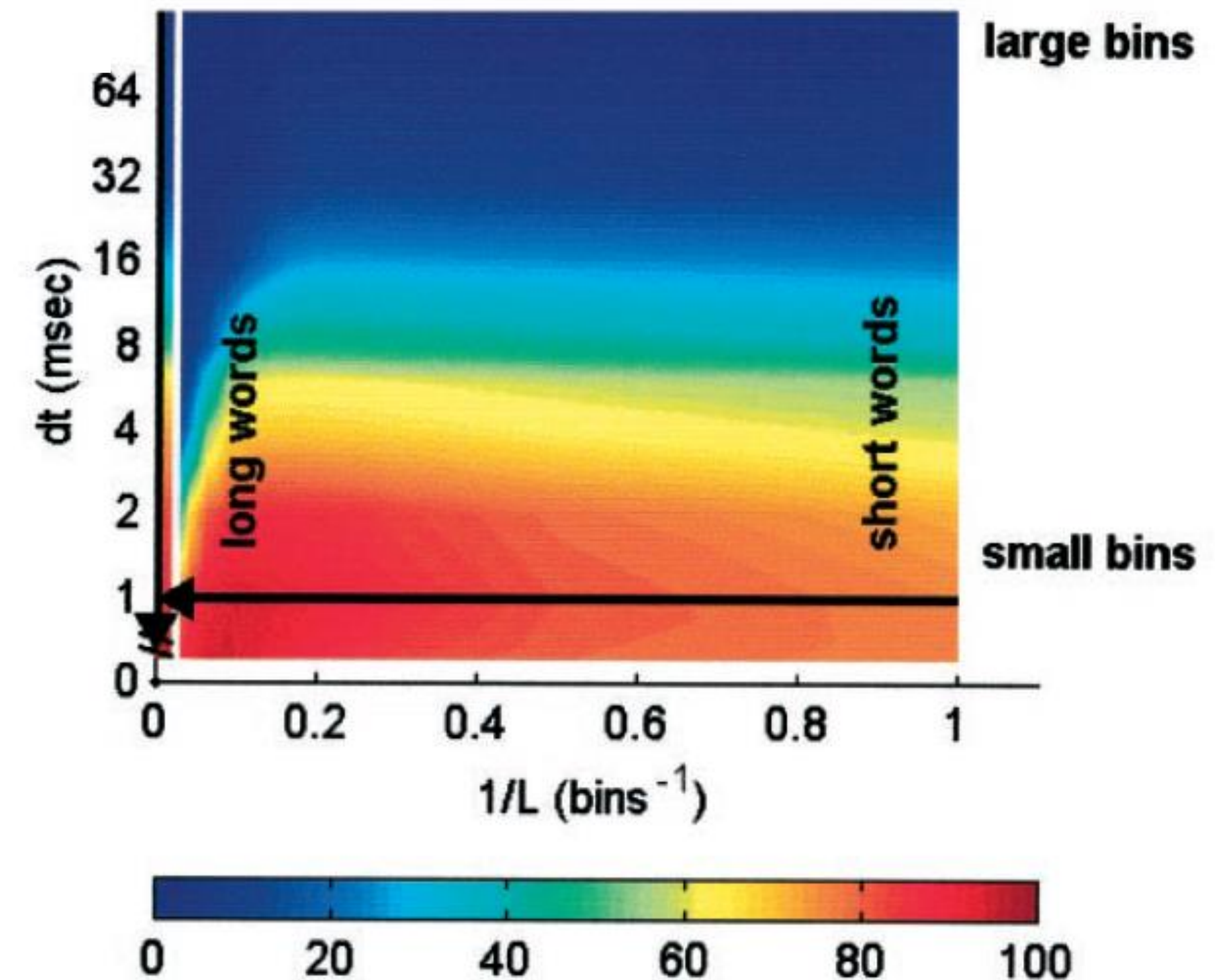
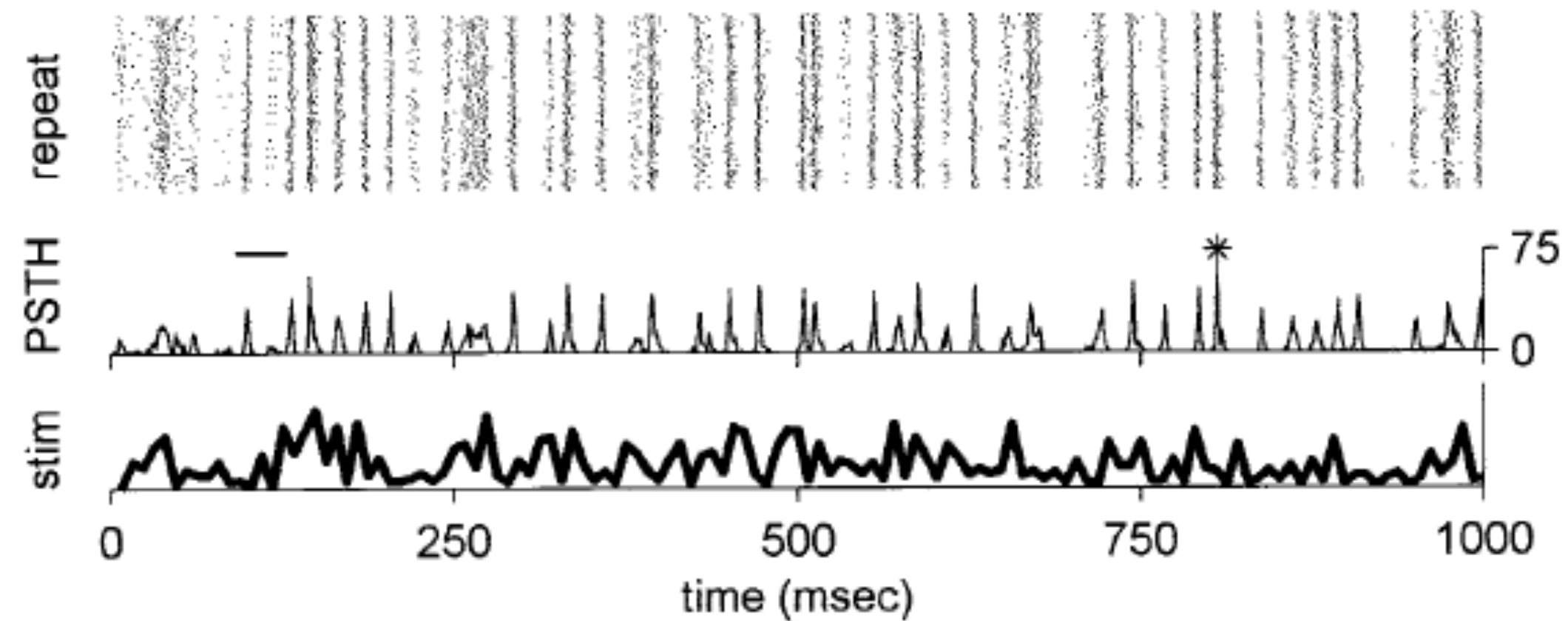
$$H = - \sum_i p(w_i | S_j) \log_2 p(w_i | S_j)$$

Spike train information

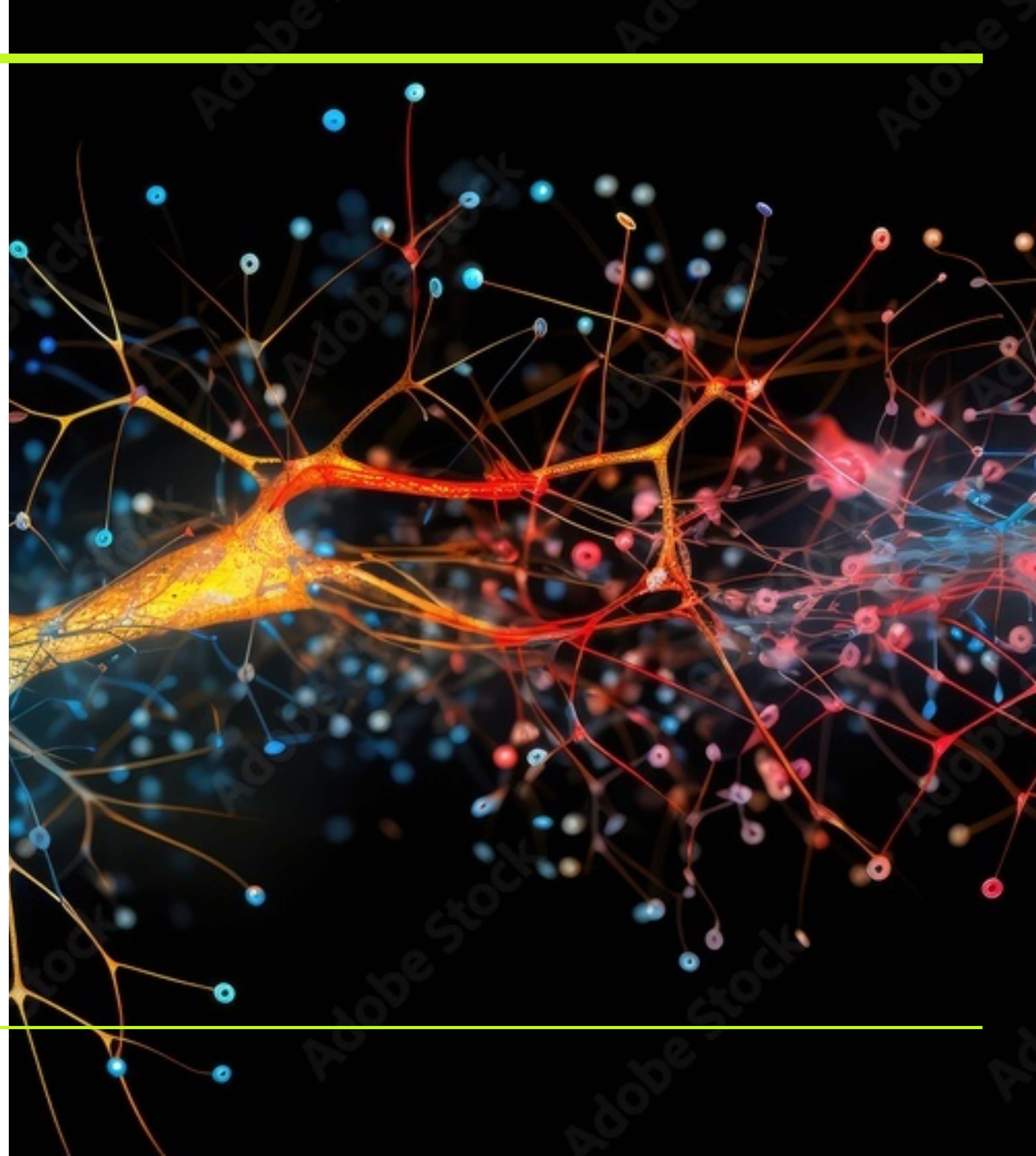
Noise



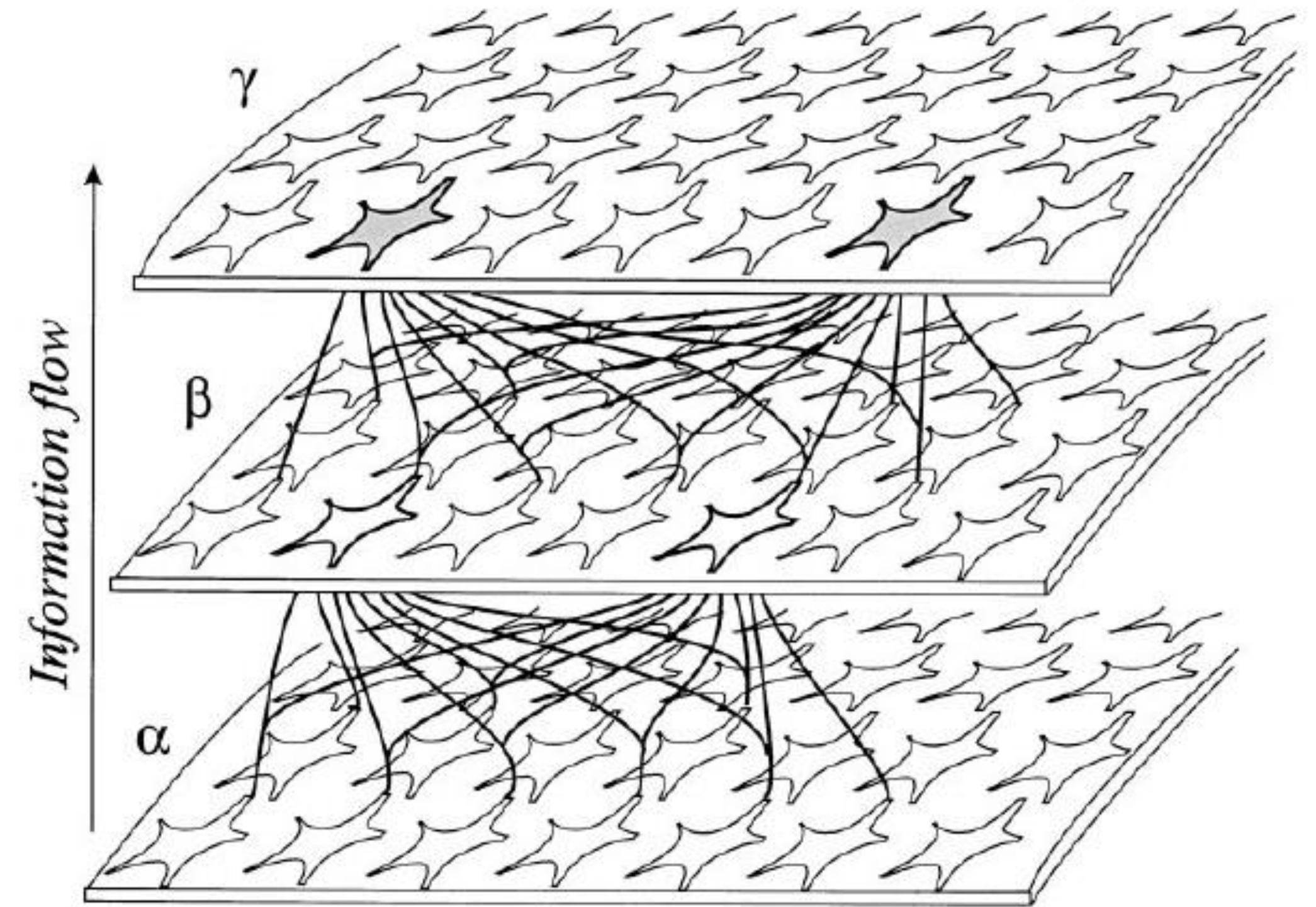
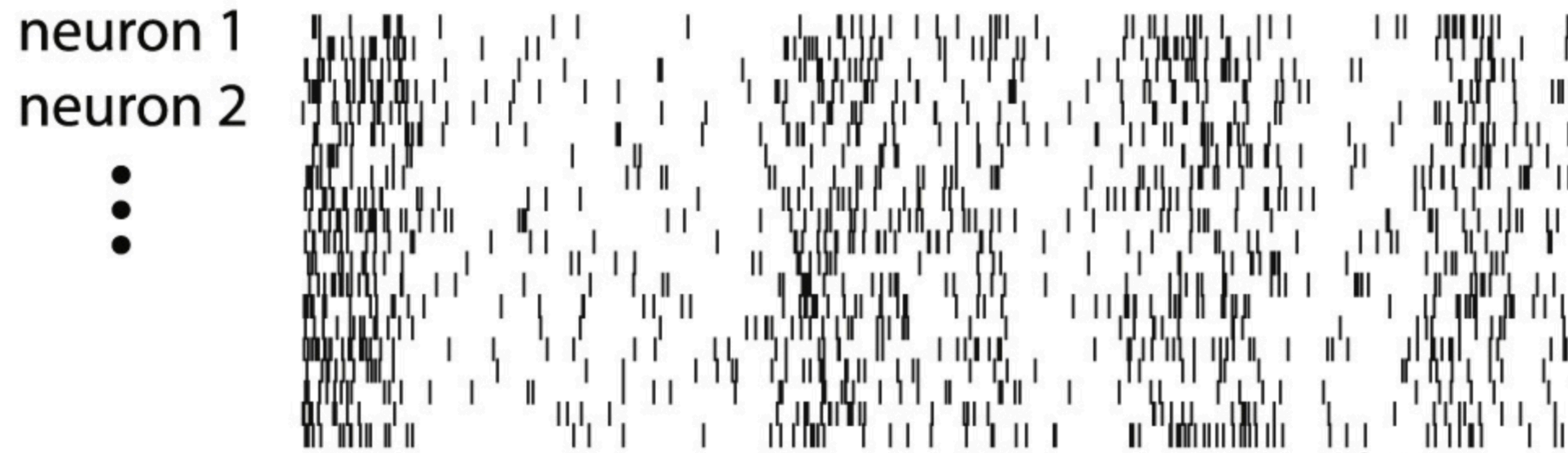
Stimuli



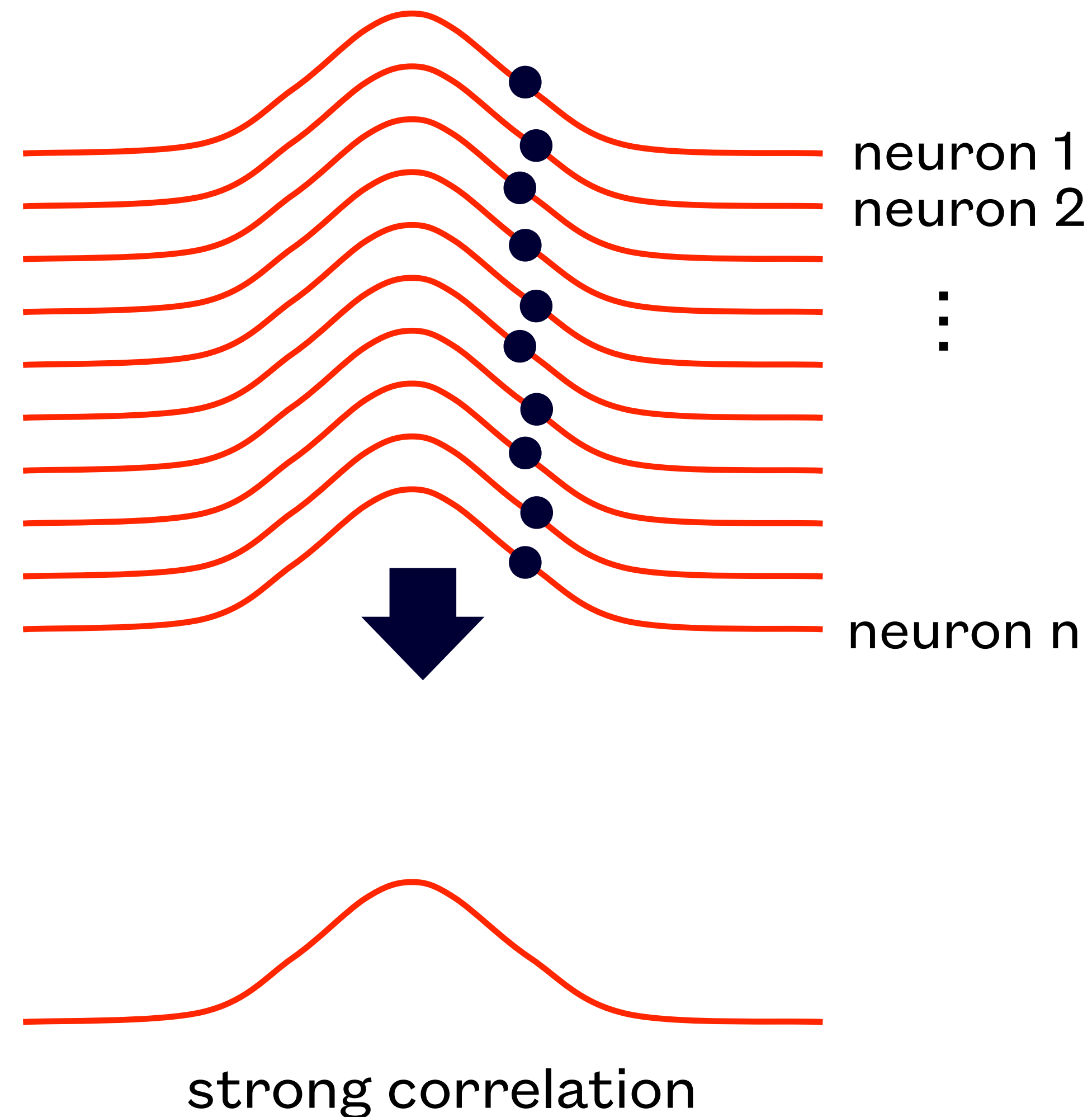
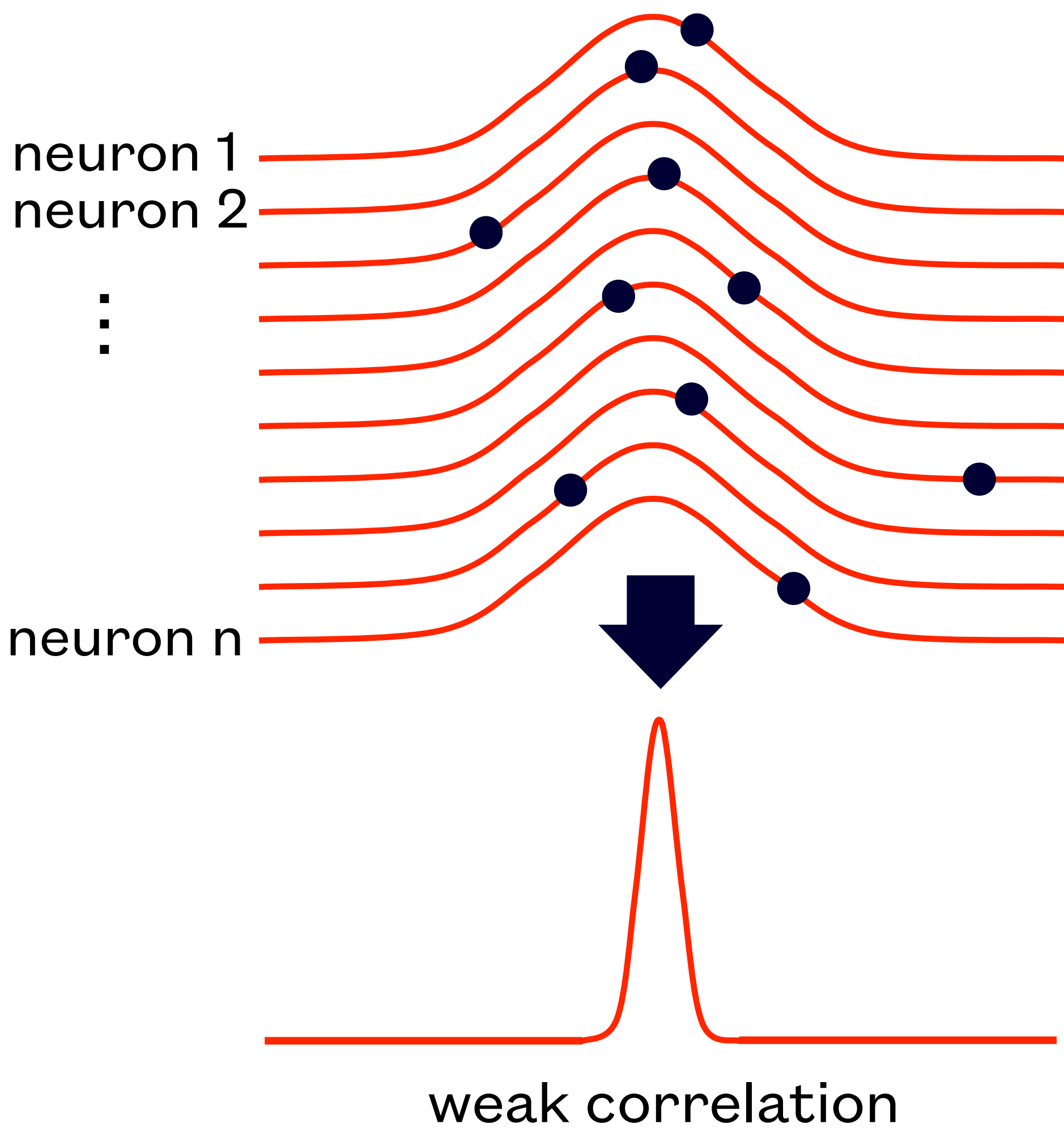
-
- ❖ Basics
 - ❖ Encoding
 - ❖ Decoding
 - ❖ Information Theory
 - ❖ **Population Analysis**
-



Neuronal responses are correlated

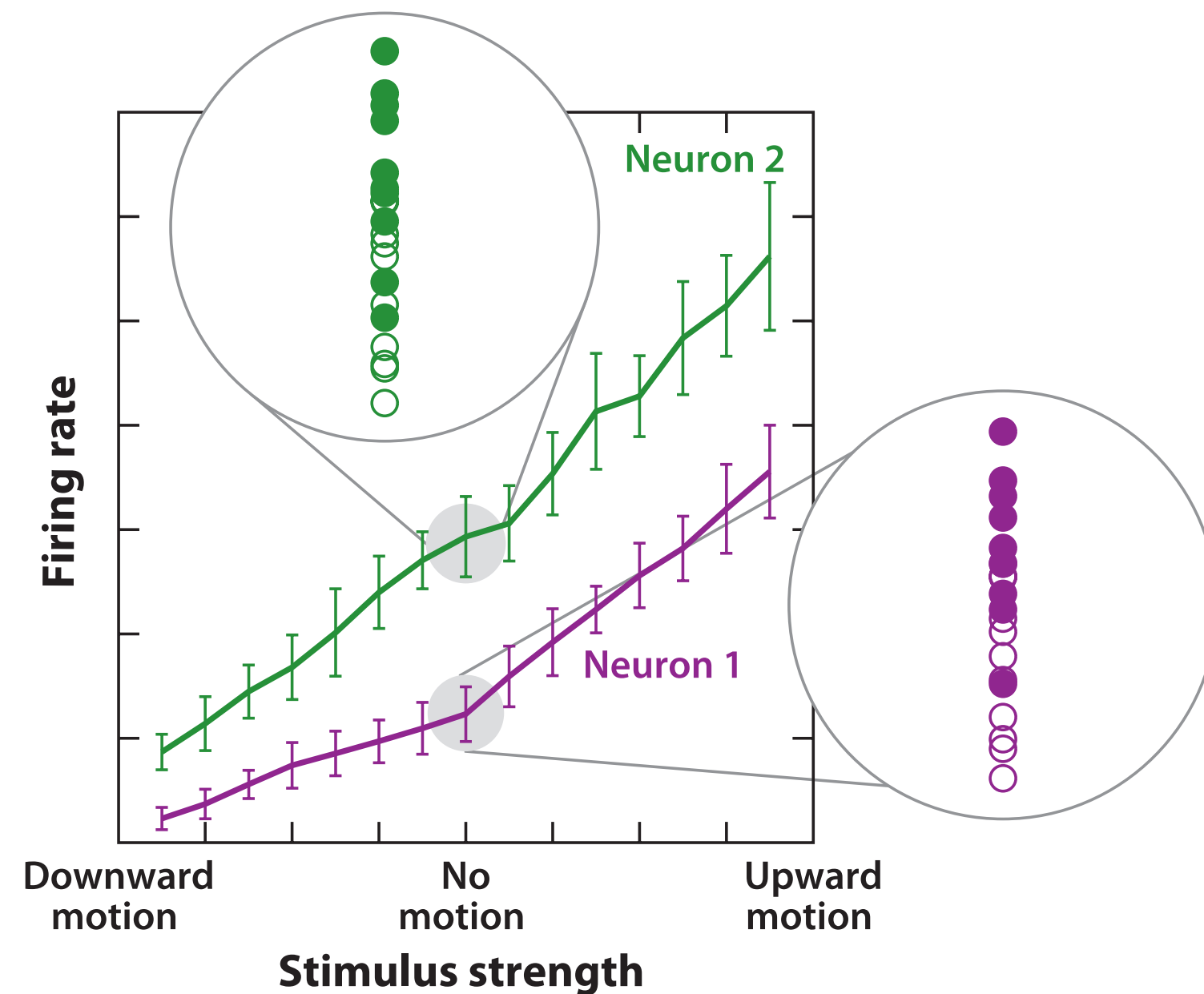


Correlation hurts

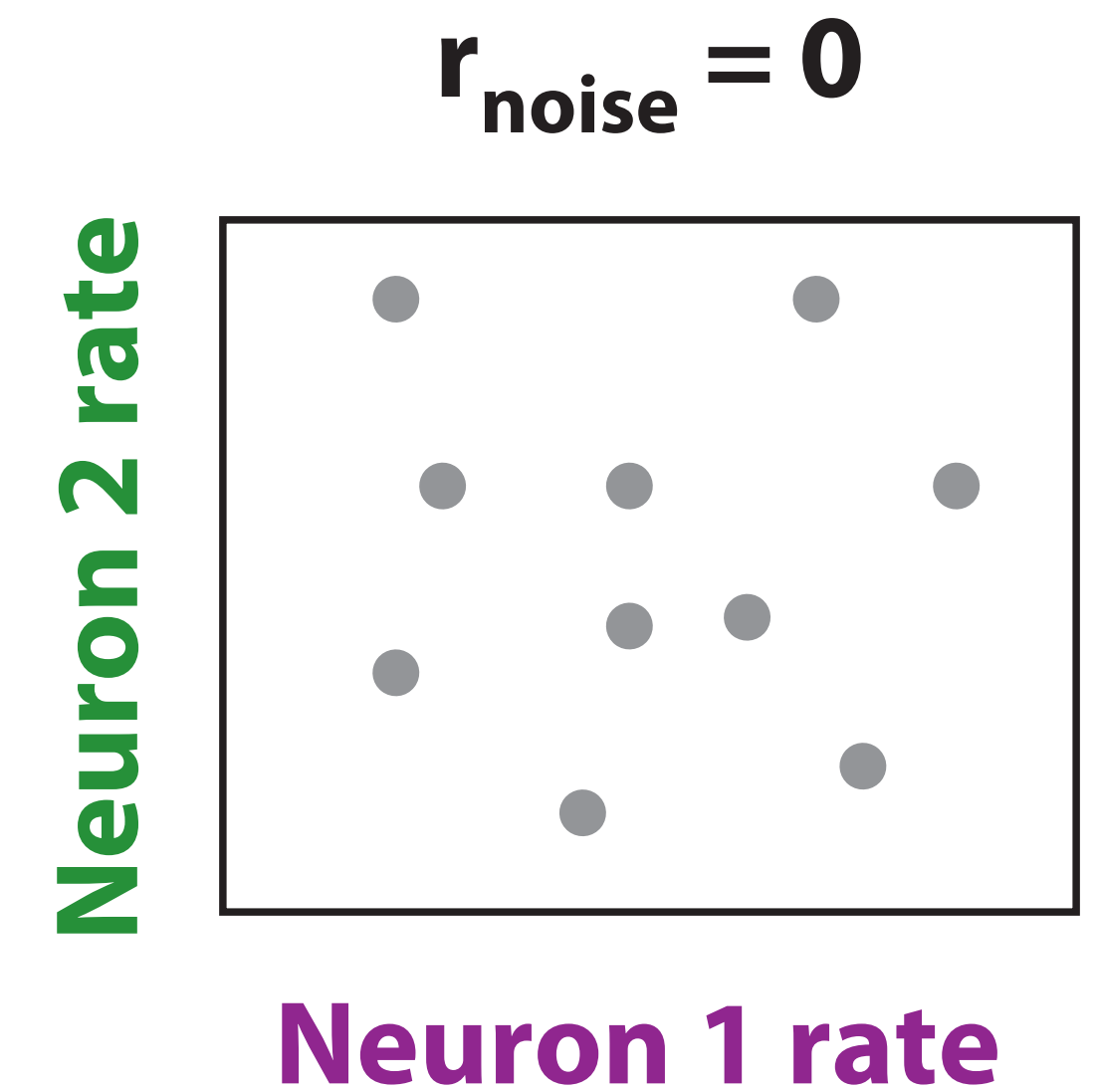
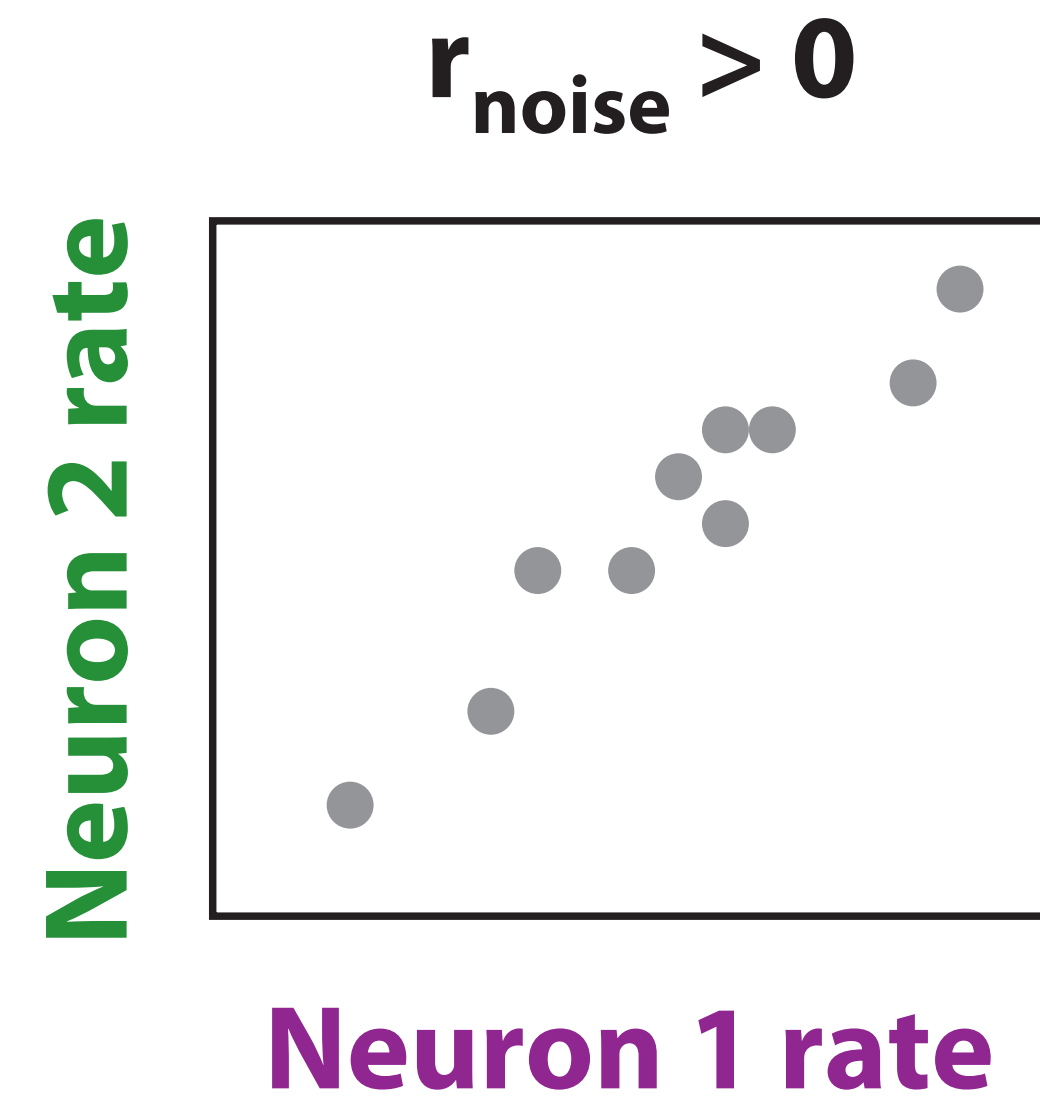


Signal vs. Noise Correlation

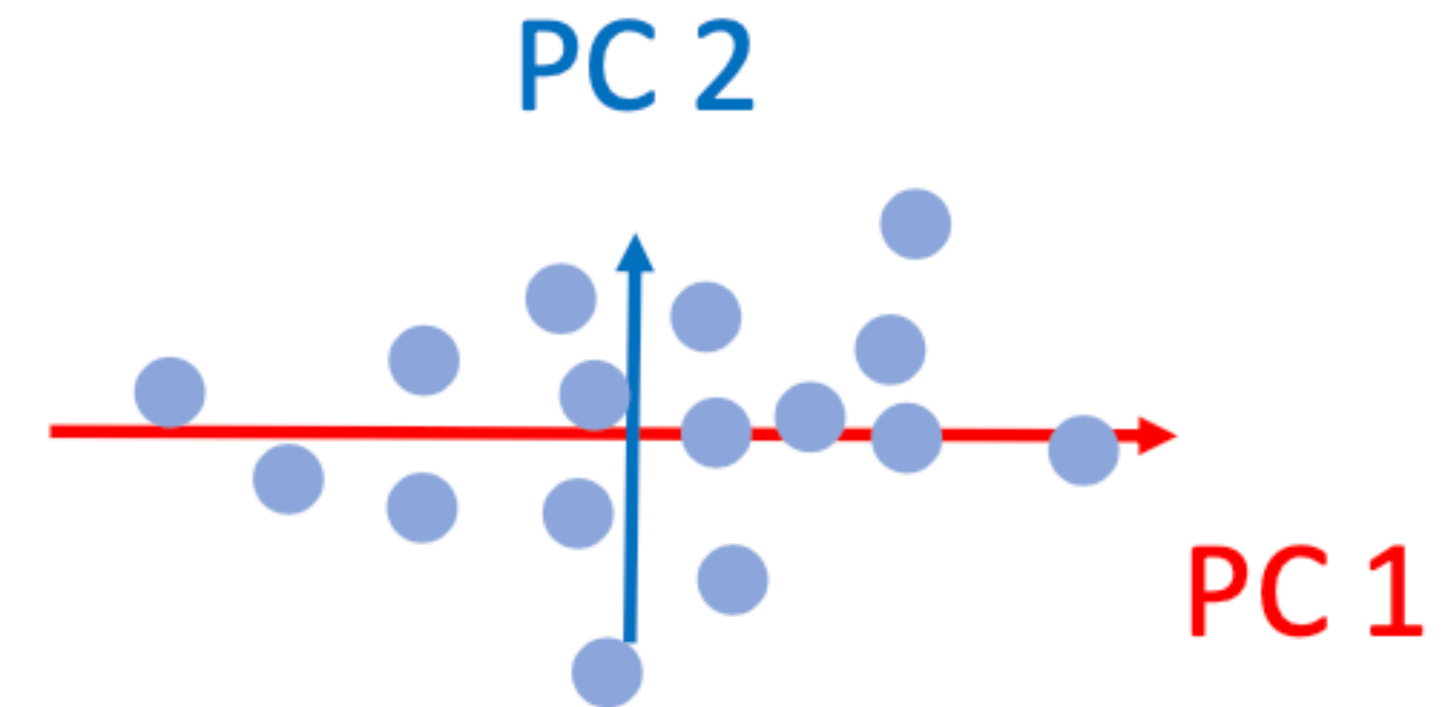
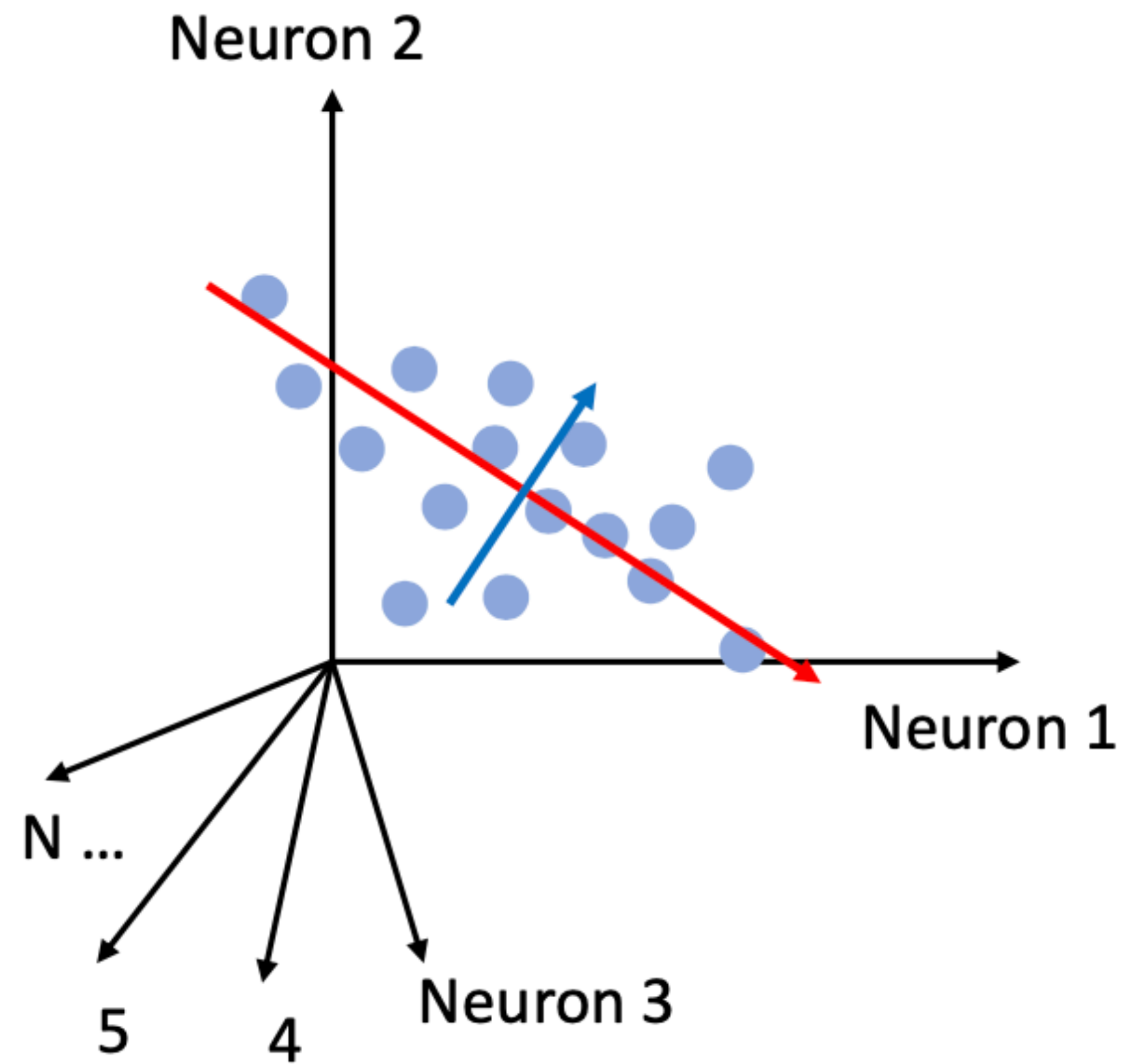
Signal Correlation



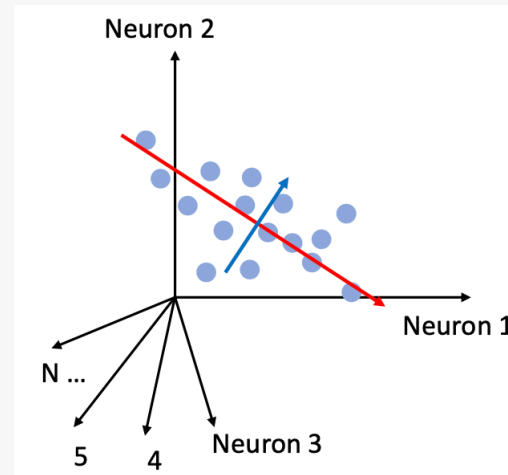
Noise Correlation



Principal Component Analysis



Principal Component Analysis

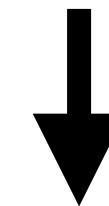


	Neuron 1	Neuron 2	Neuron 3
1	15	23	19
2	20	6	2
3	3	13	16
4	18	4	6
...			

$X \sim m \times n$

Normalize

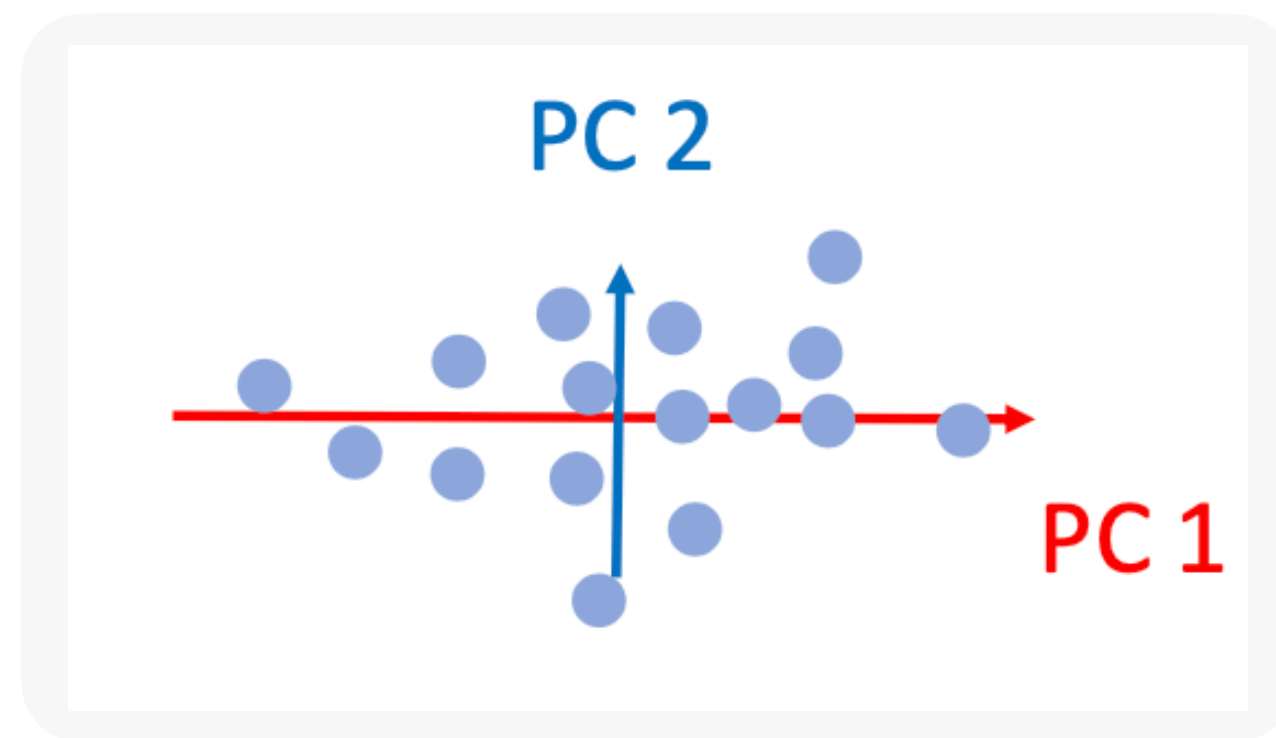
$$Cov = \frac{1}{m-1} X^T X = \begin{pmatrix} Cov(r_1, r_1) & Cov(r_1, r_2) & \dots & Cov(r_1, r_n) \\ Cov(r_2, r_1) & Cov(r_2, r_2) & \dots & Cov(r_2, r_n) \\ \vdots & \vdots & \ddots & \vdots \\ Cov(r_n, r_1) & Cov(r_n, r_2) & \dots & Cov(r_n, r_n) \end{pmatrix}$$



Eigenvalues & eigenvectors

$$C\vec{v} = \lambda\vec{v}$$

Rank by λ



Other Dimension Reduction Methods

- Linear:

PCA, FA, ICA, ...

- Nonlinear:

t-SNE, UMAP, ...

- Dynamic:

LFADS, GPFA, ...

Summary

- Basics
 - Poisson spiking, PSTH, Firing rate, Tuning curve
 - Encoding
 - Signal detection theory, Linear system
 - Decoding
 - MLE, Cramér-Rao bound, Fisher information
 - Information Theory
 - Information, Entropy, Mutual Information, Spike train information
 - Population Analysis
 - Noise correlation, PCA
-



Homework

- Complete the exercise code.

Project Idea

- Apply some of the analyses you learned today to an artificial neural network, and examine the encoding/decoding/information of its artificial neurons.
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